1.APPLICATIONS OF MATRICES AND DETERMINANTS

- If $|A| \neq 0$, then the square matrix A is called a **non-singular matrix.**
- If $|A| = 0$, then the square matrix A is called a **singular matrix.**
- **ADJOINT OF A SQUARE MATRIX:** Let A be the square matrix of order n, then the adjoint matrix of A is defined as the transpose of the matrix of cofactors of A. It is denoted by adj A. $\boldsymbol{adj\ A = [A_{ij}]^T = [(-1)^{i+j} M_{ij}]^T}$
- $A(adjA) = (adjA)A = |A|I_n$
- **INVERSE MATRIX:** Let A be a square matrix of order n. If there exists a square matrix B of order n such that $AB = BA = I_n$, then the matrix B is called an inverse of A.

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- $AA^{-1} = A^{-1}A = I_n$
- A singular matrix has no inverse.
- If A is non-singular, then
  $(i)|A^{-1}| = \frac{1}{|A|}$ $(ii)\ (A^T)^{-1} = (A^{-1})^T$ $(iii)(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$
- If A is non-singular, then $(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$, where $\boldsymbol{\lambda}$ is a non-zero scalar.
- $(AB)^{-1} = B^{-1}A^{-1}$
- If A is non-singular, then A^{-1} is also non-singular and $(A^{-1})^{-1} = A$
- If A is non-singular square matrix of order n, then
  $(i)(adjA)^{-1} = adj(A^{-1}) = \frac{1}{|A|}A \implies adj\ A = |A|A^{-1}$
  $(ii)|adjA| = |A|^{n-1}$ $(iii)adj(adjA) = |A|^{n-2}A$
  $(iv)adj(\lambda A) = \lambda^{n-1}adj(A)$ $(v)|adj(adjA)| = |A|^{(n-1)^2}$
  $(vi)(adjA)^T = adj(A^T)$ $(vii)adj(AB) = (adjB)(adjA)$
- $|adj\ A| = |A|^2 \implies |A| = \pm\sqrt{|adj\ A|}$
- $A^{-1} = \pm\frac{1}{\sqrt{|adjA|}}adjA$
- $A = \pm\frac{1}{\sqrt{|adjA|}}adj(adjA)$
- If A and B are any two non-singular square matrices of order n, then $adj(AB) = (adj\ B)(adj\ A)$
- A square matrix A is called orthogonal if $AA^T = A^TA = I$
- A is orthogonal if and only if A is non-singular and $A^{-1} = A^T$
- Application of matrices is in computer graphic and cryptography.

THEOREM 1: For every square matrix A of order n, $\boldsymbol{A(adj\ A) = (adj\ A)A = |A|I_n}$

Proof:

For simplicity, we prove the theorem for n=3 only.

Consider $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $adj\ A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$

$$A(adj\ A) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix}$$

$$= |A|\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A|I_3 \rightarrow (1)$$

$$(adj\ A)A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix}$$

$$= |A|\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A|I_3 \rightarrow (2)$$

From (1) & (2), we get $A(adj\ A) = (adj\ A)A = |A|I_n$

THEOREM 2: If a square matrix has an inverse, then it is unique.

Proof:

Let A be a square matrix of order n such that an inverse of A exists. If possible, let there be two inverse B and C of A.

By definition, $AB = BA = I_n\ and\ AC = CA = I_n$

$C = CI_n = C(AB) = (CA)B = I_nB = B \implies B = C$

Hence it is proved.

THEOREM 3: Let A be a square matrix of order n. Then $\boldsymbol{A^{-1}}$ exists if and only if A is non-singular.

Proof:

Case (i) : Suppose that A^{-1} exists, then $AA^{-1} = A^{-1}A = I_n$.

By the product rule for determinants,

$det(AA^{-1}) = det(A^{-1})det(A) = det(I_n) = 1 \implies |A| \neq 0$.

Hence A is non-singular.

Case (ii) : Conversely, suppose A is non-singular, then $|A| \neq 0$.

W.k.T. $A(\text{adj } A) = (\text{adj } A)A = |A|I_n$

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$\div by\ |A| \Longrightarrow A\left(\frac{1}{|A|}adj\ A\right) = \left(\frac{1}{|A|}adj\ A\right)A = I_n$

$\Longrightarrow AB = BA = I_n\ ; B = \frac{1}{|A|}adj\ A$

Hence the inverse of A exists and the inverse is $A^{-1} = \frac{1}{|A|}adj\ A$

PROPERTIES OF INVERSES OF MATRICES

THEOREM 4: If A is non-singular, then $\left|A^{-1}\right| = \frac{1}{|A|}$

Proof:

Let A be non-singular then $|A| \neq 0$ and A^{-1} exists.

By definition, $AA^{-1} = A^{-1}A = I_n \Longrightarrow |AA^{-1}| = |A^{-1}A| = |I_n|$

By the product rule for determinants,

$|AA^{-1}| = |I_n| \Longrightarrow |A||A^{-1}| = 1 \Longrightarrow |A^{-1}| = \frac{1}{|A|}$

THEOREM 5: If A is non-singular, then $(A^T)^{-1} = (A^{-1})^T$

Proof:

By definition, $AA^{-1} = A^{-1}A = I_n$

Taking transpose ,$(AA^{-1})^T = (A^{-1}A)^T = (I_n)^T$

By the reversal law of transpose, $(A^{-1})^T A^T = A^T(A^{-1})^T = |I_n|$

Hence $(A^T)^{-1} = (A^{-1})^T$

THEOREM 6: If A is non-singular, then $(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$, where λ is a non-zero scalar.

Proof:

By definition, $AA^{-1} = A^{-1}A = I_n$

Muliply & divide by λ, $(\lambda A)\left(\frac{1}{\lambda}A^{-1}\right) = \left(\frac{1}{\lambda}A^{-1}\right)(\lambda A) = I_n \Longrightarrow (\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$

THEOREM 7: LEFT CANCELLATION LAW

Let A,B and C be square matrices of order n. If A is non-singular and AB=AC then B=C

Proof:

Since A is non-singular, A^{-1} exists and $AA^{-1} = A^{-1}A = I_n$.

Given AB=AC. Pre multiply by A^{-1} on both sides,

$A^{-1}(AB) = A^{-1}(AC) \Longrightarrow (A^{-1}A)B = (A^{-1}A)C \Longrightarrow I_nB = I_nC \Longrightarrow B = C$

THEOREM 8: RIGHT CANCELLATION LAW

Let A,B and C be square matrices of order n. If A is non-singular and BA=CA then B=C

Proof:

Since A is non-singular, A^{-1} exists and $AA^{-1} = A^{-1}A = I_n$.

Given BA=CA. Post multiply by A^{-1} on both sides,

$(BA)A^{-1} = (CA)A^{-1} \Longrightarrow B(AA^{-1}) = C(AA^{-1}) \Longrightarrow BI_n = CI_n \Longrightarrow B = C$

THEOREM 9: REVERSAL LAW FOR INVERSES

If A and B are non-singular matrices of the same order, then the product AB is also non-singular and $(AB)^{-1} = B^{-1}A^{-1}$

Proof:

Since A and B are non-singular matrices of the same order n, then $|A| \neq 0$, $|B| \neq 0 \Longrightarrow A^{-1}\ and\ B^{-1}$ of order n exists. The product of AB and $B^{-1}A^{-1}$ can also found.

$|AB| = |A||B| \neq 0 \Longrightarrow AB$ is also non-singular and $(AB)^{-1}$ exists.

$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I_n)A^{-1} = AA^{-1} = I_n$

$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}(I_n)B = B^{-1}B = I_n$

Hence $(AB)^{-1} = B^{-1}A^{-1}$

THEOREM 10: LAW OF DOUBLE INVERSE

If A is non-singular, then A^{-1} is also non-singular and $(A^{-1})^{-1} = A$

Proof:

Since A is non-singular,then $|A| \neq 0$ and A^{-1} exists.

Now $|A^{-1}| = \frac{1}{|A|} \neq 0 \Longrightarrow A^{-1}$ is also non-singular and $AA^{-1} = A^{-1}A = I_n$.

Now $AA^{-1} = I_n \Longrightarrow (AA^{-1})^{-1} = I \Longrightarrow (A^{-1})^{-1}A^{-1} = I$

Post- multiply by A on both sides, $(A^{-1})^{-1} = A$

THEOREM 11: If A is non-singular square matrix of order n, then $(\mathbf{adjA})^{-1} = \mathbf{adj}(\mathbf{A^{-1}}) = \frac{\mathbf{1}}{|\mathbf{A}|}\mathbf{A}$

Proof:

$A^{-1} = \frac{1}{|A|}(adj\ A) \Longrightarrow adj\ A = |A|A^{-1}$

$\Longrightarrow (adj\ A)^{-1} = (|A|A^{-1})^{-1} = (A^{-1})^{-1}|A|^{-1} = \frac{1}{|A|}\mathrm{A} \rightarrow (1)$

$A^{-1} = \frac{1}{|A|}(adj\ A) \Longrightarrow adj\ A = |A|A^{-1}$

Replacing A by A^{-1}, $adj\ (A^{-1}) = |A^{-1}|(A^{-1})^{-1} = \frac{1}{|A|}\mathrm{A} \rightarrow (2)$

From (1) and (2), $(\mathrm{adjA})^{-1} = \mathrm{adj}(\mathrm{A}^{-1}) = \frac{1}{|A|}\mathrm{A}$

THEOREM 12: If A is non-singular square matrix of order n, then $|\mathbf{adjA}| = |\mathbf{A}|^{\mathbf{n-1}}$

Proof:

$\mathrm{A(adj\ A)} = \mathrm{(adj\ A)A} = |\mathrm{A}|\mathrm{I_n} \Longrightarrow det[\mathrm{A(adj\ A)}] = \det(A)\det(adjA) = \det[|\mathrm{A}|\mathrm{I_n}]$

$\Longrightarrow |A||adj\ A| = |A|^n \Longrightarrow |\mathrm{adjA}| = |\mathrm{A}|^{\mathrm{n-1}}$

THEOREM 13: If A is non-singular square matrix of order n, then $\mathbf{adj}(\mathbf{adjA}) = |\mathbf{A}|^{\mathbf{n-2}}\mathbf{A}$

Proof:

For any non-singular matrix B of order n, $\mathrm{B(adj\ B)} = \mathrm{(adj\ B)B} = |\mathrm{B}|\mathrm{I_n}$

Put $B = adjA \Longrightarrow adjA\big(\mathrm{adj}\ (adjA)\big) = |adjA|\mathrm{I_n}$

$\Longrightarrow adjA\big(\mathrm{adj}\ (adjA)\big) = |\mathrm{A}|^{\mathrm{n-1}}\mathrm{I_n}$

Pre-multiply by A on both sides,

$\Longrightarrow [A(adjA)]\big(\mathrm{adj}\ (adjA)\big) = A|\mathrm{A}|^{\mathrm{n-1}}\mathrm{I_n} \Longrightarrow |\mathrm{A}|\mathrm{I_n}\big(\mathrm{adj}\ (adjA)\big) = A|\mathrm{A}|^{\mathrm{n-1}}\mathrm{I_n}$

$\Longrightarrow \mathrm{adj}\ (adjA) = |\mathrm{A}|^{\mathrm{n-2}}A$

THEOREM 14: If A is non-singular square matrix of order n, then $\mathbf{adj}(\boldsymbol{\lambda}\mathbf{A}) = \boldsymbol{\lambda}^{\mathbf{n-1}}\mathbf{adj}(\mathbf{A})$

Proof:

W.K.T, $(\mathrm{adjA})^{-1} = \frac{1}{|\mathrm{A}|}\mathrm{A} \Longrightarrow adj\ A = |A|A^{-1}$

Replace A by $\lambda\mathrm{A} \Longrightarrow adj\ (\lambda\mathrm{A}) = |\lambda\mathrm{A}|(\lambda\mathrm{A})^{-1} = \lambda^n|A|\frac{1}{\lambda}A^{-1} = \lambda^{n-1}|A|A^{-1} = \lambda^{n-1}\mathrm{adj(A)}$

THEOREM 15: If A is non-singular square matrix of order n, then $|\mathbf{adj}(\mathbf{adjA})| = |\mathbf{A}|^{(\mathbf{n-1})^2}$

Proof:

W.K.T, $\mathrm{adj}\ (adjA) = |\mathrm{A}|^{\mathrm{n-2}}A$

$\Longrightarrow |\mathrm{adj}\ (adjA)| = ||\mathrm{A}|^{\mathrm{n-2}}A| = (|\mathrm{A}|^{\mathrm{n-2}})^n|A| = |A|^{n^2-2n+1} = |\mathrm{A}|^{(\mathrm{n-1})^2}$

THEOREM 16: If A is non-singular square matrix of order n, then$(\mathbf{adjA})^{\mathbf{T}} = \mathbf{adj}(\mathbf{A^T})$

Proof:

W.K.T, $A^{-1} = \frac{1}{|A|}(adj\ A)$

Replace A by $A^T \Longrightarrow (A^T)^{-1} = \frac{1}{|A^T|}\big(adj\ (A^T)\big)$

$\Longrightarrow adj\ (A^T) = |A^T|(A^T)^{-1} = (|A|A^{-1})^T = (\mathrm{adjA})^{\mathrm{T}}$

THEOREM 17: If A and B are any two non-singular matrices of order n, then $\mathbf{adj}(\mathbf{AB}) = (\mathbf{adjB})(\mathbf{adjA})$

Proof:

W.K.T, $adj\ A = |A|A^{-1}$

Replace A by AB $\Longrightarrow adj\ (AB) = |AB|(AB)^{-1} = |A||B|A^{-1}B^{-1}$

$= (|B|B^{-1})(|A|A^{-1}) = (\mathrm{adjB})(\mathrm{adjA})$

EXERCISE 1.1

1.Find the adjoint of the following matrices.

(i) $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$

Let $A = \begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix} \Longrightarrow adj\ A = \begin{bmatrix} 2 & -4 \\ -6 & -3 \end{bmatrix}$

(ii) $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$

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Let $A=\begin{bmatrix}2&3&1\\3&4&1\\3&7&2\end{bmatrix}$

$$\begin{matrix}4&7&3&4\\1&2&1&1\\3&3&2&3\\4&7&3&4\end{matrix} \Rightarrow adj\ A=\begin{bmatrix}8-7&7-6&3-4\\3-6&4-3&3-2\\21-12&9-14&8-9\end{bmatrix}=\begin{bmatrix}1&1&-1\\-3&1&1\\9&-5&-1\end{bmatrix}$$

(iii) $\frac{1}{3}\begin{bmatrix}2&2&1\\-2&1&2\\1&-2&2\end{bmatrix}$

$$\boxed{adj(\lambda A)=\lambda^{n-1}adj(A)}$$

$$\begin{matrix}1&-2&2&1\\2&2&1&2\\-2&1&2&-2\\1&-2&2&1\end{matrix} \Rightarrow adj\ A=\left(\frac{1}{3}\right)^{3-1}\begin{bmatrix}2+4&-2-4&4-1\\2+4&4-1&-2-4\\4-1&2+4&2+4\end{bmatrix}$$

$$=\frac{1}{9}\begin{bmatrix}6&-6&3\\6&3&-6\\3&6&6\end{bmatrix}=\frac{1}{3}\begin{bmatrix}2&-2&1\\2&1&-2\\1&2&2\end{bmatrix}$$

2. Find the inverse of the following

(i) $\begin{bmatrix}-2&4\\1&-3\end{bmatrix}$

Let $A=\begin{bmatrix}-2&4\\1&-3\end{bmatrix}$; $|A|=\begin{vmatrix}-2&4\\1&-3\end{vmatrix}=6-4=2\neq 0$; $adj\ A=\begin{bmatrix}-3&-4\\-1&-2\end{bmatrix}$

$A^{-1}=\frac{1}{|A|}adj\ A \Rightarrow A^{-1}=\frac{1}{2}\begin{bmatrix}-3&-4\\-1&-2\end{bmatrix}$

(ii) $\begin{bmatrix}5&1&1\\1&5&1\\1&1&5\end{bmatrix}$

Let $A=\begin{bmatrix}5&1&1\\1&5&1\\1&1&5\end{bmatrix}$

$$|A|=\begin{vmatrix}5&1&1\\1&5&1\\1&1&5\end{vmatrix}=5(25-1)-1(5-1)+1(1-5)$$

$$=120-4-4$$

$$=112\neq 0$$

$$\begin{matrix}5&1&1&5\\1&5&1&1\\1&1&5&1\\5&1&1&5\end{matrix} \Rightarrow adj\ A=\begin{bmatrix}25-1&1-5&1-5\\1-5&25-1&1-5\\1-5&1-5&25-1\end{bmatrix}=\begin{bmatrix}24&-4&-4\\-4&24&-4\\-4&-4&24\end{bmatrix}$$

$$A^{-1}=\frac{1}{|A|}adj\ A \Rightarrow A^{-1}=\frac{1}{112}\begin{bmatrix}24&-4&-4\\-4&24&-4\\-4&-4&24\end{bmatrix}=\frac{1}{28}\begin{bmatrix}6&-1&-1\\-1&6&-1\\-1&-1&6\end{bmatrix}$$

(iii) $\begin{bmatrix}2&3&1\\3&4&1\\3&7&2\end{bmatrix}$

Let $A=\begin{bmatrix}2&3&1\\3&4&1\\3&7&2\end{bmatrix}$

$$|A|=\begin{vmatrix}2&3&1\\3&4&1\\3&7&2\end{vmatrix}=2(8-7)-3(6-3)+1(21-12)=2-9+9=2\neq 0$$

$$\begin{matrix}4&7&3&4\\1&2&1&1\\3&3&2&3\\4&7&3&4\end{matrix} \Rightarrow adj\ A=\begin{bmatrix}8-7&7-6&3-4\\3-6&4-3&3-2\\21-12&9-14&8-9\end{bmatrix}=\begin{bmatrix}1&1&-1\\-3&1&1\\9&-5&-1\end{bmatrix}$$

$$A^{-1}=\frac{1}{|A|}adj\ A \Rightarrow A^{-1}=\frac{1}{2}\begin{bmatrix}1&1&-1\\-3&1&1\\9&-5&-1\end{bmatrix}$$

EXAMPLE 1.2 Find the inverse of $\begin{bmatrix}a&b\\c&d\end{bmatrix}$

EXAMPLE 1.3 Find the inverse of $\begin{bmatrix}2&-1&3\\-5&3&1\\-3&2&3\end{bmatrix}$

3. $If\ F(\alpha)=\begin{bmatrix}\cos\alpha&0&\sin\alpha\\0&1&0\\-\sin\alpha&0&\cos\alpha\end{bmatrix}$, $show\ that\ [F(\alpha)]^{-1}=F(-\alpha)$

$$F(-\alpha)=\begin{bmatrix}\cos(-\alpha)&0&\sin(-\alpha)\\0&1&0\\-\sin(-\alpha)&0&\cos(-\alpha)\end{bmatrix}=\begin{bmatrix}\cos\alpha&0&-\sin\alpha\\0&1&0\\\sin\alpha&0&\cos\alpha\end{bmatrix}\rightarrow(1)$$

$\left(\because \begin{matrix}\cos(-\alpha)=\cos\alpha\\ \sin(-\alpha)=-\sin\alpha\end{matrix}\right)$

$$|F(\alpha)| = \begin{vmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{vmatrix} = \cos\alpha\,(\cos\alpha) - 0 + \sin\alpha\,(\sin\alpha)$$

$$= cos^2\alpha + sin^2\alpha = 1$$

$$\begin{matrix} 1 & 0 & 0 & 1 \\ 0 & \cos\alpha & \sin\alpha & 0 \\ 0 & -\sin\alpha & \cos\alpha & 0 \\ 1 & 0 & 0 & 1 \end{matrix} \Longrightarrow$$

$$adjF(\alpha) = \begin{bmatrix} \cos\alpha - 0 & 0 - 0 & 0 - \sin\alpha \\ 0 - 0 & cos^2\alpha + sin^2\alpha & 0 - 0 \\ 0 + \sin\alpha & 0 - 0 & \cos\alpha - 0 \end{bmatrix} = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix}$$

$$[F(\alpha)]^{-1} = \frac{1}{|F(\alpha)|} adj\,F(\alpha) \Longrightarrow [F(\alpha)]^{-1} = \frac{1}{1}\begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix}$$

$$[F(\alpha)]^{-1} = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix} \rightarrow (2)$$

From (1) & (2), $[F(\alpha)]^{-1} = F(-\alpha)$

4. $If\ A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$, show that $A^2 - 3A - 7I_2 = 0_2$. Hence find A^{-1}

$$A^2 = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 25-3 & 15-6 \\ -5+2 & -3+4 \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

$$-3A = -3\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -15 & -9 \\ 3 & 6 \end{bmatrix}$$

$$-7I_2 = -7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix}$$

$$A^2 - 3A - 7I_2 = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} + \begin{bmatrix} -15 & -9 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_2$$

$A^2 - 3A - 7I_2 = 0_2$ pre-multiply by A^{-1} on both sides, we get

$$A - 3I_2 - 7A^{-1} = 0_2 \Longrightarrow 7A^{-1} = A - 3I_2 \Longrightarrow A^{-1} = \frac{1}{7}[A - 3I_2]$$

$$A^{-1} = \frac{1}{7}\left\{\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\} = \frac{1}{7}\left\{\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} + \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}\right\} = \frac{1}{7}\begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

EXAMPLE 1.10 If $A = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$, find x and y such that $A^2 + xA + yI_2 = 0_2$.

Hence find A^{-1}

$$A^2 = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 16+6 & 12+15 \\ 8+10 & 6+25 \end{bmatrix} = \begin{bmatrix} 22 & 27 \\ 18 & 31 \end{bmatrix}$$

$$xA = x\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 4x & 3x \\ 2x & 5x \end{bmatrix} \quad ; \quad yI_2 = y\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} y & 0 \\ 0 & y \end{bmatrix}$$

$$A^2 + xA + yI_2 = 0_2 \Longrightarrow \begin{bmatrix} 22 & 27 \\ 18 & 31 \end{bmatrix} + \begin{bmatrix} 4x & 3x \\ 2x & 5x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 0 & y \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Longrightarrow \begin{bmatrix} 22+4x+y & 27+3x \\ 18+2x & 31+5x+y \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Longrightarrow \begin{cases} 27 + 3x = 0 \Longrightarrow 3x = -27 \Longrightarrow \boxed{x = -9} \\ 22 + 4x + y = 0 \Longrightarrow 22 + 4(-9) + y = 0 \Longrightarrow 22 - 36 + y = 0 \Longrightarrow \boxed{y = 14} \end{cases}$$

$A^2 + xA + yI_2 = 0_2 \Longrightarrow A^2 - 9A + 14I_2 = 0_2$

pre-multiply by A^{-1} on both sides, we get

$$A - 9I_2 + 14A^{-1} = 0_2 \Longrightarrow 14A^{-1} = 9I_2 - A \Longrightarrow A^{-1} = \frac{1}{14}[9I_2 - A]$$

$$A^{-1} = \frac{1}{14}\left\{9\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}\right\} = \frac{1}{14}\left\{\begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}\right\} = \frac{1}{14}\begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}$$

5.If $A = \frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, prove that $A^{-1} = A^T$

$$A^T = \frac{1}{9}\begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & 8 \\ 4 & 7 & 4 \end{bmatrix} \rightarrow (1)$$

$$|A| = \left(\frac{1}{9}\right)^3[-8(16-56) - 1(16-7) + 4(7-16)] \qquad \boxed{(\because |KA| = K^n|A|)}$$

$$= \frac{1}{729}[-576 - 9 - 144] = \frac{1}{729}[-729] = -1$$

$$\begin{matrix} 4 & -8 & 1 & 4 \\ 7 & 4 & 4 & 7 \\ 4 & 1 & -8 & 4 \\ 4 & -8 & 1 & 4 \end{matrix} \Longrightarrow \qquad \boxed{adj(\lambda A) = \lambda^{n-1} adj(A)}$$

$$adj\,A = \left(\frac{1}{9}\right)^{3-1}\begin{bmatrix} 16+56 & -32-4 & 7-16 \\ 7-16 & -32-4 & 16+56 \\ -32-4 & 1-64 & -32-4 \end{bmatrix} = \frac{1}{81}\begin{bmatrix} 72 & -36 & -9 \\ -9 & -36 & 72 \\ -36 & -63 & -36 \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} 8 & -4 & -1 \\ -1 & -4 & 8 \\ -4 & -7 & -4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} adj\,A \Longrightarrow A^{-1} = \frac{1}{-1}.\frac{1}{9}\begin{bmatrix} 8 & -4 & -1 \\ -1 & -4 & 8 \\ -4 & -7 & -4 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & 8 \\ 4 & 7 & 4 \end{bmatrix} \rightarrow (2)$$

From (1) & (2), $A^{-1} = A^T$

6. If $A = \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$, verify that $A(adjA) = (adjA)A = |A|I_2$

$|A| = 24 - 20 = 4$

$|A|I_2 = 4\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \rightarrow (1)$

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$adjA = \begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix}$

$A(adjA) = \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \rightarrow (2)$

$(adjA)A = \begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix}\begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \rightarrow (3)$

From (1),(2) & (3) , $A(adjA) = (adjA)A = |A|I_2$

EXAMPLE 1.1 If $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, verify that $A(adjA) = (adjA)A = |A|I_3$

$|A| = 8(21 - 16) - (-6)(-18 + 8) + 2(24 - 14) = 40 - 60 + 20 = 0$

$|A|I_3 = 0\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow (1)$

$\begin{matrix} 7 & -4 & -6 & 7 \\ -4 & 3 & 2 & -4 \\ -6 & 2 & 8 & -6 \\ 7 & -4 & -6 & 7 \end{matrix} \Rightarrow$

$adj\ A = \begin{bmatrix} 21-16 & -8+18 & 24-14 \\ -8+18 & 24-4 & -12+32 \\ 24-14 & -12+32 & 56-36 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}$

$A(adjA) = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}\begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow (2)$

$(adjA)A = \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow (3)$

From (1),(2) & (3) , $A(adjA) = (adjA)A = |A|I_3$

7.If $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ *and* $B = \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$ verify that $(AB)^{-1} = B^{-1}A^{-1}$

$AB = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}\begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -3+10 & -9+4 \\ -7+25 & -21+10 \end{bmatrix} = \begin{bmatrix} 7 & -5 \\ 18 & -11 \end{bmatrix}$

$|AB| = -77 + 90 = 13$; $adj(AB) = \begin{bmatrix} -11 & 5 \\ -18 & 7 \end{bmatrix}$

$(AB)^{-1} = \frac{1}{|AB|}adj(AB) = \frac{1}{13}\begin{bmatrix} -11 & 5 \\ -18 & 7 \end{bmatrix} \rightarrow (1)$

$|B| = -2 + 15 = 13$; $adj\ B = \begin{bmatrix} 2 & -2 \\ -7 & -1 \end{bmatrix}$; $B^{-1} = \frac{1}{|B|}adj\ B = \frac{1}{13}\begin{bmatrix} 2 & 3 \\ -5 & -1 \end{bmatrix}$

$|A| = 15 - 14 = 1$; $adj\ A = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|}adj\ A = \frac{1}{1}\begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$

$B^{-1}A^{-1} = \frac{1}{13}\begin{bmatrix} 2 & 3 \\ -5 & -1 \end{bmatrix}\begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = \frac{1}{13}\begin{bmatrix} 10-21 & -4+9 \\ -25+7 & 10-3 \end{bmatrix}$

$= \frac{1}{13}\begin{bmatrix} -11 & 5 \\ -18 & 7 \end{bmatrix} \rightarrow (2)$

From (1) & (2), $(AB)^{-1} = B^{-1}A^{-1}$

EXAMPLE 1.9 If $A = \begin{bmatrix} 0 & -3 \\ 1 & 4 \end{bmatrix}$ *and* $B = \begin{bmatrix} -2 & -3 \\ 0 & -1 \end{bmatrix}$ verify that $(AB)^{-1} = B^{-1}A^{-1}$

8. If $adj\ A = \begin{bmatrix} 2 & -4 & 2 \\ -3 & 12 & -7 \\ -2 & 0 & 2 \end{bmatrix}$, find A.

$$\boxed{A = \pm\frac{1}{\sqrt{|adjA|}}adj(adjA)}$$

$|adjA| = 2(24 - 0) - (-4)(-6 - 14) + 2(0 + 24) = 48 - 80 + 48 = 16$

$\sqrt{|adjA|} = \sqrt{16} = 4$

$\begin{matrix} 12 & 0 & -4 & 12 \\ -7 & 2 & 2 & -7 \\ -3 & -2 & 2 & -3 \\ 12 & 0 & -4 & 12 \end{matrix} \Rightarrow adj(adjA) = \begin{bmatrix} 24-0 & 0+8 & 28-24 \\ 14+6 & 4+4 & -6+14 \\ 0+24 & 8-0 & 24-12 \end{bmatrix}$

$= \begin{bmatrix} 24 & 8 & 4 \\ 20 & 8 & 8 \\ 24 & 8 & 12 \end{bmatrix}$

$A = \pm\frac{1}{\sqrt{|adjA|}}adj(adjA) = \pm\frac{1}{4}\begin{bmatrix} 24 & 8 & 4 \\ 20 & 8 & 8 \\ 24 & 8 & 12 \end{bmatrix} = \pm\begin{bmatrix} 6 & 2 & 1 \\ 5 & 2 & 2 \\ 6 & 2 & 3 \end{bmatrix}$

EXAMPLE1.5 Find a matrix A if $adj\ A = \begin{bmatrix} 7 & 7 & -7 \\ -1 & 11 & 7 \\ 11 & 5 & 7 \end{bmatrix}$

9. If $adj\ A = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$, find A^{-1}.

$$\boxed{A^{-1} = \pm \frac{1}{\sqrt{|adjA|}} adjA}$$

$|adjA| = 0(12-0) - (-2)(36-18) + 0(0+6) = 0 + 36 + 0 = 36$

$\sqrt{|adjA|} = \sqrt{36} = 6$

$A^{-1} = \pm \frac{1}{\sqrt{|adjA|}} adjA = \pm \frac{1}{6} \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$

EXAMPLE 1.6 If $adj\ A = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.

10.Find adj(adjA) if $adj\ A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$$\begin{matrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 2 & 0 & 0 & 2 \end{matrix} \Longrightarrow \text{adj(adjA)} = \begin{bmatrix} 2-0 & 0-0 & 0-2 \\ 0-0 & 1+1 & 0-0 \\ 0+2 & 0-0 & 2-0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

11.$A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, show that $A^T A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$

$|A| = 1 + tan^2 x = sec^2 x = \frac{1}{cos^2 x}$

$\text{adjA} = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} adj\ A = \frac{1}{\frac{1}{cos^2 x}} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} = cos^2 x \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$

$$= \begin{bmatrix} cos^2 x & -sinxcosx \\ sinxcosx & cos^2 x \end{bmatrix}$$

$A^T A^{-1} = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \begin{bmatrix} cos^2 x & -sinxcosx \\ sinxcosx & cos^2 x \end{bmatrix}$

$= \begin{bmatrix} cos^2 x - sin^2 x & -sinxcosx - sinxcosx \\ sinxcosx + sinxcosx & cos^2 x - sin^2 x \end{bmatrix} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$

12.Find the matrix A for which $A \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix}$

$A \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix} \Longrightarrow AB = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix} \Longrightarrow A = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix} B^{-1} \rightarrow (1)$

$B = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \Longrightarrow |B| = -10 + 3 = -7\ ;\ adj\ B = \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix}$

$B^{-1} = \frac{1}{|B|} adjB \Longrightarrow B^{-1} = \frac{1}{-7} \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix}$

$(1) \Longrightarrow A = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix} \frac{1}{-7} \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix} = -\frac{1}{7} \begin{bmatrix} -28+7 & -42+35 \\ -14+7 & -21+35 \end{bmatrix}$

$= -\frac{1}{7} \begin{bmatrix} -21 & -7 \\ -7 & 14 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix} \qquad \therefore A = \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix}$

13. Given $A = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, B = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, find a matrix X such that $AXB = C$.

$$AXB = C$$

Pre multiplying by A^{-1} and post multiplying by $B^{-1} \Longrightarrow X = A^{-1} C B^{-1} \rightarrow (1)$

$|A| = 0 + 2 = 2\ ;\ adj\ A = \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix};\ A^{-1} = \frac{1}{|A|} adj\ A = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}$

$|B| = 3 + 2 = 5\ ;\ adj\ B = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix};\ A^{-1} = \frac{1}{|A|} adj\ A = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

$(1) \Longrightarrow X = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 0+2 & 0+2 \\ -2+2 & -2+2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

$= \frac{1}{10} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 2-2 & 4+6 \\ 0 & 0 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 0 & 10 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

14.If $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, show that $A^{-1} = \frac{1}{2}(A^2 - 3I)$

$|A| = 0 - 1(0-1) + 1(1-0) = 1 + 1 = 2$

$$\begin{matrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{matrix} \Longrightarrow adj\ A = \begin{bmatrix} 0-1 & 1-0 & 1-0 \\ 1-0 & 0-1 & 1-0 \\ 1-0 & 1-0 & 0-1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

$A^{-1} = \frac{1}{|A|} adj\ A = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \rightarrow (1)$

$A^2=\begin{bmatrix}0&1&1\\1&0&1\\1&1&0\end{bmatrix}\begin{bmatrix}0&1&1\\1&0&1\\1&1&0\end{bmatrix}=\begin{bmatrix}0+1+1&0+0+1&0+1+0\\0+0+1&1+0+1&1+0+0\\0+1+0&1+0+0&1+1+0\end{bmatrix}=\begin{bmatrix}2&1&1\\1&2&1\\1&1&2\end{bmatrix}$

$\frac{1}{2}(A^2-3I)=\frac{1}{2}\left(\begin{bmatrix}2&1&1\\1&2&1\\1&1&2\end{bmatrix}-3\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\right)$

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$=\frac{1}{2}\left(\begin{bmatrix}2&1&1\\1&2&1\\1&1&2\end{bmatrix}+\begin{bmatrix}-3&0&0\\0&-3&0\\0&0&-3\end{bmatrix}\right)=\frac{1}{2}\begin{bmatrix}-1&1&1\\1&-1&1\\1&1&-1\end{bmatrix}\rightarrow(2)$

From (1) & (2), $A^{-1}=\frac{1}{2}(A^2-3I)$

15. Decrypt the received encoded message $[2\quad -3],[20\quad 4]$ with the encryption matrix $\begin{bmatrix}-1&-1\\2&1\end{bmatrix}$ and the decryption matrix as its inverse, where the system of code are described by the numbers 1-26 to the letters A-Z respectively, and the number 0 to a blank space.

$A=[2\quad -3], B=[20\quad 4]$

Encoded matrix C= $\begin{bmatrix}-1&-1\\2&1\end{bmatrix}\Longrightarrow$ Decoded matrix $=\mathrm{C}^{-1}$

$|C|=-1+2=1\ ;adj\ C=\begin{bmatrix}1&1\\-2&-1\end{bmatrix}\ ;\ \mathrm{C}^{-1}=\frac{1}{|C|}adj\ C=\begin{bmatrix}1&1\\-2&-1\end{bmatrix}$

Received message= $(A\mathrm{C}^{-1})(B\mathrm{C}^{-1})$

$=\left\{[2\quad -3]\quad\begin{bmatrix}1&1\\-2&-1\end{bmatrix}\right\}\left\{[20\quad 4]\quad\begin{bmatrix}1&1\\-2&-1\end{bmatrix}\right\}$

$=[2+6\quad 2+3][20-8\quad 20-4]=[8\quad 5][12\quad 16]=HELP$

(∵ In alphabetical order, 8=H, 5=E, 12=L, 16=P)

EXAMPLE 1.8 Verify the property $(A^T)^{-1}=(A^{-1})^T$ with $A=\begin{bmatrix}2&9\\1&7\end{bmatrix}$

$A^T=\begin{bmatrix}2&1\\9&7\end{bmatrix}\ ;\ |A^T|=14-9=5\ ;adj(A^T)=\begin{bmatrix}7&-1\\-9&2\end{bmatrix}$

$(A^T)^{-1}=\frac{1}{|A^T|}adj(A^T)=\frac{1}{5}\begin{bmatrix}7&-1\\-9&2\end{bmatrix}\rightarrow(1)$

$|A|=14-9=5\ ;adj\ A=\begin{bmatrix}7&-9\\-1&2\end{bmatrix};A^{-1}=\frac{1}{|A|}adj\ A=\frac{1}{5}\begin{bmatrix}7&-9\\-1&2\end{bmatrix}$

$(A^{-1})^T=\frac{1}{5}\begin{bmatrix}7&-1\\-9&2\end{bmatrix}\rightarrow(2)$

From (1) & (2), $(A^T)^{-1}=(A^{-1})^T$

EXAMPLE 1.11 Prove that $\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$ is orthogonal.

A square matrix A is called orthogonal if $\boxed{AA^T=A^TA=I_n}$

$A=\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}\Longrightarrow A^T=\begin{bmatrix}\cos\theta&\sin\theta\\-\sin\theta&\cos\theta\end{bmatrix}$

$AA^T=\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}\begin{bmatrix}\cos\theta&\sin\theta\\-\sin\theta&\cos\theta\end{bmatrix}$

$=\begin{bmatrix}cos^2\theta+sin^2\theta&\sin\theta\cos\theta-\sin\theta\cos\theta\\\sin\theta\cos\theta-\sin\theta\cos\theta&sin^2\theta+cos^2\theta\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I_2\rightarrow(1)$

$A^TA=\begin{bmatrix}\cos\theta&\sin\theta\\-\sin\theta&\cos\theta\end{bmatrix}\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$

$=\begin{bmatrix}cos^2\theta+sin^2\theta&\sin\theta\cos\theta-\sin\theta\cos\theta\\\sin\theta\cos\theta-\sin\theta\cos\theta&sin^2\theta+cos^2\theta\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I_2\rightarrow(2)$

From (1) & (2), $AA^T=A^TA=I_2$, hence A is orthogonal

EXAMPLE 1.12 If $A=\frac{1}{7}\begin{bmatrix}6&-3&a\\b&-2&6\\2&c&3\end{bmatrix}$ is orthogonal, find a, b and c and hence A^{-1}

A square matrix A is called orthogonal if $\boxed{AA^T=A^TA=I_n}$

$A=\frac{1}{7}\begin{bmatrix}6&-3&a\\b&-2&6\\2&c&3\end{bmatrix}\Longrightarrow A^T=\frac{1}{7}\begin{bmatrix}6&b&2\\-3&-2&c\\a&6&3\end{bmatrix}$

$AA^T=I_3\Longrightarrow\frac{1}{7}\begin{bmatrix}6&-3&a\\b&-2&6\\2&c&3\end{bmatrix}\frac{1}{7}\begin{bmatrix}6&b&2\\-3&-2&c\\a&6&3\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

$\Longrightarrow\frac{1}{49}\begin{bmatrix}36+9+a^2&6b+6+6a&12-3c+3a\\6b+6+6a&b^2+4+36&2b-2c+18\\12-3c+3a&2b-2c+18&4+c^2+9\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

$\Longrightarrow\begin{bmatrix}a^2+45&6a+6b+6&3a-3c+12\\6a+6b+6&b^2+40&2b-2c+18\\3a-3c+12&2b-2c+18&c^2+13\end{bmatrix}=\begin{bmatrix}49&0&0\\0&49&0\\0&0&49\end{bmatrix}$

$$\Rightarrow \left\{\begin{array}{c} a^2+45=49 \Rightarrow a^2=4 \Rightarrow a=\pm 2 \\ b^2+40=49 \Rightarrow b^2=9 \Rightarrow b=\pm 3 \\ c^2+13=49 \Rightarrow c^2=36 \Rightarrow c=\pm 6 \\ 6a+6b+6=0 \Rightarrow a+b=-1 \\ 3a-3c+12=0 \Rightarrow a-c=-4 \\ 2b-2c+18=0 \Rightarrow b-c=-9 \end{array}\right\} \Rightarrow \boxed{a=2, b=-3, c=6}$$

Since A is orthogonal, $A^{-1}=A^T \Rightarrow A^{-1}=\frac{1}{7}\begin{bmatrix} 6 & -3 & 2 \\ -3 & -2 & 6 \\ 2 & 6 & 3 \end{bmatrix}$

RANK OF A MATRIX

- **Rank of a matrix in minor method:** The rank of a matrix A is defined as the order of a highest order non-vanishing minor of the matrix A. It is denoted by $\rho(A)$.
- The rank of zero matrix is 0.
- The rank of the identity matrix I_n is n.
- If a matrix contains at-least one non-zero element then $\rho(A) \geq 1$.
- If A is an $m \times n$ matrix then $\rho(A) \leq min\{m, n\}$
- A square matrix A order n is invertible $\Leftrightarrow \rho(A) = n$
- If all the entries below the leading diagonal are zero, use minor method for finding the rank.
- **Rank of a matrix in row echelon form:** The rank of a matrix in row echelon form is the number of non-zero rows in it.
- The rank of the non-zero matrix is equal to the number of non-zero rows in a row-echelon form of the matrix.
- **Elementary matrix:** An elementary matrix is defined as a matrix which is obtained from an identity matrix by applying only one elementary transformation.
- **Gauss-Jordan Method:** Transforming a non-singular matrix A to the form I_n by applying elementary row operations is called Gauss-Jordan method.
- **Steps in finding A^{-1} by Gauss-Jordan method:**

- Write matrix of the form $[A|I_n]$
- Apply elementary row operations and convert $[A|I_n]$ to $[I_n|A^{-1}]$ and find A^{-1}.

RANK OF A MATRIX IN MINOR METHOD

EXERCISE 1.2

1.Find the rank of the following matrices by minor method:

(i) $\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$

Let $A=\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$. Order of A is 2×2. So $\rho(A) \leq min\{2,2\} = 2$

$|A| = 4-4 = 0 \Rightarrow \rho(A) < 2$

Since A contains at-least one non-Zero element, $\rho(A) = 1$.

(ii) $\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$

Let $A=\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$. Order of A is 3×2. So $\rho(A) \leq min\{3,2\} = 2$

2×2 Minor:

$\begin{vmatrix} -1 & 3 \\ 4 & -7 \end{vmatrix} = 7-12 = 5 \neq 0 \Rightarrow \rho(A) = 2$

(iii) $\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$

Let $A=\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$. Order of A is 2×4. So $\rho(A) \leq min\{2,4\} = 2$

2×2 Minors:

$\begin{vmatrix} 1 & -2 \\ 3 & -6 \end{vmatrix} = 6-6 = 0$; $\begin{vmatrix} 1 & -1 \\ 3 & -3 \end{vmatrix} = -3+3 = 0$; $\begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} = 1-0 = 1 \neq 0$

$\therefore \rho(A) = 2$

(iv) $\begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -6 \\ 5 & 1 & -1 \end{bmatrix}$

Let $A=\begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -6 \\ 5 & 1 & -1 \end{bmatrix}$. Order of A is 3×3. So $\rho(A) \leq min\{3,3\} = 3$

$|A| = 1(-4+6)-(-2)(-2+30)+3(2-20) = 2+56-54 = 4 \neq 0$

$\therefore \rho(A) = 3$

(v) $\begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ 8 & 1 & 0 & 2 \end{bmatrix}$

Let $A=\begin{bmatrix}0 & 1 & 2 & 1\\ 0 & 2 & 4 & 3\\ 8 & 1 & 0 & 2\end{bmatrix}$. Order of A is 3×4. So $\rho(A)\le min\{3,4\}=3$

3×3 Minors:

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$\begin{vmatrix}0 & 1 & 2\\ 0 & 2 & 4\\ 8 & 1 & 0\end{vmatrix}=0-0+8(4-4)=0$; $\begin{vmatrix}0 & 1 & 1\\ 0 & 2 & 3\\ 8 & 1 & 2\end{vmatrix}0-0+8(3-2)=8\neq 0$

$\therefore \rho(A)=3$

EXAMPLE 1.15 $\begin{bmatrix}3 & 2 & 5\\ 1 & 1 & 2\\ 3 & 3 & 6\end{bmatrix}$

Let $A=\begin{bmatrix}3 & 2 & 5\\ 1 & 1 & 2\\ 3 & 3 & 6\end{bmatrix}$. Order of A is 3×3. So $\rho(A)\le min\{3,3\}=3$

$|A|=0 \quad (\because R_2:R_3)$

$\therefore \rho(A)<3$

2×2 *Minor*:

$\begin{vmatrix}3 & 2\\ 1 & 1\end{vmatrix}=3-2=1\neq 0$. So $\rho(A)=2$

EXAMPLE1.15 $\begin{bmatrix}4 & 3 & 1 & -2\\ -3 & -1 & -2 & 4\\ 6 & 7 & -1 & 2\end{bmatrix}$

Let $A=\begin{bmatrix}4 & 3 & 1 & -2\\ -3 & -1 & -2 & 4\\ 6 & 7 & -1 & 2\end{bmatrix}$. Order of A is 3×4. So $\rho(A)\le min\{3,4\}=3$

3×3 Minors:

$\begin{vmatrix}4 & 3 & 1\\ -3 & -1 & -2\\ 6 & 7 & -1\end{vmatrix}=0$; $\begin{vmatrix}4 & 3 & -2\\ -3 & -1 & 4\\ 6 & 7 & 2\end{vmatrix}=0$;

$\begin{vmatrix}4 & 1 & -2\\ -3 & -2 & 4\\ 6 & -1 & 2\end{vmatrix}=0$; $\begin{vmatrix}3 & 1 & -2\\ -1 & -2 & 4\\ 6 & -1 & 2\end{vmatrix}=0$

So $\rho(A)<3$

2×2 Minor:

$\begin{vmatrix}4 & 3\\ -3 & -1\end{vmatrix}=-4+9=5\neq 0$. So $\rho(A)=2$

EXAMPLE 1.16 Find the rank of the following matrices which are in row-echelon form:

(i) $\begin{bmatrix}2 & 0 & -7\\ 0 & 3 & 1\\ 0 & 0 & 1\end{bmatrix}$

Let $A=\begin{bmatrix}2 & 0 & -7\\ 0 & 3 & 1\\ 0 & 0 & 1\end{bmatrix}$. Order of A is 3×3. So $\rho(A)\le min\{3,3\}=3$

3×3 Minor:

$\begin{vmatrix}2 & 0 & -7\\ 0 & 3 & 1\\ 0 & 0 & 1\end{vmatrix}=2(3-0)-0+0=6\neq 0$. So $\rho(A)=3$

(ii) $\begin{bmatrix}-2 & 2 & -1\\ 0 & 5 & 1\\ 0 & 0 & 0\end{bmatrix}$

Let $A=\begin{bmatrix}-2 & 2 & -1\\ 0 & 5 & 1\\ 0 & 0 & 0\end{bmatrix}$. Order of A is 3×3. So $\rho(A)\le min\{3,3\}=3$

3×3 Minor:

$\begin{vmatrix}-2 & 2 & -1\\ 0 & 5 & 1\\ 0 & 0 & 0\end{vmatrix}=0$. So $\rho(A)<3$

2×2 Minor:

$\begin{vmatrix}-2 & 2\\ 0 & 5\end{vmatrix}=-10-0=-10\neq 0$. So $\rho(A)=2$

(iii) $\begin{bmatrix}6 & 0 & -9\\ 0 & 2 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$

Let $A=\begin{bmatrix}6 & 0 & -9\\ 0 & 2 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$. Order of A is 4×3. So $\rho(A)\le min\{4,3\}=3$

3×3 Minor:

All the 3×3 minors are equal to zero. So $\rho(A)<3$

2×2 Minor:

$\begin{vmatrix}6 & 0\\ 0 & 2\end{vmatrix}=12-0=12\neq 0.$

So $\rho(A) = 2$

RANK OF A MATRIX IN ROW REDUCTION METHOD

2.Find the rank of the following matrices by row reduction (row-echelon) method:

(i) $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

Let $A = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & -6 & 2 & -4 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 5R_1 \end{matrix}$

$\sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3 \to R_3 - 2R_2$

The equivalent matrix is in row-echelon form and the number of non-zero rows is 2. So $\rho(A) = 2$

(ii) $\begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$

Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & -4 & 4 \\ 0 & -3 & 2 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - R_1 \\ R_4 \to R_4 - R_1 \end{matrix}$

$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & -4 \end{bmatrix} \begin{matrix} R_3 \to 7R_3 - 4R_2 \\ R_4 \to 4R_4 - 3R_3 \end{matrix}$

$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix} R_4 \to R_4 + 2R_3$

The equivalent matrix is in row-echelon form and the number of non-zero rows is 3. So $\rho(A) = 3$

(iii) $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

Let $A = \begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

$\sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 4 \\ 3 & -8 & 5 & 2 \end{bmatrix} R_1 \to R_3$

$\sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 0 & -1 & 7 & 0 \\ 0 & -2 & 14 & -4 \end{bmatrix} \begin{matrix} R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 + 3R_1 \end{matrix}$

$\sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 0 & -1 & 7 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix} R_3 \to R_3 - 2R_2$

The equivalent matrix is in row-echelon form and the number of non-zero rows is 3. So $\rho(A) = 3$

EXAMPLE 1.17 Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ by reducing it to a row - echelon form.

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -2 \\ 0 & -6 & -4 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$

$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -2 \\ 0 & 0 & 0 \end{bmatrix} R_3 \to R_3 - 2R_2$

The equivalent matrix is in row-echelon form and the number of non-zero rows is

2. So $\rho(A) = 2$

EXAMPLE 1.18 Find the rank of the matrix $\begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix}$ by reducing it to a row -echelon form.

$$A = \begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -2 & 4 & 3 \\ -6 & 8 & -4 & -2 \\ 6 & 2 & -1 & 7 \end{bmatrix} R_2 \to 2R_2$$

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$$\sim \begin{bmatrix} 2 & -2 & 4 & 3 \\ 0 & 2 & 8 & 7 \\ 0 & 8 & -13 & -2 \end{bmatrix} \begin{matrix} R_2 \to R_2 + 3R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 2 & -2 & 4 & 3 \\ 0 & 2 & 8 & 7 \\ 0 & 0 & -45 & -30 \end{bmatrix} R_3 \to R_3 - 4R_2$$

The equivalent matrix is in row-echelon form and the number of non-zero rows is

3. So $\rho(A) = 3$

GAUSS-JORDAN METHOD FOR FINDING INVERSE OF THE MATRIX

EXAMPLE 1.19 Show that the matrix $\begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & -1 \\ 5 & 2 & 1 \end{bmatrix}$ is non-singular and reduce it to the identity matrix by elementary row transformation.

Let $A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & -1 \\ 5 & 2 & 1 \end{bmatrix}$

$|A| = 3(0+2) - 1(2+5) + 4(4-0) = 6 - 7 + 16 = 15 \neq 0$

∴ A is non-singular.

$$\begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & -1 \\ 5 & 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{3} & \frac{4}{3} \\ 2 & 0 & -1 \\ 5 & 2 & 1 \end{bmatrix} R_1 \to \frac{1}{3}R_1$$

$$\sim \begin{bmatrix} 1 & \frac{1}{3} & \frac{4}{3} \\ 0 & -\frac{2}{3} & -\frac{11}{3} \\ 0 & \frac{1}{3} & -\frac{17}{3} \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 5R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & \frac{1}{3} & \frac{4}{3} \\ 0 & 1 & \frac{11}{2} \\ 0 & \frac{1}{3} & -\frac{17}{3} \end{bmatrix} R_2 \to \left(-\frac{3}{2}\right) R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{11}{2} \\ 0 & 0 & -\frac{15}{2} \end{bmatrix} \begin{matrix} R_1 \to R_1 - \frac{1}{3}R_2 \\ R_3 \to R_3 - \frac{1}{3}R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{11}{2} \\ 0 & 0 & 1 \end{bmatrix} R_3 \to \left(-\frac{2}{15}\right) R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1 \to R_1 - \frac{1}{2}R_3 \\ R_2 \to R_2 - \frac{11}{2}R_3 \end{matrix}$$

3.Find the inverse of each of the following by Gauss-Jordan method:

(i) $\begin{bmatrix} 2 & -1 \\ 5 & -2 \end{bmatrix}$

$$[A|I_2] = \left(\begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 5 & -2 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 5 & -2 & 0 & 1 \end{array}\right) R_2 \to \frac{1}{2}R_2$$

$$\sim \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{5}{2} & 1 \end{array}\right) R_2 \to R_2 - 5R_1$$

$$\sim \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & -5 & 2 \end{array}\right) R_2 \to 2R_2$$

$$\sim \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & -5 & 2 \end{array}\right) R_1 \to R_1 + \frac{1}{2}R_2$$

$$= [I_2 | A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -2 & 1 \\ -5 & 2 \end{bmatrix}$$

(ii) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 6 & -2 & -3 \end{bmatrix}$

$$[A|I_3] = \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 6 & -2 & -3 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 4 & -3 & -6 & 0 & 1 \end{array}\right) \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 6R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -4 & 1 \end{array}\right) R_3 \to R_3 - 4R_2$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & -3 & 1 \\ 0 & 0 & 1 & -2 & -4 & 1 \end{array}\right) R_2 \to R_2 + R_3$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -3 & 1 \\ 0 & 1 & 0 & -3 & -3 & 1 \\ 0 & 0 & 1 & -2 & -4 & 1 \end{array}\right) R_1 \to R_1 + R_2$$

$$= [I_3 | A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -2 & -3 & 1 \\ -3 & -3 & 1 \\ -2 & -4 & 1 \end{bmatrix}$$

(iii) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

$$[A|I_3] = \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right) \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right) \begin{matrix} R_1 \to R_1 - 2R_2 \\ R_3 \to R_3 + 2R_2 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 14 & -5 & -3 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right) \begin{matrix} R_1 \to R_1 + 9R_3 \\ R_2 \to R_2 - 3R_3 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -44 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) R_3 \to (-)R_3$$

$$= [I_3 | A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

EXAMPLE 1.20 Find the inverse of the non-singular matrix $A = \begin{bmatrix} 0 & 5 \\ -1 & 6 \end{bmatrix}$, by Gauss-Jordan method.

$$[A|I_2] = \left(\begin{array}{cc|cc} 0 & 5 & 1 & 0 \\ -1 & 6 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{cc|cc} -1 & 6 & 0 & 1 \\ 0 & 5 & 1 & 0 \end{array}\right) R_1 \leftrightarrow R_2$$

$$\sim \left(\begin{array}{cc|cc} 1 & -6 & 0 & -1 \\ 0 & 5 & 1 & 0 \end{array}\right) R_1 \to (-)R_1$$

$$\sim \left(\begin{array}{cc|cc} 1 & -6 & 0 & -1 \\ 0 & 1 & \frac{1}{5} & 0 \end{array}\right) R_2 \to \frac{1}{5}R_2$$

$$\sim \left(\begin{array}{cc|cc} 1 & 0 & \frac{6}{5} & -1 \\ 0 & 1 & \frac{1}{5} & 0 \end{array}\right) R_1 \to R_1 + 6R_2$$

$$= [I_2 | A^{-1}] \qquad \Rightarrow A^{-1} = \begin{bmatrix} \frac{6}{5} & -1 \\ \frac{1}{5} & 0 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 6 & -5 \\ 1 & 0 \end{bmatrix}$$

EXAMPLE 1.21 Find the inverse of $A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$, by Gauss-Jordan method.

$$[A|I_3] = \left(\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|ccc}1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0\\ 3 & 2 & 1 & 0 & 1 & 0\\ 2 & 1 & 2 & 0 & 0 & 1\end{array}\right) R_1 \to \frac{1}{2}R_1$$

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$$\sim\left(\begin{array}{ccc|ccc}1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0\\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} & 1 & 0\\ 0 & 0 & 1 & -1 & 0 & 1\end{array}\right) \begin{matrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 2R_1 \end{matrix}$$

$$\sim\left(\begin{array}{ccc|ccc}1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0\\ 0 & 1 & -1 & -3 & 2 & 0\\ 0 & 0 & 1 & -1 & 0 & 1\end{array}\right) R_2 \to 2R_2$$

$$\sim\left(\begin{array}{ccc|ccc}1 & 0 & 1 & 2 & -1 & 0\\ 0 & 1 & -1 & -3 & 2 & 0\\ 0 & 0 & 1 & -1 & 0 & 1\end{array}\right) R_1 \to R_1 - \frac{1}{2}R_2$$

$$\sim\left(\begin{array}{ccc|ccc}1 & 0 & 0 & 3 & -1 & -1\\ 0 & 1 & 0 & -4 & 2 & 1\\ 0 & 0 & 1 & -1 & 0 & 1\end{array}\right) \begin{matrix} R_1 \to R_1 - R_3 \\ R_2 \to R_2 + R_3 \end{matrix}$$

$$= [I_3|A^{-1}] \quad \Longrightarrow A^{-1} = \begin{bmatrix}3 & -1 & -1\\ -4 & 2 & 1\\ -1 & 0 & 1\end{bmatrix}$$

MATRIX INVERSION METHOD

EXERCISE 1.3

1.Solve the following system of linear equations by matrix inversion method:

$(i) 2x + 5y = -2, x + 2y = -3$

$$\begin{bmatrix}2 & 5\\ 1 & 2\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}-2\\ -3\end{bmatrix} \Longrightarrow AX = B \Longrightarrow X = A^{-1}B$$

$$|A| = 4 - 5 = -1 \ ; adj\ A = \begin{bmatrix}2 & -5\\ -1 & 2\end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}adjA = \frac{1}{-1}\begin{bmatrix}2 & -5\\ -1 & 2\end{bmatrix} = \begin{bmatrix}-2 & 5\\ 1 & -2\end{bmatrix}$$

$$X = A^{-1}B \Longrightarrow \begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}-2 & 5\\ 1 & -2\end{bmatrix}\begin{bmatrix}-2\\ -3\end{bmatrix} = \begin{bmatrix}4 - 15\\ -2 + 6\end{bmatrix} = \begin{bmatrix}-11\\ 4\end{bmatrix}$$

Solution: $(x, y) = (-11,4)$

$(ii) 2x - y = 8, 3x + 2y = -2$

$$\begin{bmatrix}2 & -1\\ 3 & 2\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}8\\ -2\end{bmatrix} \Longrightarrow AX = B \Longrightarrow X = A^{-1}B$$

$$|A| = 4 + 3 = 7 \ ; adj\ A = \begin{bmatrix}2 & 1\\ -3 & 2\end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}adjA = \frac{1}{7}\begin{bmatrix}2 & 1\\ -3 & 2\end{bmatrix}$$

$$X = A^{-1}B \Longrightarrow \begin{bmatrix}x\\ y\end{bmatrix} = \frac{1}{7}\begin{bmatrix}2 & 1\\ -3 & 2\end{bmatrix}\begin{bmatrix}8\\ -2\end{bmatrix} = \frac{1}{7}\begin{bmatrix}16 - 2\\ -24 - 4\end{bmatrix} = \frac{1}{7}\begin{bmatrix}14\\ -28\end{bmatrix} = \begin{bmatrix}2\\ -4\end{bmatrix}$$

Solution: $(x, y) = (2, -4)$

EXAMPLE 1.22 Solve the following system of linear equations by matrix inversion method: $5x + 2y = 3, 3x + 2y = 5$

$(iii) 2x + 3y - z = 9, x + y + z = 9, 3x - y - z = -1$

$$\begin{bmatrix}2 & 3 & -1\\ 1 & 1 & 1\\ 3 & -1 & -1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}9\\ 9\\ -1\end{bmatrix} \Longrightarrow AX = B \Longrightarrow X = A^{-1}B$$

$$|A| = 2(-1 + 1) - 3(-1 - 3) - 1(-1 - 3) = 12 + 4 = 16$$

$$\begin{matrix}1 & -1 & 3 & 1\\ 1 & -1 & -1 & 1\\ 1 & 3 & 2 & 1\\ 1 & -1 & 3 & 1\end{matrix} \Longrightarrow \text{adjA} = \begin{bmatrix}-1 + 1 & 1 + 3 & 3 + 1\\ 3 + 1 & -2 + 3 & -1 - 2\\ -1 - 3 & 9 + 2 & 2 - 3\end{bmatrix} = \begin{bmatrix}0 & 4 & 4\\ 4 & 1 & -3\\ -4 & 11 & -1\end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}adjA = \frac{1}{16}\begin{bmatrix}0 & 4 & 4\\ 4 & 1 & -3\\ -4 & 11 & -1\end{bmatrix}$$

$$X = A^{-1}B \Longrightarrow \begin{bmatrix}x\\ y\\ z\end{bmatrix} = \frac{1}{16}\begin{bmatrix}0 & 4 & 4\\ 4 & 1 & -3\\ -4 & 11 & -1\end{bmatrix}\begin{bmatrix}9\\ 9\\ -1\end{bmatrix} = \frac{1}{16}\begin{bmatrix}0 + 36 - 4\\ 36 + 9 + 3\\ -36 + 99 + 1\end{bmatrix} = \frac{1}{16}\begin{bmatrix}32\\ 48\\ 64\end{bmatrix}$$

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}2\\ 3\\ 4\end{bmatrix}$$ Solution : $(x, y, z) = (2,3,4)$

$(iv) x + y + z - 2 = 0, 6x - 4y + 5z - 31 = 0, 5x + 2y + 2z = 13$

$$\begin{bmatrix}1 & 1 & 1\\ 6 & -4 & 5\\ 5 & 2 & 2\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}2\\ 31\\ 13\end{bmatrix} \Longrightarrow AX = B \Longrightarrow X = A^{-1}B$$

$$|A| = 1(-8 - 10) - 1(12 - 25) + 1(12 + 20) = -18 + 13 + 32 = 27$$

$$\begin{matrix}-4 & 2 & 1 & -4\\ 5 & 2 & 1 & 5\\ 6 & 5 & 1 & 6\\ -4 & 2 & 1 & -4\end{matrix} \Longrightarrow \text{adjA} = \begin{bmatrix}-8 - 10 & 2 - 2 & 5 + 4\\ 25 - 12 & 2 - 5 & 6 - 5\\ 12 + 20 & 5 - 2 & -4 - 6\end{bmatrix} = \begin{bmatrix}-18 & 0 & 9\\ 13 & -3 & 1\\ 32 & 3 & -10\end{bmatrix}$$

$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{27}\begin{bmatrix} -18 & 0 & 9 \\ 13 & -3 & 1 \\ 32 & 3 & -10 \end{bmatrix}$

$X = A^{-1}B \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{27}\begin{bmatrix} -18 & 0 & 9 \\ 13 & -3 & 1 \\ 32 & 3 & -10 \end{bmatrix}\begin{bmatrix} 2 \\ 31 \\ 13 \end{bmatrix} = \frac{1}{27}\begin{bmatrix} -36+0+117 \\ 26-93+13 \\ 64+93-130 \end{bmatrix}$

$= \frac{1}{27}\begin{bmatrix} 81 \\ -54 \\ 27 \end{bmatrix} \Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ Solution : $(x, y, z) = (3, -2, 1)$

EXAMPLE 1.23 Solve the following system of linear equations by matrix inversion method: $2x_1 + 3x_2 + 3x_3 = 5, x_1 - 2x_2 + x_3 = -4, 3x_1 - x_2 - 2x_3 = 3$

$\begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix} \Rightarrow AX = B \Rightarrow X = A^{-1}B$

$|A| = 2(4+1) - 3(-2-3) + 3(-1+6) = 10 + 15 + 15 = 40$

$\begin{matrix} -2 & -1 & 3 & -2 \\ 1 & -2 & 3 & 1 \\ 1 & 3 & 2 & 1 \\ -2 & -1 & 3 & -2 \end{matrix} \Rightarrow \text{adjA} = \begin{bmatrix} 4+1 & -3+6 & 3+6 \\ 3+2 & -4-9 & 3-2 \\ -1+6 & 9+2 & -4-3 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{40}\begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$

$X = A^{-1}B \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40}\begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}\begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix} = \frac{1}{40}\begin{bmatrix} 25-12+27 \\ 25+52+3 \\ 25-44-21 \end{bmatrix} = \frac{1}{40}\begin{bmatrix} 40 \\ 80 \\ -40 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ Solution : $(x, y, z) = (1, 2, -1)$

2.If $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$, find the products AB and BA and hence solve the system of equations $x + y + 2z = 1, 3x + 2y + z = 7, 2x + y + 3z = 2$

$AB = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -5+3+6 & -5+2+3 & -10+1+9 \\ 7+3-10 & 7+2-5 & 14+1-15 \\ 1-3+2 & 1-2+1 & 2-1+3 \end{bmatrix}$

$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4I_3$

$BA = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4I_3$

$AB = BA = 4I_3 \Rightarrow B^{-1} = \frac{1}{4}A \rightarrow (1)$

$\begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix} \Rightarrow BX = C \Rightarrow X = B^{-1}C \rightarrow (2)$

From $(1) \Rightarrow B^{-1} = \frac{1}{4}A = \frac{1}{4}\begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$

From $(2) \Rightarrow X = B^{-1}C \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -5+7+6 \\ 7+7-10 \\ 1-7+2 \end{bmatrix}$

$= \frac{1}{4}\begin{bmatrix} 8 \\ 4 \\ -4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$ Solution : $(x, y, z) = (2, 1, -1)$

EXAMPLE 1.24 If $A = \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$, find the products AB and BA and hence solve the system of equations $x - y + z = 4, x - 2y - 2z = 9, 2x + y + 3z = 1$

3. A man is appointed in a job with a monthly salary of certain amount and a fixed amount of annual increment. If his salary was Rs.19800 per month at the end of the first month after 3 years of service and Rs.23400 per month at the end of the first month after 9 years of service, find his starting salary and his annual increment. (Use matrix inversion method)

Let his starting salary be x and the annual increment be y.

From the given data, $x + 3y = 19800 \rightarrow (1)$; $x + 9y = 23400 \rightarrow (2)$

$\begin{bmatrix} 1 & 3 \\ 1 & 9 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19800 \\ 23400 \end{bmatrix} \Rightarrow AX = B \Rightarrow X = A^{-1}B$

$|A| = 9 - 3 = 6$; $adj\ A = \begin{bmatrix} 9 & -3 \\ -1 & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{6}\begin{bmatrix} 9 & -3 \\ -1 & 1 \end{bmatrix}$

$X = A^{-1}B \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 9 & -3 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 19800 \\ 23400 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 178200 - 70200 \\ -19800 + 23400 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 108000 \\ 3600 \end{bmatrix}$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 18000 \\ 600 \end{bmatrix}$

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Solution: His starting salary = Rs.18000 ; Annual increment = Rs.600

4. 4 men and 4 women can finish a piece of work jointly in 3 days while 2 men and 5 women can finish the same work jointly in 4 days. Find the time taken by one man alone and that of one woman alone to finish the same work by using matrix inversion method.

Let the number of days taken by a man and woman be a and y respectively.

Work finished by a man in one day $= \frac{1}{x}$

Work finished by a woman in one day $= \frac{1}{y}$

From the given data, $\frac{4}{x} + \frac{4}{y} = \frac{1}{3}$; $\frac{2}{x} + \frac{5}{y} = \frac{1}{4}$

Let $\frac{1}{x} = a$, $\frac{1}{y} = b$

$\frac{4}{x} + \frac{4}{y} = \frac{1}{3} \Rightarrow 4a + 4b = \frac{1}{3} \Rightarrow 12a + 12b = 1 \rightarrow (1)$

$\frac{2}{x} + \frac{5}{y} = \frac{1}{4} \Rightarrow 2a + 5b = \frac{1}{4} \Rightarrow 8a + 20b = 1 \rightarrow (2)$

$\begin{bmatrix} 12 & 12 \\ 8 & 20 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow AX = B \Rightarrow X = A^{-1}B$

$|A| = 240 - 96 = 144$; $adj\ A = \begin{bmatrix} 20 & -12 \\ -8 & 12 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{144}\begin{bmatrix} 20 & -12 \\ -8 & 12 \end{bmatrix}$

$X = A^{-1}B \Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{144}\begin{bmatrix} 20 & -12 \\ -8 & 12 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{144}\begin{bmatrix} 20 - 12 \\ -8 + 12 \end{bmatrix} = \frac{1}{144}\begin{bmatrix} 8 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{18} \\ \frac{1}{36} \end{bmatrix}$

$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \frac{1}{18} \\ \frac{1}{36} \end{bmatrix} \Rightarrow x = \frac{1}{a} = 18 ; y = \frac{1}{b} = 36$

Solution: A man can finish the work in 18 days and a woman can finish the work in 36 days.

5. The prices of three commodities A, B and C are Rs.x, y and z per units respectively. A person P purchases 4 units of B and sells 2 units of A and 5 units of C. Person Q purchase 2 unit of C and sells 3 units of A and one unit of B. Person R purchases one unit of A and sells 3 unit of B and one unit of C. In the process P, Q and R earn Rs.15000, Rs.1000 and Rs.4000 respectively. Find the prices per unit of A, B and C. (Use matrix inversion method)

Let the prices of one unit of A, B and C are Rs. x, y and z respectively.

From the given data,

$2x - 4y + 5z = 15000 \rightarrow (1)$

$3x + y - 2z = 1000 \rightarrow (2)$

$-x + 3y + z = 4000 \rightarrow (3)$

$\begin{bmatrix} 2 & -4 & 5 \\ 3 & 1 & -2 \\ -1 & 3 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15000 \\ 1000 \\ 4000 \end{bmatrix} \Rightarrow AX = B \Rightarrow X = A^{-1}B$

$|A| = 2(1 + 6) - (-4)(3 - 2) + 5(9 + 1) = 14 + 4 + 50 = 68$

$\begin{matrix} 1 & 3 & -4 & 1 \\ -2 & 1 & 5 & -2 \\ 3 & -1 & 2 & 3 \\ 1 & 3 & -4 & 1 \end{matrix} \Rightarrow \text{adjA} = \begin{bmatrix} 1+6 & 15+4 & 8-5 \\ 2-3 & 2+5 & 15+4 \\ 9+1 & 4-6 & 2+12 \end{bmatrix} = \begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{68}\begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix}$

$X = A^{-1}B \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{68}\begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix}\begin{bmatrix} 15000 \\ 1000 \\ 4000 \end{bmatrix}$

$= \frac{1000}{68}\begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix}\begin{bmatrix} 15 \\ 1 \\ 4 \end{bmatrix}$

$= \frac{1000}{68}\begin{bmatrix} 105 + 19 + 12 \\ -15 + 7 + 76 \\ 150 - 2 + 56 \end{bmatrix}$

$= \frac{1000}{68}\begin{bmatrix} 136 \\ 68 \\ 204 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2000 \\ 1000 \\ 3000 \end{bmatrix}$$

Solution: The price of one unit of A, B and C are Rs.2000, 1000 and 3000 respectively.

CRAMER'S RULE
EXERCISE 1.4

1. Solve the following systems of linear equations by Cramer's rule:

$(i) 5x - 2y + 16 = 0, x + 3y - 7 = 0$

$\Delta = \begin{vmatrix} 5 & -2 \\ 1 & 3 \end{vmatrix} = 15 + 2 = 17$ $\quad \Delta_x = \begin{vmatrix} -16 & -2 \\ 7 & 3 \end{vmatrix} = -48 + 14 = -34$

$\Delta_y = \begin{vmatrix} 5 & -16 \\ 1 & 7 \end{vmatrix} = 35 + 16 = 51$ $\quad x = \frac{\Delta_x}{\Delta} = \frac{-34}{17} = -2 \quad ; y = \frac{\Delta_y}{\Delta} = \frac{51}{17} = 3$

Solution: $(x, y) = (-2,3)$

$(ii) \frac{3}{x} + 2y = 12, \frac{2}{x} + 3y = 13$

Let $\frac{1}{x} = a, y = b \Rightarrow 3a + 2b = 12 ; 2a + 3b = 13$

$\Delta = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = 9 - 4 = 5$ $\quad \Delta_a = \begin{vmatrix} 12 & 2 \\ 13 & 3 \end{vmatrix} = 36 - 26 = 10$

$\Delta_y = \begin{vmatrix} 3 & 12 \\ 2 & 13 \end{vmatrix} = 39 - 24 = 15$

$a = \frac{\Delta_a}{\Delta} = \frac{10}{5} = 2 \Rightarrow x = \frac{1}{a} \Rightarrow x = \frac{1}{2}$ $\quad ; \quad b = \frac{\Delta_b}{\Delta} = \frac{15}{5} = 3 \Rightarrow y = 3$

Solution: $(x, y) = \left(\frac{1}{2}, 3\right)$

$(iii) 3x + 3y - z = 11, 2x - y + 2z = 9, 4x + 3y + 2z = 25$

$\Delta = \begin{vmatrix} 3 & 3 & -1 \\ 2 & -1 & 2 \\ 4 & 3 & 2 \end{vmatrix} = 3(-2-6) - 3(4-8) - 1(6+4) = -24 + 12 - 10 = -22$

$\Delta_x = \begin{vmatrix} 11 & 3 & -1 \\ 9 & -1 & 2 \\ 25 & 3 & 2 \end{vmatrix} = 11(-2-6) - 3(18-50) - 1(27+25) = -88 + 96 - 52 = -44$

$\Delta_y = \begin{vmatrix} 3 & 11 & -1 \\ 2 & 9 & 2 \\ 4 & 25 & 2 \end{vmatrix} = 3(18-50) - 11(4-8) - 1(50-36) = -96 + 44 - 14 = -66$

$\Delta_z = \begin{vmatrix} 3 & 3 & 11 \\ 2 & -1 & 9 \\ 4 & 3 & 25 \end{vmatrix} = 3(-25-27) - 3(50-36) + 11(6+4) = -156 - 42 + 110 = -88$

$x = \frac{\Delta_x}{\Delta} = \frac{-44}{-22} = 2 \ ; y = \frac{\Delta_y}{\Delta} = \frac{-66}{-22} = 3 \ ; z = \frac{\Delta_z}{\Delta} = \frac{-88}{-22} = 4$

Solution : $(x, y, z) = (2,3,4)$

$(iv) \frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 1 = 0, \frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0, \frac{2}{x} - \frac{5}{y} - \frac{4}{z} + 1 = 0$

Let $\frac{1}{x} = a, \frac{1}{y} = b, \frac{1}{z} = c$

$\Rightarrow 3a - 4b - 2c = 1 \rightarrow (1), a + 2b + c = 2 \rightarrow (2), 2a - 5b - 4c = -1 \rightarrow (3)$

$\Delta = \begin{vmatrix} 3 & -4 & -2 \\ 1 & 2 & 1 \\ 2 & -5 & -4 \end{vmatrix} = 3(-8+5) - (-4)(-4-2) - 2(-5-4) = -9 - 24 + 18 = -15$

$\Delta_a = \begin{vmatrix} 1 & -4 & -2 \\ 2 & 2 & 1 \\ -1 & -5 & -4 \end{vmatrix} = 1(-8+5) - (-4)(-8+1) - 2(-10+2) = -3 - 28 + 16 = -15$

$\Delta_b = \begin{vmatrix} 3 & 1 & -2 \\ 1 & 2 & 1 \\ 2 & -1 & -4 \end{vmatrix} = 3(-8+1) - 1(-4-2) - 2(-1-4) = -21 + 6 + 10 = -5$

$\Delta_c = \begin{vmatrix} 3 & -4 & 1 \\ 1 & 2 & 2 \\ 2 & -5 & -1 \end{vmatrix} = 3(-2+10) - (-4)(-1-4) + 1(-5-4) = 24 - 20 - 9 = -5$

$a = \frac{\Delta_a}{\Delta} = \frac{-15}{-15} = 1 \Rightarrow x = \frac{1}{a} = 1$

$b = \frac{\Delta_b}{\Delta} = \frac{-5}{-15} = \frac{1}{3} \Rightarrow y = \frac{1}{b} = 3$

$c = \frac{\Delta_c}{\Delta} = \frac{-5}{-15} = \frac{1}{3} \Rightarrow z = \frac{1}{c} = 3$

Solution : $(x, y, z) = (1,3,3)$

EXAMPLE 1.25 Solve by Cramer's rule, the system of equations $x_1 - x_2 = 3, 2x_1 + 3x_2 + 4x_3 = 17, x_2 + 2x_3 = 7$

$\Delta = \begin{vmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{vmatrix} = 1(6-4) - (-1)(4-0) + 0 = 2 + 4 = 6$

$\Delta_1 = \begin{vmatrix} 3 & -1 & 0 \\ 17 & 3 & 4 \\ 7 & 1 & 2 \end{vmatrix} = 3(6-4) - (-1)(34-28) + 0 = 6 + 6 = 12$

$\Delta_2 = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 17 & 4 \\ 0 & 7 & 2 \end{vmatrix} = 1(34-28) - 3(4-0) + 0 = 6 - 12 = -6$

$\Delta_3 = \begin{vmatrix} 1 & -1 & 3 \\ 2 & 3 & 17 \\ 0 & 1 & 7 \end{vmatrix} = 1(21-17) - (-1)(14-0) + 3(2-0) = 4 + 14 + 6 = 24$

$x_1 = \frac{\Delta_1}{\Delta} = \frac{12}{6} = 2$; $x_2 = \frac{\Delta_2}{\Delta} = \frac{-6}{6} = -1$; $x_3 = \frac{\Delta_3}{\Delta} = \frac{24}{6} = 4$

Solution : $(x_1, x_2, x_3) = (2, -1, 4)$

2. In a competitive examination, one mark is awarded for every correct answer while $\frac{1}{4}$ mark is deducted for every wrong answer. A student answered 100 questions and get 80 marks. How many questions did he answer correctly? (Use Cramer's rule)

Let the number questions answered correctly be x and y be the number of questions answered wrong.

From the given data, $x + y = 100 \rightarrow (1)$; $x - \frac{1}{4}y = 80 \Longrightarrow 4x - y = 320 \rightarrow (2)$

$\Delta = \begin{vmatrix} 1 & 1 \\ 4 & -1 \end{vmatrix} = -1 - 4 = -5$

$\Delta_x = \begin{vmatrix} 100 & 1 \\ 320 & -1 \end{vmatrix} = -100 - 320 = -420$

$\Delta_y = \begin{vmatrix} 1 & 100 \\ 4 & 320 \end{vmatrix} = 320 - 400 = -80$

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$x = \frac{\Delta_x}{\Delta} = \frac{-420}{-5} = 84$; $y = \frac{\Delta_y}{\Delta} = \frac{-80}{-5} = 16$

Solution:

The number questions answered correctly= 84

The number questions answered wrong= 16

3. A chemist has one solution which is 50% acid and another solution which is 25% acid. How much each should be mixed to make 10 litres of 40% acid solution? (Use Cramer's rule)

Let x and y be the amount of solution containing 50% and 25% acid respectively.

From the given data,

$x + y = 10 \rightarrow (1)$

$50\%\ of\ x\ + 25\%\ of\ y = 40\%\ of\ 10 \Longrightarrow \frac{50}{100}x + \frac{25}{100}y = \frac{40}{100}.10$

$\Longrightarrow 50x + 25y = 400 \rightarrow (2)$

$\Delta = \begin{vmatrix} 1 & 1 \\ 50 & 25 \end{vmatrix} = 25 - 50 = -25$

$\Delta_x = \begin{vmatrix} 10 & 1 \\ 400 & 25 \end{vmatrix} = 250 - 400 = -150$

$\Delta_y = \begin{vmatrix} 1 & 100 \\ 50 & 400 \end{vmatrix} = 400 - 500 = -100$

$x = \frac{\Delta_x}{\Delta} = \frac{-150}{-25} = 6$; $y = \frac{\Delta_y}{\Delta} = \frac{-100}{-25} = 4$

Solution:

6 litres of solution containing 50% acid and 4 litres of solution containing 25% acid must be mixed to make 40% acid solution.

4. A fish tank can be filled in 10 minutes using both pumps A and B simultaneously. However, pump B can pump water in or out at the same rate. If pump B is inadvertently run in reverse, then the tank will be filled in 30 minutes. How long would it take each pump to fill the tank by itself? (Use Cramer's rule)

Let x and y be the time taken by pumps A and B respectively.

Amount of water filled by pump A in one minute= $\frac{1}{x}$

Amount of water filled by pump B in one minute= $\frac{1}{y}$

From the given data, $\frac{1}{x} + \frac{1}{y} = \frac{1}{10} \rightarrow (1)$; $\frac{1}{x} - \frac{1}{y} = \frac{1}{30} \rightarrow (2)$

Let $\frac{1}{x} = a$, $\frac{1}{y} = b$

$(1) \Longrightarrow a + b = \frac{1}{10} \Longrightarrow 10a + 10b = 1 \rightarrow (3)$

$(2) \Longrightarrow a - b = \frac{1}{30} \Longrightarrow 30a - 30b = 1 \rightarrow (4)$

Solving (3) & (4) ,

$\Delta = \begin{vmatrix} 10 & 10 \\ 30 & -30 \end{vmatrix} = -300 - 300 = -600$

$\Delta_a = \begin{vmatrix} 1 & 10 \\ 1 & -30 \end{vmatrix} = -30 - 10 = -40$

$\Delta_b = \begin{vmatrix} 10 & 1 \\ 30 & 1 \end{vmatrix} = 10 - 30 = -20$

$a = \frac{\Delta_a}{\Delta} = \frac{-40}{-600} = \frac{1}{15}$; $b = \frac{\Delta_b}{\Delta} = \frac{-20}{-600} = \frac{1}{30} \Longrightarrow x = \frac{1}{a} = 15$, $y = \frac{1}{b} = 30$

Solution:

Time taken by pump A= 15 minutes ; Time taken by pump B= 30 minutes

5.A family of 3 people went out for dinner in a restaurant. The cost of two dosai, three idlies and two vadais is Rs. 150. The cost of two dosai, two idlies and four vadais is Rs. 200. The cost of five dosai, four idlies and two vadais is Rs. 250. The family has Rs. 350 in hand and they ate 3 dosai and six idlies and six vadais. Will they be able to manage to pay the bill within the amount they had? (Use Cramer's

rule)

Let Rs.x, y and z be the cost of I dosai, 1 idly and I vadai respectively.

From the given data,

$2x+3y+2z=150 \rightarrow (1)$

$2x+2y+4z=200, \div 2 \Longrightarrow x+y+2z=100 \rightarrow (2)$

$5x+4y+2z=250 \rightarrow (3)$

$\Delta = \begin{vmatrix} 2 & 3 & 2 \\ 1 & 1 & 2 \\ 5 & 4 & 2 \end{vmatrix} = 2(2-8)-3(2-10)+2(4-5) = -12+24-2 = 10$

$\Delta_x = \begin{vmatrix} 150 & 3 & 2 \\ 100 & 1 & 2 \\ 250 & 4 & 2 \end{vmatrix} = 150(2-8)-3(200-500)+2(400-250)$

$= -900+900+300 = 300$

$\Delta_y = \begin{vmatrix} 2 & 150 & 2 \\ 1 & 100 & 2 \\ 5 & 250 & 2 \end{vmatrix} = 2(200-500)-150(2-10)+2(250-500)$

$= -600+1200-500 = 100$

$\Delta_z = \begin{vmatrix} 2 & 3 & 150 \\ 1 & 1 & 100 \\ 5 & 4 & 250 \end{vmatrix} = 2(250-400)-3(250-500)+150(4-5)$

$-300+750-150 = 300$

$x = \frac{\Delta_x}{\Delta} = \frac{300}{10} = 30 \ ; y = \frac{\Delta_y}{\Delta} = \frac{100}{10} = 10 \ ; z = \frac{\Delta_z}{\Delta} = \frac{300}{10} = 30$

The cost of 3 dosai , 6 idlies and 6 vadais $= 3x+6y+6z$

$= 3(30)+6(10)+6(30) = 90+60+180 = 330 < 350$

Hence they can manage to pay the bill.

EXAMPLE 1.26 In a T20 match, Chennai Super Kings needed just 6 runs to win with 1 ball left to go in the last over. The last ball was bowled and the batsman at the crease hit it high up. The ball travelled along the path in a vertical plane and the equation of the path is $y=ax^2+bx+c$ with respect to a xy-coordinate ystem in the vertical plane and the ball travelled through the points $(10,8), (20,16),$ $(40,22)$, can you conclude that , Chennai Super Kings won the match ? Justify your answer. (All the distances are measured in metres and the meeting point of the plane of the path with the farthest boundary line is $(70,0)$.)

$$y=ax^2+bx+c$$

$(10,8) \Longrightarrow 100a+10b+c=8 \rightarrow (1)$

$(20,16) \Longrightarrow 400a+20b+c=16 \rightarrow (2)$

$(40,22) \Longrightarrow 1600a+40b+c=22 \rightarrow (3)$

$\Delta = \begin{vmatrix} 100 & 10 & 1 \\ 400 & 20 & 1 \\ 1600 & 40 & 1 \end{vmatrix} = 100 \times 10 \begin{vmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 16 & 4 & 1 \end{vmatrix}$

$= 1000[1(2-4)-1(4-16)+1(16-32)] = 1000[-2+12-16] = -6000$

$\Delta_a = \begin{vmatrix} 8 & 10 & 1 \\ 16 & 20 & 1 \\ 22 & 40 & 1 \end{vmatrix} = 2 \times 10 \begin{vmatrix} 4 & 1 & 1 \\ 8 & 2 & 1 \\ 11 & 4 & 1 \end{vmatrix} = 20[4(2-4)-1(8-11)+1(32-22)]$

$= 20[-8+3+10] = 100$

$\Delta_b = \begin{vmatrix} 100 & 8 & 1 \\ 400 & 16 & 1 \\ 1600 & 22 & 1 \end{vmatrix} = 100 \times 2 \begin{vmatrix} 1 & 4 & 1 \\ 4 & 8 & 1 \\ 16 & 11 & 1 \end{vmatrix}$

$= 200[1(8-11)-4(4-16)+1(44-128)] = 200[-3+48-84] = -7800$

$\Delta_c = \begin{vmatrix} 100 & 10 & 8 \\ 400 & 20 & 16 \\ 1600 & 40 & 22 \end{vmatrix} = 100 \times 10 \times 2 \begin{vmatrix} 1 & 1 & 4 \\ 4 & 2 & 8 \\ 16 & 4 & 11 \end{vmatrix}$

$= 2000[1(22-32)-1(44-128)+4(16-32)] = 2000[-10+84-64] = 20000$

$a = \frac{\Delta_a}{\Delta} = \frac{100}{-6000} = -\frac{1}{60} \ ; b = \frac{\Delta_b}{\Delta} = \frac{-7800}{-6000} = \frac{13}{10} \ ; c = \frac{c}{\Delta} = \frac{20000}{-6000} = -\frac{10}{3}$

$\therefore$ The equation of the path is $y = -\frac{1}{60}x^2 + \frac{13}{10}x - \frac{10}{3}$

Sub. $x=70$, we get $y = -\frac{1}{60}(70)^2 + \frac{13}{10}(70) - \frac{10}{3} = -\frac{4900}{60} + \frac{910}{10} - \frac{10}{3} = 6$

$\Longrightarrow$The ball went 6 m high over the boundary line and it is impossible for a fielder to catch the ball. Hence the ball went for six and the Chennai Super Kings won the match.

GAUSSIAN ELIMINATION METHOD

EXERCISE 1.5

1.Solve the following systems of linear equations by Gaussian elimination method:

$(i) 2x-2y+3z=2, x+2y-z=3, 3x-y+2z=1$

$[A|B] = \left(\begin{array}{ccc|c} 2 & -2 & 3 & 2 \\ 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \end{array}\right)$

$\sim \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -2 & 3 & 2 \\ 3 & -1 & 2 & 1 \end{array}\right) R_1 \leftrightarrow R_2$

$$\sim\left(\begin{array}{ccc|c}1 & 2 & -1 & 3\\ 0 & -6 & 5 & -4\\ 0 & -7 & 5 & -8\end{array}\right)\begin{matrix}R_2\to R_2-2R_1\\ R_3\to R_3-3R_1\end{matrix}$$

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$$\sim\left(\begin{array}{ccc|c}1 & 2 & -1 & 3\\ 0 & -6 & 5 & -4\\ 0 & 0 & -5 & -20\end{array}\right)R_3\to 6R_3-7R_2$$

$\Rightarrow x+2y-z=3\to(1)\,;\,-6y+5z=-4\to(2);\,-5z=-20\Rightarrow z=4$

Put $z=4$ in $(2)\Rightarrow -6y+5(4)=-4\Rightarrow -6y+20=-4$

$\Rightarrow -6y=-24\Rightarrow y=4$

Put $y=16,z=4$ in $(1)\Rightarrow x+2(4)-4=3\Rightarrow x+8-4=3\Rightarrow x=-1$

Solution: $(x,y,z)=(-1,4,4)$

$(ii)2x+4y+6z=22,3x+8y+5z=27,-x+y+2z=2$

$2x+4y+6z=22\,,\div 2\Rightarrow x+2y+3z=11$

$$[A|B]=\left(\begin{array}{ccc|c}1 & 2 & 3 & 11\\ 3 & 8 & 5 & 27\\ -1 & 1 & 2 & 2\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}1 & 2 & 3 & 11\\ 0 & 2 & -4 & -6\\ 0 & 3 & 5 & 13\end{array}\right)\begin{matrix}R_2\to R_2-3R_1\\ R_3\to R_3+R_1\end{matrix}$$

$$\sim\left(\begin{array}{ccc|c}1 & 2 & 3 & 11\\ 0 & 2 & -4 & -6\\ 0 & 0 & 22 & 44\end{array}\right)R_3\to 2R_3-3R_2$$

$\Rightarrow x+2y+3z=11\to(1)\,;2y-4z=-6\to(2)\,;22z=44\Rightarrow z=2$

Put $z=2\ in\ (2)\Rightarrow 2y-4(2)=-6\ \Rightarrow 2y=-6+8\Rightarrow 2y=2\Rightarrow y=1$

Put $y=1,z=2\ in\ (1)\Rightarrow x+2(1)+3(2)=11\Rightarrow x+8=11\Rightarrow x=3$

Solution: $(x,y,z)=(3,1,2)$

EXAMPLE 1.27 Solve the following systems of linear equations by Gaussian elimination method: $4x+3y+6z=25,x+5+7z=13,2x+9y+z=1$

2. If ax^2+bx+c is divided by $x+3,x-5$ and $x-1$ the remainders are 21, 61 and 9 respectively. Find a,b and c. (Use Gaussian elimination method)

Let $p(x)=ax^2+bx+c$

$p(-3)=21\Rightarrow=a(-3)^2+b(-3)+c=21\Rightarrow 9a-3b+c=21\to(1)$

$p(5)=61\Rightarrow=a(5)^2+b(5)+c=61\Rightarrow 25a+5b+c=61\to(2)$

$p(1)=9\Rightarrow=a(1)^2+b(1)+c=9\Rightarrow a+b+c=9\to(3)$

$$[A|B]=\left(\begin{array}{ccc|c}9 & -3 & 1 & 21\\ 25 & 5 & 1 & 61\\ 1 & 1 & 1 & 9\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 1 & 9\\ 25 & 5 & 1 & 61\\ 9 & -3 & 1 & 21\end{array}\right)R_1\leftrightarrow R_3$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 1 & 9\\ 0 & -20 & -24 & -164\\ 0 & -12 & -8 & -60\end{array}\right)\begin{matrix}R_2\to R_2-25R_1\\ R_3\to R_3-9R_1\end{matrix}$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 1 & 9\\ 0 & 5 & 6 & 41\\ 0 & 3 & 2 & -15\end{array}\right)\begin{matrix}R_2\to\left(-\frac{1}{4}\right)R_2\\ R_3\to\left(-\frac{1}{4}\right)R_3\end{matrix}$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 1 & 9\\ 0 & 5 & 6 & 41\\ 0 & 0 & -8 & -48\end{array}\right)R_3\to 5R_3-3R_2$$

$\Rightarrow a+b+c=9\to(1)\,;5b+6c=41\to(2)\,;-8c=-48\Rightarrow c=6$

Put $c=6\ in\ (2)\Rightarrow 5b+6(6)=41\ \Rightarrow 5b=41-36\Rightarrow 5b=5\Rightarrow b=1$

Put $b=1,c=6\ in\ (1)\Rightarrow a+1+6=9\Rightarrow a+7=9\Rightarrow a=2$

Solution: $(a,b,c)=(2,1,6)$

3. An amount of Rs.65,000 is invested in three bonds at the rates of $6\%,8\%$ and 10% per annum respectively. The total annual income is Rs.5,000. The income from the third bond is Rs.800 more than that from the second bond. Determine the price of each bond. (Use Gaussian elimination method)

Let the price of three bond be x, y and z respectively.

From the given data,

$x+y+z=65000\to(1)$

Total annual income = Rs.5,000$\Rightarrow 6\%\ of\ x+8\%\ of\ y+10\%\ of\ z=5000$

$\Rightarrow\frac{6}{100}x+\frac{8}{100}y+\frac{10}{100}z=5000\ \Rightarrow 6x+8y+10z=500000$

$\Rightarrow\div 2,3x+4y+5z=250000\to(2)$

Income from third bond = income from second bond + Rs.800

$\Rightarrow 10\%\ of\ z=8\%\ of\ y+800\Rightarrow\frac{10}{100}z=\frac{8}{100}y+800$

$\Rightarrow\frac{10}{100}z=\frac{8y+80000}{100}\Rightarrow 10z=8y+80000\Rightarrow -8y+10z=80000$

$\Rightarrow\div 2,-4y+5z=40000\to(3)$

$$[A|B]=\left(\begin{array}{ccc|c}1 & 1 & 1 & 65000\\ 3 & 4 & 5 & 250000\\ 0 & -4 & 5 & 40000\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 1 & 65000\\ 0 & 1 & 2 & 55000\\ 0 & -4 & 5 & 40000\end{array}\right) R_2 \to R_2 - 3R_1$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 1 & 65000\\ 0 & 1 & 2 & 55000\\ 0 & 0 & 13 & 260000\end{array}\right) R_3 \to R_3 + 4R_2$$

$\Rightarrow x+y+z=65000 \to (1)\,; y+2z=55000 \to (2)\,;$

$13z=260000 \Rightarrow z=20000$

Put $z=20000\ in\ (2) \Rightarrow y+2(20000)=55000 \Rightarrow y=55000-40000 \Rightarrow y=15000$

Put $y=15000, z=20000\ in\ (1) \Rightarrow x+15000+20000=65000$

$\Rightarrow x+35000=65000 \Rightarrow x=30000$

Solution:

The price of $6\%, 8\%$ and 10% are Rs.30000, Rs.15000 and Rs.20000 respectively.

4. A boy is walking along the path $y=ax^2+bx+c$ through the points$(-6,8), (-2,-12), (3,8)$. He wants to meet his friend at$P(7,60)$. Will he meet his friend? (Use Gaussian elimination method)

$$y=ax^2+bx+c$$

$(-6,8) \Rightarrow 8=a(-6)^2+b(-6)+c \Rightarrow 36a-6b+c=8 \to (1)$

$(-2,-12) \Rightarrow -12=a(-2)^2+b(-2)+c \Rightarrow 4a-2b+c=-12 \to (2)$

$(3,8) \Rightarrow 8=a(3)^2+b(3)+c \Rightarrow 9a+3b+c=8 \to (3)$

$$[A|B]=\left(\begin{array}{ccc|c}36 & -6 & 1 & 8\\ 4 & -2 & 1 & -12\\ 9 & 3 & 1 & 8\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}36 & -6 & 1 & 8\\ 0 & -12 & 8 & -116\\ 0 & 18 & 3 & 24\end{array}\right) \begin{array}{l}R_2 \to 9R_2 - R_1\\ R_3 \to 4R_3 - R_1\end{array}$$

$$\sim\left(\begin{array}{ccc|c}36 & -6 & 1 & 8\\ 0 & -6 & 4 & -58\\ 0 & 6 & 1 & 8\end{array}\right) \begin{array}{l}R_2 \to \left(\frac{1}{2}\right)R_2\\ R_3 \to \left(\frac{1}{3}\right)R_3\end{array}$$

$$\sim\left(\begin{array}{ccc|c}36 & -6 & 1 & 8\\ 0 & -6 & 4 & -58\\ 0 & 0 & 5 & -50\end{array}\right) R_3 \to R_3 + R_2$$

$\Rightarrow 36a-6b+c=8 \to (1)\,; -6b+4c=-58 \to (2)\,; 5c=-50 \Rightarrow c=-10$

Put $c=-10\ in\ (2) \Rightarrow -6b+4(-10)=-58 \Rightarrow -6b-40=-58$

$\Rightarrow -6b=-58+40 \Rightarrow -6b=-18 \Rightarrow b=3$

Put $b=3, c=-10\ in\ (1) \Rightarrow 36a-6(3)-10=8 \Rightarrow 36a-18-10=8$

$\Rightarrow 36a=8+28 \Rightarrow 36a=36 \Rightarrow a=1$

Solution: $(a,b,c)=(1,3,-10) \Rightarrow \boxed{y=x^2+3x-10}$

Put $x=7 \Rightarrow y=(7)^2+3(7)-10=49+21-10=60 \Rightarrow y=60$

The point $P(7,60)$ satisfies the equation $y=x^2+3x-10$, hence the boy will meet friend at $P(7,60)$.

EXAMPLE 1.28 The upward speed $v(t)$ a rocket at time t is approximated by $v(t)=at^2+bt+c, 0\le t\le 100$ where a, b and c are constant. It has been found that the speed at times $t=3, t=6$ and $t=9$ seconds are respectively 64, 133 and 208 miles per second respectively. Find the speed at time t=15 seconds. (Use Gaussian elimination method)

RANK METHOD – NON-HOMOGENEOUS LINEAR EQUATIONS

- If $\rho(A)=\rho([A|B])=3$, then the system of equations is consistent and has unique solution.
- If $\rho(A)=\rho([A|B])=2$, then the system of equations is consistent and has infinitely many solutions.(Put z=k and find x, y)
- If $\rho(A)=\rho([A|B])=1$, then the system of equations is consistent and has infinitely many solutions.(Put y=s and z=t and find x)
- If $\rho(A)\neq\rho([A|B])$, then the system of equations is inconsistent and has no solution.

EXERCISE 1.6

1.Test for consistency and if possible, solve the following systems of equations by rank method:

$(i) x-y+2z=2, 2x+y+4z=7, 4x-y+z=4$

$$AX=B \Rightarrow \begin{bmatrix}1 & -1 & 2\\ 2 & 1 & 4\\ 4 & -1 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}2\\ 7\\ 4\end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 2 & 1 & 4 & 7 \\ 4 & -1 & 1 & 4 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 3 & 0 & 3 \\ 0 & 3 & -7 & -4 \end{array}\right) \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 4R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 3 & 0 & 3 \\ 0 & 0 & -7 & -7 \end{array}\right) R_3 \to R_3 - R_2$$

The last equivalent matrix is in echelon form.

$\rho([A|B]) = 3 = \rho(A) \Longrightarrow$ The system of equations is consistent and has unique solution

$$AX = B \Longrightarrow \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 0 \\ 0 & 0 & -7 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -7 \end{pmatrix}$$

$$x - y + 2z = 2 \to (1)$$

$$\Longrightarrow \quad 3y = 3 \Longrightarrow y = 1$$

$$-7z = -7 \Longrightarrow z = 1$$

Put $y = 1, z = 1$ in $(1) \Longrightarrow x - 1 + 2 = 2 \Longrightarrow x = 1$

Solution: $(x, y, z) = (1,1,1)$

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EXAMPLE 1.29 Test for consistency of the following systems of equations and if possible solve: $x + 2y - z = 3, 3x - y + 2z = 1, x - 2y + 3z = 3, x - y + z + 1 = 0$

$(ii) 3x + y + z = 2, x - 3y + 2z = 1, 7x - y + 4z = 5$

$$AX = B \Longrightarrow \begin{bmatrix} 3 & 1 & 1 \\ 1 & -3 & 2 \\ 7 & -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 3 & 1 & 1 & 2 \\ 1 & -3 & 2 & 1 \\ 7 & -1 & 4 & 5 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 3 & 1 & 1 & 2 \\ 7 & -1 & 4 & 5 \end{array}\right) R_1 \leftrightarrow R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 0 & 10 & -5 & -1 \\ 0 & 20 & -10 & -2 \end{array}\right) \begin{matrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 7R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 0 & 10 & -5 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right) R_3 \to R_3 - 2R_2$$

The last equivalent matrix is in echelon form.

$\rho([A|B]) = 2 = \rho(A) \Longrightarrow$ The system of equations is consistent and has unique solution

$$AX = B \Longrightarrow \begin{pmatrix} 1 & -3 & 2 \\ 0 & 10 & -5 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\Longrightarrow \begin{matrix} x - 3y + 2z = 1 \to (1) \\ 10y - 5z = -1 \to (2) \end{matrix}$$

Put $z = t$ in $(2) \Longrightarrow 10y - 5t = -1 \Longrightarrow 10y = 5t - 1 \Longrightarrow y = \frac{1}{10}(5t - 1)$

Put $y = \frac{1}{10}(5t - 1), z = t$ in $(1) \Longrightarrow x - 3\frac{1}{10}(5t - 1) + 2t = 1$

$\Longrightarrow x = 1 - 2t + 3\frac{1}{10}(5t - 1) = \frac{10 - 20t + 15t - 3}{10} = \frac{1}{10}(7 - 5t)$

Solution: $(x, y, z) = \left(\frac{1}{10}(7 - 5t), \frac{1}{10}(5t - 1), t\right); t \in R$

EXAMPLE 1.30 Test for consistency of the following systems of equations and if possible solve: $4x - 2y + 6z = 8, x + y - 3z = -1, 15x - 3y + 9z = 21$

$(iii) 2x + 2y + z = 5, x - y + z = 1, 3x + y + 2z = 4$

$$AX = B \Longrightarrow \begin{bmatrix} 2 & 2 & 1 \\ 1 & -1 & 1 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 2 & 2 & 1 & 5 \\ 1 & -1 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 2 & 2 & 1 & 5 \\ 3 & 1 & 2 & 4 \end{array}\right) R_1 \leftrightarrow R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 4 & -1 & 1 \end{array}\right) \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{array}\right) R_3 \to R_3 - R_2$$

The last equivalent matrix is in echelon form.

$\rho([A|B]) = 3, \rho(A) = 2 \Longrightarrow \rho([A|B]) \neq \rho(A)$, The system of equations is inconsistent and has no solution

EXAMPLE 1.32 Test for consistency of the following systems of equations and if possible solve: $x - y + z = -9, 2x - y + z = 4, 3x - y + z = 6, 4x - y + 2z = 7$

$(iv) 2x - y + z = 2, 6x - 3y + 3z = 6, 4x - 2y + 2z = 4$

$$AX = B \Longrightarrow \begin{bmatrix} 2 & -1 & 1 \\ 6 & -3 & 3 \\ 4 & -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 6 & -3 & 3 & 6 \\ 4 & -2 & 2 & 4 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \begin{matrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 2R_1 \end{matrix}$$

The last equivalent matrix is in echelon form.

$\rho([A|B]) = 1, \rho(A) = 1 \Longrightarrow$ The system of equations is consistent and has infinitely many solutions.

$$AX = B \Longrightarrow \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$$

$\Longrightarrow 2x - y + z = 2$

Put y= and z=t in the above equation, $2x - s + t = 2 \Longrightarrow 2x = s - t + 2$

$\Longrightarrow x = \frac{1}{2}(s - t + 2)$

Solution: $(x, y, z) = \left(\frac{1}{2}(s - t + 2), s, t\right); s, t \in R$

EXAMPLE 1.31 Test for consistency of the following systems of equation and if possible solve: $x - y + z = -9, 2x - 2y + 2z = -18, 3x - 3y + 3z + 27 = 0$

2.Find the value of k for which the equations $kx - 2y + z = 1, x - 2ky + z = -2, x - 2y + kz = 1$ have (i) no solution (ii) unique solution (iii) infinitely many solution

$$AX = B \Longrightarrow \begin{bmatrix} k & -2 & 1 \\ 1 & -2k & 1 \\ 1 & -2 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} k & -2 & 1 & 1 \\ 1 & -2k & 1 & -2 \\ 1 & -2 & k & 1 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 1 & -2k & 1 & -2 \\ k & -2 & 1 & 1 \end{array}\right) R_1 \leftrightarrow R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2-2k & 1-k & -3 \\ 0 & 2k-2 & 1-k^2 & 1-k \end{array}\right) \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - kR_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2(1-k) & 1-k & -3 \\ 0 & 0 & 2-k-k^2 & -k-2 \end{array}\right) R_3 \to R_3 + R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2(1-k) & 1-k & -3 \\ 0 & 0 & (k+2)(1-k) & -(k+2) \end{array}\right)$$

(i) When $k = 1, k \neq -2, \rho([A|B]) = 3, \rho(A) = 1 \Longrightarrow \rho([A|B]) \neq \rho(A)$, the system of equations is inconsistent and has no solution.

(ii) When $k \neq 1,\ k \neq -2, \rho([A|B]) = 3 = \rho(A)$, the system of equations is consistent and has unique solution.

(iii) When $k = -2, k \neq 1, \rho([A|B]) = 2 = \rho(A)$, the system of equations is consistent and has infinitely many solutions.

3.Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9, 7x + 3y - 5z = 8, 2x + 3y + \lambda z = \mu$, have (i) no solution (ii) unique solution (iii) infinitely many solution

$$AX = B \Longrightarrow \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -5 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -5 & 8 \\ 2 & 3 & \lambda & \mu \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -25 & --9 \\ 0 & 0 & \lambda - 5 & \mu - 9 \end{array}\right) \begin{matrix} R_2 \to 2R_2 - 7R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

(i) When $\lambda = 5, \mu \neq 9, \rho([A|B]) = 3, \rho(A) = 2 \Longrightarrow \rho([A|B]) \neq \rho(A)$, the system of equations is inconsistent and has no solution.

(ii) When $\lambda \neq 5, \mu \neq 9, \rho([A|B]) = 3 = \rho(A)$, the system of equations is consistent and has unique solution.

(iii) When $\lambda = 5, \mu = 9, \rho([A|B]) = 2 = \rho(A)$, the system of equations is consistent and has infinitely many solutions.

EXAMPLE 1.34 Investigate for what values of λ *and* μ the system of linear equations $x + 2y + z = 7, x + y + \lambda z = \mu, x + 3y - 5z = 5$, have (i) no solution (ii) unique solution (iii) infinitely many solution

$$AX = B \Longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & \lambda \\ 1 & 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ \mu \\ 5 \end{bmatrix}$$

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Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 1 & 1 & \lambda & \mu \\ 1 & 3 & -5 & 5 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 1 & 3 & -5 & 5 \\ 1 & 1 & \lambda & \mu \end{array}\right) R_2 \leftrightarrow R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 0 & 1 & -6 & -2 \\ 0 & -1 & \lambda - 1 & \mu - 7 \end{array}\right) \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 7 \\ 0 & 1 & -6 & -2 \\ 0 & 0 & \lambda - 7 & \mu - 9 \end{array}\right) R_3 \to R_3 + R_2$$

(i) When $\lambda = 7, \mu \neq 9, \rho([A|B]) = 3, \rho(A) = 2 \Longrightarrow \rho([A|B]) \neq \rho(A)$, the system of equations is inconsistent and has no solution.

(ii) When $\lambda \neq 7, \mu \neq 9, \rho([A|B]) = 3 = \rho(A)$, the system of equations is consistent and has unique solution.

(iii) When $\lambda = 7, \mu = 9, \rho([A|B]) = 2 = \rho(A)$, the system of equations is consistent and has infinitely many solutions.

EXAMPLE 1.33 Find the condition on a, b and c so that the following system of linear equations has one parameter family of solutions: $x + y + z = a, x + 2y + 3z = b, 3x + 5y + 7z = c$

$$AX = B \Longrightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Augmented matrix $[A|B] = \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & 2 & 3 & b \\ 3 & 5 & 7 & c \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b - a \\ 0 & 2 & 4 & c - 3a \end{array}\right) \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b - a \\ 0 & 0 & 0 & (c - 3a) - 2(b - a) \end{array}\right) R_3 \to R_3 - 2R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b - a \\ 0 & 0 & 0 & c - 2b - a \end{array}\right)$$

Since given that the system have one parameter family of solutions, $\rho([A|B]) = \rho(A) = 2$. So the third row must be a zero row. So $c - 2b - a = 0 \Longrightarrow c = a + 2b$

RANK METHOD – HOMOGENEOUS LINEAR EQUATIONS

- If $\rho(A) = \rho([A|O]) = 3$, then the system of equations is consistent and has trivial solution.ie., $(x, y, z) = (0,0,0)$
- If $\rho(A) = \rho([A|O]) = 2$ *or* $1 < n$, then the system of equations is consistent and has non-trivial solutions.
- If the system of equations has non-trivial solution, $\rho([A|O]) < n$, so the determinant of the coefficient matrix is 0.

EXERCISE 1.7

1. Solve the following system of homogenous equations:

$(i) 3x + 2y + 7z = 0, 4x - 3y - 2z = 0, 5x + 9y + 23z = 0$

$$AX = O \Longrightarrow \begin{bmatrix} 3 & 2 & 7 \\ 4 & -3 & -2 \\ 5 & 9 & 23 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Augmented matrix $[A|O] = \left(\begin{array}{ccc|c} 3 & 2 & 7 & 0 \\ 4 & -3 & -2 & 0 \\ 5 & 9 & 23 & 0 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 3 & 2 & 7 & 0 \\ 0 & -17 & -34 & 0 \\ 0 & 17 & 34 & 0 \end{array}\right) \begin{matrix} R_2 \to 3R_2 - 4R_1 \\ R_3 \to 3R_3 - 5R_1 \end{matrix}$$

$$\sim\left(\begin{array}{ccc|c}3 & 2 & 7 & 0\\ 0 & -17 & -34 & 0\\ 0 & 0 & 0 & 0\end{array}\right) R_3 \to R_3 + R_2$$

$\Longrightarrow \rho([A|B]) = 2 = \rho(A)$, the system of equations is consistent and has non-trivial solution.

$$AX = O \Longrightarrow \begin{pmatrix}3 & 2 & 7\\ 0 & -17 & -34\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix} \Longrightarrow \begin{matrix}3x + 2y + 7z = 0 \to (1)\\ -17y - 34z = 0 \to (2)\end{matrix}$$

Put z=t in $(2) \Longrightarrow -17y - 34t = 0 \Longrightarrow y = -2t$

Put $y = -2t, z = t\ in\ (1) \Longrightarrow 3x + 2(-2t) + 7t = 0 \Longrightarrow 3x = -3t \Longrightarrow x = -t$

Solution : $(x, y, z) = (-t, -2t, t); t \in R$

EXAMPLE 1.36 Solve the system : $x + 3y - 2z = 0, 2x - y + 4z = 0, x - 11y + 14z = 0$

$1(ii) 2x + 3y - z = 0, x - y - 2z = 0, 3x + y + 3z = 0$

$$AX = O \Longrightarrow \begin{bmatrix}2 & 3 & -1\\ 1 & -1 & -2\\ 3 & 1 & 3\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

Augmented matrix $[A|O] = \left(\begin{array}{ccc|c}2 & 3 & -1 & 0\\ 1 & -1 & -2 & 0\\ 3 & 1 & 3 & 0\end{array}\right)$

$$\sim\left(\begin{array}{ccc|c}1 & -1 & -2 & 0\\ 2 & 3 & -1 & 0\\ 3 & 1 & 3 & 0\end{array}\right) R_1 \leftrightarrow R_2$$

$$\sim\left(\begin{array}{ccc|c}1 & -1 & -2 & 0\\ 0 & 5 & 3 & 0\\ 0 & 4 & 9 & 0\end{array}\right) \begin{matrix}R_2 \to R_2 - 2R_1\\ R_3 \to R_3 - 3R_1\end{matrix}$$

$$\sim\left(\begin{array}{ccc|c}1 & -1 & -2 & 0\\ 0 & 5 & 3 & 0\\ 0 & 0 & 33 & 0\end{array}\right) R_3 \to 5R_3 - 4R_2$$

$\Longrightarrow \rho([A|B]) = 3 = \rho(A)$, the system of equations is consistent and has trivial solution. $(x, y, z) = (0,0,0)$

EXAMPLE 1.35 Solve the system : $x + 2y + 3z = 0, 3x + 4y + 4z = 0, 7x + 10y + 12z = 0$

EXAMPLE 1.37 Solve the system : $x + y - 2z = 0, 2x - 3y + z = 0, 3x - 7y + 10z = 0, 6x - 9y + 10z = 0$

2.Determine the values of **λ** for which the following system of equations $x + y + 3z = 0, 4x + 3y + \lambda z = 0, 2x + y + 2z = 0$ has (i) a unique solution (ii) a non- trivial solution

$$AX = O \Longrightarrow \begin{bmatrix}1 & 1 & 3\\ 4 & 3 & \lambda\\ 2 & 1 & 2\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

Augmented matrix $[A|O] = \left(\begin{array}{ccc|c}1 & 1 & 3 & 0\\ 4 & 3 & \lambda & 0\\ 2 & 1 & 2 & 0\end{array}\right)$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 3 & 0\\ 2 & 1 & 2 & 0\\ 4 & 3 & \lambda & 0\end{array}\right) R_2 \leftrightarrow R_3$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 3 & 0\\ 0 & -1 & -4 & 0\\ 0 & -1 & \lambda - 12 & 0\end{array}\right) \begin{matrix}R_2 \to R_2 - 2R_1\\ R_3 \to R_3 - 4R_1\end{matrix}$$

$$\sim\left(\begin{array}{ccc|c}1 & 1 & 3 & 0\\ 0 & -1 & -4 & 0\\ 0 & 0 & \lambda - 8 & 0\end{array}\right) R_3 \to R_3 - R_2$$

(i)When $\lambda \neq 8 \Longrightarrow \rho(A) = \rho([A|O]) = 3$, then the system of equations is consistent and has trivial solution. $(x, y, z) = (0,0,0)$

(ii)When $\lambda = 8 \Longrightarrow \rho(A) = \rho([A|O]) = 2$, then the system of equations is consistent and has non- trivial solution.

EXAMPLE 1.38 Determine the values of **λ** for which the following system of equations $(3\lambda - 8)x + 3y + 3z = 0, 3x + (3\lambda - 8)y + 3z = 0, 3x + 3y + (3\lambda - 8)z = 0$ has a non- trivial solution.

Here the number of unknowns is 3.

Since the system of equations has non-trivial solution, $\rho([A|O]) < 3$, so the determinant of the coefficient matrix is 0.

$$\begin{vmatrix}3\lambda - 8 & 3 & 3\\ 3 & 3\lambda - 8 & 3\\ 3 & 3 & 3\lambda - 8\end{vmatrix} = 0$$

$$\Longrightarrow \begin{vmatrix}3\lambda - 2 & 3\lambda - 2 & 3\lambda - 2\\ 3 & 3\lambda - 8 & 3\\ 3 & 3 & 3\lambda - 8\end{vmatrix} = 0 \quad (R_1 \to R_1 + R_2 + R_3)$$

$\Rightarrow (3\lambda-2)\begin{vmatrix}1 & 1 & 1\\ 3 & 3\lambda-8 & 3\\ 3 & 3 & 3\lambda-8\end{vmatrix}=0$

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$\Rightarrow (3\lambda-2)\begin{vmatrix}1 & 1 & 1\\ 0 & 3\lambda-11 & 0\\ 0 & 0 & 3\lambda-11\end{vmatrix}=0 \quad \begin{pmatrix}R_2\to R_2-3R_1\\ R_3\to R_3-3R_1\end{pmatrix}$

Expanding through C_1,

$\Rightarrow (3\lambda-2)(3\lambda-11)^2=0 \Rightarrow \lambda=\frac{2}{3}, \lambda=\frac{11}{3}$

EXAMPLE 1.40 If the system of equations $px+by+cz=0, ax+qy+cz=0, ax+by+rz=0$ has a non-trivial solution and $p\neq a, q\neq b, r\neq c$, prove that $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=2$

Since the system of equations has non-trivial solution, $\rho([A|O])<3$, so the determinant of the coefficient matrix is 0.

$\begin{vmatrix}p & b & c\\ a & q & c\\ a & b & r\end{vmatrix}=0$

$\Rightarrow \begin{vmatrix}p & b & c\\ a-p & q-b & 0\\ a-p & 0 & r-c\end{vmatrix}=0 \quad \begin{pmatrix}R_2\to R_2-R_1\\ R_3\to R_3-R_1\end{pmatrix}$

$\Rightarrow \begin{vmatrix}p & b & c\\ -(p-a) & q-b & 0\\ -(p-a) & 0 & r-c\end{vmatrix}=0$

$\Rightarrow (p-a)(q-b)(r-c)\begin{vmatrix}\frac{p}{p-a} & \frac{q}{q-b} & \frac{r}{r-c}\\ -1 & 1 & 0\\ -1 & 0 & 1\end{vmatrix}=0 \Rightarrow \begin{vmatrix}\frac{p}{p-a} & \frac{q}{q-b} & \frac{r}{r-c}\\ -1 & 1 & 0\\ -1 & 0 & 1\end{vmatrix}=0$

$\Rightarrow \frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=0$

$\Rightarrow \frac{p}{p-a}+\frac{q-(q-b)}{q-b}+\frac{r-(r-c)}{r-c}=0$

$\Rightarrow \frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}-1-1=0 \Rightarrow \frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=2$

3.By using Gaussian elimination method, balance the chemical reaction equation :

$C_2H_6+O_2\to H_2O+CO_2$

$C_2H_6+O_2\to H_2O+CO_2 \Rightarrow x_1C_2H_6+x_2O_2\to x_3H_2O+x_4CO_2\to(1)$

x_1			x_2				x_3			x_4		
C	H	O	C	H	O	→	C	H	O	C	H	O
2	6	0	0	0	2		0	2	1	1	0	2

Equating the number of atom of C on both sides with x_1, x_2, x_3, x_4 , we get

$2x_1+0x_2=0x_3+1x_4 \Rightarrow 2x_1+0x_2-0x_3-1x_4=0\to(2)$

Equating the number of atom of H on both sides with x_1, x_2, x_3, x_4 , we get

$6x_1+0x_2=2x_3+0x_4 \Rightarrow 6x_1+0x_2-2x_3-0x_4=0$

$\div 2 \Rightarrow 3x_1+0x_2-x_3-0x_4=0\to(3)$

Equating the number of atom of O on both sides with x_1, x_2, x_3, x_4 , we get

$0x_1+2x_2=1x_3+2x_4 \Rightarrow 0x_1+2x_2-1x_3-2x_4=0\to(4)$

From $(2),(3),(4) \Rightarrow$ Augmented matrix $[A|O]=\left(\begin{array}{cccc|c}2 & 0 & 0 & -1 & 0\\ 3 & 0 & -1 & 0 & 0\\ 0 & 2 & -1 & -2 & 0\end{array}\right)$

$\sim\left(\begin{array}{cccc|c}2 & 0 & 0 & -1 & 0\\ 0 & 0 & -2 & 3 & 0\\ 0 & 2 & -1 & -2 & 0\end{array}\right) \sim R_2\to 2R_2-3R_1$

$\sim\left(\begin{array}{cccc|c}2 & 0 & 0 & -1 & 0\\ 0 & 2 & -1 & -2 & 0\\ 0 & 0 & -2 & 3 & 0\end{array}\right) R_2\leftrightarrow R_3$

$\rho(A)=\rho([A|O])=3<4$,the systems of equations is consistent and has infinite solution.

$2x_1-1x_4=0\to(5)\ ;\ 2x_2-x_3-2x_4=0\to(6)\ ;\ -2x_3+3x_4=0\to(7)$

Put $x_4=t$ in $(5) \Rightarrow 2x_1-t=0 \Rightarrow x_1=\frac{t}{2}$

Put $x_4=t$ in $(7) \Rightarrow -2x_3+3x_4=0 \Rightarrow -2x_3+3t=0 \Rightarrow x_3=\frac{3t}{2}$

Put $x_3=\frac{3t}{2}, x_4=t$ in $(6) \Rightarrow 2x_2-\frac{3t}{2}-2t=0 \Rightarrow x_2=\frac{7t}{4}$

$(x_1,x_2,x_3,x_4)=\left(\frac{t}{2},\frac{7t}{4},\frac{3t}{2},t\right)$

Put $t=4 \Rightarrow (x_1,x_2,x_3,x_4)=(2,7,6,4)$

Sub. The above values in $(1) \Rightarrow 2C_2H_6+7O_2\to 6H_2O+4CO_2$

EXAMPLE 1.38 By using Gaussian elimination method, balance the chemical reaction equation : $C_5H_8+O_2\to CO_2+H_2O$

2.COMPLEX NUMBERS

POWERS OF IMAGINARY UNIT

- $i^2 = -1, i^3 = -i, i^4 = 1$
- Division algorithm : $n = 4(q) + r$
- $i^n = (i)^{4q+r} = (i)^{4q}(i)^r = (i^4)^q(i)^r \Longrightarrow if\ r = 0, i^n = 1\ ; if\ r = 1\ , i^n = i;$ $if\ r = 2, i^n = -1; if\ r = 3, i^n = -i$
- $\sqrt{ab} = \sqrt{a}\sqrt{b}$ is valid only if at least one of a, b is non-negative.
- For $y \in R, y^2 > 0$

EXERCISE 2.1

Simplify the following:

1.$i^{1947} + i^{1950}$

$1947 \div 4 = 4(486) + 3\ ; 1950 = 4(487) + 2$

$i^{1947} + i^{1950} = (i^4)^{486}(i)^3 + (i^4)^{487}(i)^2 = -i - 1 = -1 - i$

2.$i^{1948} - i^{-1869}$

$1948 \div 4 = 4(487) + 0\ ; 1869 \div 4 = 4(467) + 1$

$i^{1948} - i^{-1869} = i^{1948} - \frac{1}{i^{1869}} = (i^4)^{487}(i)^0 - \frac{1}{(i^4)^{467}i^1} = 1 - \frac{1}{i} = 1 - \left(\frac{1}{i} \times \frac{-i}{-i}\right)$

$= 1 - \left(\frac{-i}{-i^2}\right) = 1 + i\ \ (\because -i^2 = 1)$

3.$\sum_{n=1}^{12} i^n$

$\sum_{n=1}^{12} i^n = i^1 + i^2 + i^3 + i^4 + i^5 + i^6 + i^7 + +i^8 + i^9 + i^{10} + i^{11} + i^{12}$

$= i^1 + i^2 + i^3 + i^4 + i^4 i + (i^2)^3 + (i^2)^3 i + (i^2)^4 + (i^3)^3 + (i^2)^5 + (i^2)^5 i + (i^2)^6$

$= i - 1 - i + 1 + i - 1 - i + 1 + i - 1 - i + 1$

$= 0$

4.$i^{59} + \frac{1}{i^{59}}$

$59 \div 4 = 4(14) + 3$

$i^{59} + \frac{1}{i^{59}} = (i^4)^{14} i^3 + \frac{1}{(i^4)^{14} i^3}$

$= -i + \frac{1}{-i} = i + \frac{1}{-i}\frac{i}{i} = -i + \frac{i}{1} = i - i = 0$

5.$i.\, i^2.\, i^3.\, i^4 \ldots\ldots i^{2000}$

$= i^{(1+2+3\ldots\ldots+2000)}$

$= i^{\frac{2000\times2001}{2}} = i^{1000\times2001} = i^{1000}.\, i^{2001}$

$= (i^4)^{250}(i^4)^{500} i^1 \qquad (\because 2001 \div 4 = 4(500) + 1)$

$= 1.1.\, i$

$= i$

6.$\sum_{n=1}^{10} i^{n+50}$

$\sum_{n=1}^{10} i^{n+50} = i^{51} + i^{52} + i^{53} + \cdots + i^{60}$

$= i^{50}(i^1 + i^2 + i^3 + i^4 + i^5 + i^6 + i^7 + i^8 + i^9 + i^{10})$

$= (i^4)^{12} i^2 (i - 1 - i + 1 + i - 1 - i + 1 + i - 1)$

$= -1(i - 1)$

$= 1 - i$

EXAMPLE 2.1 Simplify the following:

(i) i^7 (ii)i^{1729} (iii)$i^{-1924} + i^{2018}$ (iv) $\sum_{n=1}^{102} i^n$ (v)$i.\, i^2.\, i^3.\, i^4 \ldots\ldots i^{40}$

COMPLEX NUMBERS

- **Rectangular Form Of a Complex Number:** A complex number is of the form $z = x + iy, x, y \in R$. x is called the real part and y is called the imaginary part of the complex number.
- Two complex numbers $z_1 = a + ib, z_2 = c + id$ are said to be equal if and only if $Re(z_1) = Re(z_2)$ and $Im(z_1) = Im(z_2)$. ie. $a = c\ and\ b = d$
- **Scalar multiplication of complex numbers:** $If\ z = x + iy\ and\ k \in R\ then\ kz = k(x + iy) = kx + kyi$
- **Addition of complex numbers:** If $z_1 = a + ib, z_2 = c + id, then\ z_1 + z_2 = (a + c) + i(b + d)$
- **Subtraction of complex numbers**: If $z_1 = a + ib, z_2 = c + id, then\ z_1 - z_2 = (a - c) + i(b - d)$
- **Multiplication of complex numbers:** If $z_1 = a + ib, z_2 = c + id, then\ z_1 z_2 = (ac - bd) + i(ad + bc)$
- Multiplication of a complex z by i successively gives a 90^0counter clockwise rotation successively about the origin.

EXERCISE 2.2

1.Evaluate the following if $z = 5 - 2i$ and $w = -1 + 3i$

$(i)z + w\ \ (ii)z - iw\ \ (iii)2z + 3w\ \ (iv)zw\ \ (v)z^2 + 2zw + w^2\ (vi)(z + w)^2$

$(i)z + w\ = 5 - 2i - 1 + 3i = 4 + i$

$(ii)z - iw\ = 5 - 2i - i(-1 + 3i) = 5 - 2i + i - 3i^2 = 5 - 2i + i + 3 = 8 - i$

$\because i^2 = -1$

$(iii)2z + 3w = 2(5 - 2i\ ) + 3(-1 + 3i) = 10 - 4i - 3 + 9i = 7 + 5i$

$(iv)\ z_1z_2 = (ac - bd) + i(ad + bc); a = 5, b = -2, c = -1, d = 3$

$zw = (5 - 2i)(-1 + 3i) = [(5)(-1) - (-2)(3)] + i[(5)(3) + (-2)(-1)]$

$= [-5 + 6] + i[15 + 2] = 1 + 17i$

$(v) z^2 = (5 - 2i)^2 = 25 - 20i + 4i^2 = 25 - 20i - 4 = 21 - 20i$

$2zw = 2(5 - 2i\ )(-1 + 3i) = 2[1 + 17i]$ from (iv)

$2zw = 2 + 34i$

$w^2 = (-1 + 3i)^2 = 1 - 6i + 9i^2 = 1 - 6i - 9 = -8 - 6i$

$z^2 + 2zw + w^2 = 21 - 20i + 2 + 34i - 8 - 6i = 15 + 8i$

$(vi)(z + w)^2 = (4 + i)^2$ from(i)

$= 16 + 8i + i^2 = 16 + 8i - 1 = 15 + 8i$

2. Given the complex number $z = 2 + 3i$, represent the complex numbers in Argand diagram.

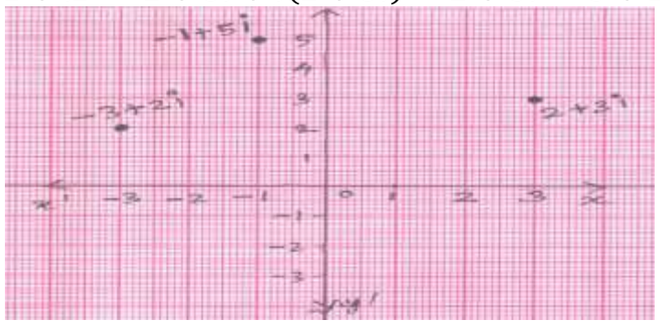
$(i) z, iz\ and\ z + iz \quad (ii) z, -iz\ and\ z - iz$

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$(i) z = 2 + 3i = (2,3)$

$iz = i(2 + 3i) = 2i + 3i^2 = -3 + 2i = (-3,2)$

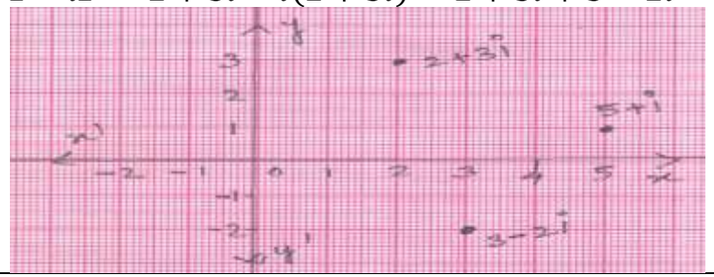
$z + iz = 2 + 3i + i(2 + 3i) = 2 + 3i\ - 3 + 2i = -1 + 5i = (-1,5)$



$(i) z = 2 + 3i = (2,3)$

$-iz = -i(2 + 3i) = -2i - 3i^2 = 3 - 2i = (3, -2)$

$z - iz = 2 + 3i - i(2 + 3i) = 2 + 3i + 3 - 2i = 5 + i = (5,1)$



3. Find the values of the real numbers $x\ and\ y$, if the complex numbers $(3 - i)x - (2 - i)y + 2i + 5$ and $2x + (-1 + 2i)y + 3 + 2i$ are equal.

$z_1 = (3 - i)x - (2 - i)y + 2i + 5$

$= 3x - xi - 2y + yi + 2i + 5$

$= (3x - 2y + 5) + i(-x + y + 2)$

$z_2 = 2x + (-1 + 2i)y + 3 + 2i$

$= 2x - y + 2yi + 3 + 2i$

$= (2x - y + 3) + i(2y + 2)$

Given : $z_1 = z_2 \Longrightarrow (3x - 2y + 5) + i(-x + y + 2) = (2x - y + 3) + i(2y + 2)$

Equating the real and imaginary parts on both sides,

$3x - 2y + 5 = 2x - y + 3$	$-x + y + 2 = 2y + 2$
$3x - 2y + 5 - 2x + y - 3 = 0$	$-x + y + 2 - 2y - 2 = 0$
$x - y = -2 \rightarrow (1)$	$-x - y = 0 \rightarrow (2)$

$(1) + (2) \Longrightarrow -2y = -2 \Longrightarrow \boxed{y = 1}$

Sub. $y = -1$ in $(1) \Longrightarrow x - 1 = -2 \Longrightarrow \boxed{x = -1}$

EXAMPLE 2.2 Find the values of the real numbers $x\ and\ y$, if the complex numbers $(2 + i)x + (1 - i)y + 2i - 3$ and $x + (-1 + 2i)y + 1 + i$ are equal.

$[Ans: x = 2\ and\ y = 1]$

PROPERTIES OF COMPLEX NUMBERS

Properties of complex numbers under addition:

I. Closure property: $\forall\ z_1, z_2 \in \mathbb{C}, (z_1 + z_2) \in \mathbb{C}$

II. Commutative property: $\forall\ z_1, z_2 \in \mathbb{C}, z_1 + z_2 = z_2 + z_1$

III. Associative property: $\forall\ z_1, z_2, z_3 \in \mathbb{C}, (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

IV. Additive identity: $\forall\ z \in \mathbb{C}, z + 0 = 0 + z = z \Longrightarrow\ 0 = 0 + 0i$ is an additive identity.

V. Additive inverse: $\forall\ z \in \mathbb{C}, -z \in \mathbb{C}\ \ni z + (-z) = (-z) + z = 0 \in \mathbb{C}, so - z$ is an additive identity.

Properties of complex numbers under multiplication:

I. Closure property: $\forall\ z_1, z_2 \in \mathbb{C}, (z_1.z_2) \in \mathbb{C}$

II. Commutative property: $\forall\ z_1, z_2 \in \mathbb{C}, z_1.z_2 = z_2.z_1$

III. Associative property: $\forall\ z_1, z_2, z_3 \in \mathbb{C}, (z_1.z_2).z_3 = z_1.(z_2.z_3)$

IV. Multiplicative identity: $\forall\ z \in \mathbb{C}, z.1 = 1.z = z \Longrightarrow\ 1 = 1 + 0i$ is a multiplicative identity.

V. Multiplicative inverse: $\forall\, z \in \mathbb{C}\ \exists \frac{1}{z}\ z \in \mathbb{C}\ \ni z.\frac{1}{z} = \frac{1}{z}.z = 1 + i0 \in \mathbb{C}$, $so\ \frac{1}{z}$ is multiplicative identity.

- If $z = x + iy$ then its multiplicative inverse is $z^{-1} = \left(\frac{x}{x^2+y^2}\right) + i\left(-\frac{y}{x^2+y^2}\right)$
- Distributive property: $\forall\, z_1, z_2, z_3 \in \mathbb{C}, (z_1 + z_2).z_3 = z_1z_3 + z_2z_3$ (or)
  - $\forall\, z_1, z_2, z_3 \in \mathbb{C}, z_1.(z_2 + z_3) = z_1z_2 + z_1z_3$
- Complex numbers obey the laws of indices:

$(i)z^m z^n = z^{m+n}$ $\quad (ii)\frac{z^m}{z^n} = z^{m-n}$

$(iii)(z^m)^n = z^m z^n$ $\quad (iv)(z_1z_2)^m = {z_1}^m {z_2}^m$

EXERCISE 2.3

1.If $z_1 = 1 - 3i, z_2 = -4i$ and $z_3 = 5$, show that

$(i)(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ $\quad (ii)(z_1z_2)z_3 = z_1(z_2z_2)$

$(i)(z_1 + z_2) + z_3 = (1 - 3i - 4i) + 5 = 1 - 7i + 5 = 6 - 7i \rightarrow (1)$

$z_1 + (z_2 + z_3) = 1 - 3i + (-4i + 5) = 1 - 3i - 4i + 5 = 6 - 7i \rightarrow (2)$

From (1) & (2), $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

$(ii)(z_1z_2)z_3 = [(1 - 3i)(-4i)]5 = (-4i + 12i^2)5$

$= (-4i - 12)5 = -60 - 20i \rightarrow (1)$

$z_1(z_2z_2) = (1 - 3i)[(-4i)(5)] = (1 - 3i)[-20i]$

$= -20i + 60i^2 = -60 - 20i \rightarrow (2)$

From (1) & (2), $(z_1z_2)z_3 = z_1(z_2z_2)$

2.If $z_1 = 3, z_2 = -7i$ and $z_3 = 5 + 4i$, show that

$(i)z_1.(z_2 + z_3) = z_1z_2 + z_1z_3$ $\quad (ii)(z_1 + z_2).z_3 = z_1z_3 + z_2z_3$

$(i)z_1.(z_2 + z_3) = 3(-7i + 5 + 4i) = 3(5 - 3i) = 15 - 9i \rightarrow (1)$

$z_1z_2 + z_1z_3 = (3)(-7i) + (3)(5 + 4i) = -21i + 15 + 12i = 15 - 9i \rightarrow (2)$

From (1) & (2), $z_1.(z_2 + z_3) = z_1z_2 + z_1z_3$

$(ii)(z_1 + z_2).z_3 = (3 - 7i)(5 + 4i)$

$= 15 + 12i - 35i - 28i^2 = 15 + 12i - 35i + 28 = 43 - 23i \rightarrow (1)$

$z_1z_3 + z_2z_3 = (3)(5 + 4i) + (-7i)(5 + 4i)$

$= 15 + 12i - 35i - 28i^2 = 15 + 12i - 35i + 28 = 43 - 23i \rightarrow (2)$

From (1) & (2), $(z_1 + z_2).z_3 = z_1z_3 + z_2z_3$

3.If $z_1 = 2 + 5i, z_2 = -3 - 4i$ and $z_3 = 1 + i$, find the additive and multiplicative inverse of z_1, z_2 and z_3

Additive inverse of z is $-z$.

So, Additive inverse of $z_1 = 2 + 5i$ is $-z_1 = -2 - 5i$,

Additive inverse of $z_2 = -3 - 4i$ is- $z_2 = 3 + 4i$

Additive inverse of $z_3 = 1 + i$ is $-z_3 = -1 - i$

Multiplicative inverse of $z = x + iy$ is $z^{-1} = \left(\frac{x}{x^2+y^2}\right) + i\left(-\frac{y}{x^2+y^2}\right)$

Multiplicative inverse of $z_1 = 2 + 5i$ is ${z_1}^{-1} = \left(\frac{2}{2^2+5^2}\right) + i\left(-\frac{5}{2^2+5^2}\right) = \frac{2}{29} - \frac{5}{29}i$

Multiplicative inverse of $z_2 = -3 - 4i$ is ${z_2}^{-1} = \left(\frac{-3}{(-3)^2+(-4)^2}\right) + i\left(-\frac{-4}{(-3)^2+(-4)^2}\right)$

$= -\frac{3}{25} + \frac{4}{25}i$

Multiplicative inverse of $z_3 = 1 + i$ is $z^{-1} = \left(\frac{1}{1^2+1^2}\right) + i\left(-\frac{1}{1^2+1^2}\right) = \frac{1}{2} - \frac{1}{2}i$

CONJUGATE OF A COMPLEX NUMBER

- The conjugate of the complex number $z = x + iy$ is $\bar{z} = x - iy$.
- $z\bar{z} = x^2 + y^2$
- The conjugate is useful in division of complex numbers.
- $\frac{z_1}{z_2} = \frac{a+ib}{c+id} = \left[\frac{ac+bd}{c^2+d^2}\right] + i\left[\frac{bc-ad}{c^2+d^2}\right]$

PROPERTIES OF COMPLEX CONJUGATES:

- ✓ $\overline{z_1 + z_2} = \bar{z_1} + \bar{z_2}$
- ✓ $\overline{z_1 - z_2} = \bar{z_1} - \bar{z_2}$
- ✓ $\overline{z_1.z_2} = \bar{z_1}.\bar{z_2}$
- ✓ $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z_1}}{\bar{z_2}}, z_2 \neq 0$
- ✓ $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$
- ✓ $\mathrm{Im}(z) = \frac{z-\bar{z}}{2i}$
- ✓ $\overline{(z^n)} = (\bar{z})^n$
- ✓ z is real if and only if $z = \bar{z}$
- ✓ z is purely imaginary if and only if $z = -\bar{z}$
- ✓ $\bar{\bar{z}} = z$

EXERCISE 2.4

1.Write the following in the rectangular form:

$(i)\overline{(5 + 9i) + (2 - 4i)}$ $\quad (ii)\frac{10-5i}{6+2i}$ $\quad (iii)\bar{3i} + \frac{1}{2-i}$

$(i)\overline{(5 + 9i) + (2 - 4i)} = \overline{5 + 9i} + \overline{2 - 4i} = 5 - 9i + 2 + 4i = 7 - 5i$

$(ii)\frac{10-5i}{6+2i} = \frac{10-5i}{6+2i} \times \frac{6-2i}{6-2i} = \frac{60-20i-30i+10i^2}{6^2+2^2}$

$= \frac{60-50i-10}{40} = \frac{50-50i}{40} = \frac{5}{4} - \frac{5}{4}i$

$(iii)\bar{3i} + \frac{1}{2-i} = -3i + \frac{1}{2-i} \times \frac{2+i}{2+i} = -3i + \frac{2+i}{2^2+1^2} = -3i + \frac{2}{5} + \frac{1}{5}i$

$= \frac{2}{5} + \frac{-15i+i}{5} = \frac{2}{5} - \frac{14}{5}i$

2.If z = x + iy, find the following in rectangular form:

$(i)Re\left(\frac{1}{z}\right)$ $(ii)Re(i\bar{z})$ $(iii)Im(3z + 4\bar{z} - 4i)$

$(i)\frac{1}{z} = \frac{1}{x+iy} \times \frac{x-iy}{x-iy} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i \Rightarrow Re\left(\frac{1}{z}\right) = \frac{x}{x^2+y^2}$ (or)

$\therefore Re\left(\frac{1}{z}\right) = Re(z^{-1}) = Re\left(\frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i\right) = \frac{x}{x^2+y^2}$

$(ii)i\bar{z} = i(\overline{x+iy}) = i(x - iy) = ix - i^2y = y + ix$

$\therefore Re(i\bar{z}) = Re(-y + ix) = y$

$(iii)3z + 4\bar{z} - 4i = 3(x + iy) + 4\overline{x+iy} - 4i = 3x + 3yi + 4(x - iy) - 4i$

$= 3x + 3yi + 4x - 4yi - 4i = 7x - yi - 4i = 7x + i(-y - 4)$

$\therefore Im(3z + 4\bar{z} - 4i) = Im\big(7x + i(-y - 4)\big) = -y - 4$

3. If $z_1 = 2 - i$ and $z_2 = -4 + 3i$, find the inverse of z_1z_2 and $\frac{z_1}{z_2}$

$z_1z_2 = (2 - i)(-4 + 3i) = (-8 + 3) + i(6 + 4) = -5 + 10i$

$(\because z_1z_2 = (ac - bd) + i(ad + bc))$

Inverse of $z_1z_2 = \left(\frac{x}{x^2+y^2}\right) + i\left(-\frac{y}{x^2+y^2}\right) = \left(\frac{-5}{(-5)^2+10^2}\right) + i\left(-\frac{10}{(-5)^2+10^2}\right)$

$= \frac{-5}{125} - \frac{10}{125}i = -\frac{1}{25} - \frac{2}{25}i$

$\frac{z_1}{z_2} = \frac{2-i}{-4+3i} = \left[\frac{-8-3}{(-4)^2+3^2}\right] + i\left[\frac{6-4}{(-4)^2+3^2}\right] = -\frac{11}{25} + \frac{2}{25}i$

$(\because \frac{z_1}{z_2} = \frac{a+ib}{c+id} = \left[\frac{ac+bd}{c^2+d^2}\right] + i\left[\frac{bc-ad}{c^2+d^2}\right])$

Inverse of $\frac{z_1}{z_2} = \left(\frac{x}{x^2+y^2}\right) + i\left(-\frac{y}{x^2+y^2}\right)$

$= \left(\frac{-11/25}{(-11/25)^2+(-2/25)^2}\right) + i\left(-\frac{-2/25}{(-11/25)^2+(-2/25)^2}\right)$

$= -\frac{11}{25\times\frac{125}{625}} + \frac{2}{25\times\frac{125}{625}}i = -\frac{11}{5} + \frac{2}{5}i$

4. The complex numbers u, v and w are related by $\frac{1}{u} = \frac{1}{v} + \frac{1}{w}$. If $v = 3 - 4i$ and $w = 4 + 3i$, find u in the rectangular form.

$\frac{1}{u} = \frac{1}{v} + \frac{1}{w} \Rightarrow \frac{1}{u} = \frac{1}{3-4i} + \frac{1}{4+3i}$

$= \left(\frac{1}{3-4i} \times \frac{3+4i}{3+4i}\right) + \left(\frac{1}{4+3i} \times \frac{4-3i}{4-3i}\right)$

$= \frac{3+4i}{3^2+4^2} + \frac{4-3i}{4^2+3^2}$

$= \frac{1}{25}(3 + 4i + 4 - 3i)$

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$\Rightarrow \frac{1}{u} = \frac{7+i}{25}$

$u = \frac{25}{7+i} = \frac{25}{7+i} \times \frac{7-i}{7-i} = \frac{25(7-i)}{7^2+1^2} = \frac{25(7-i)}{50} = \frac{7-i}{2} = \frac{7}{2} - \frac{1}{2}i$

5.Prove the following properties:

(i)z is real if and only if $z = \bar{z}$ $(ii)\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$ $(iii)\mathrm{Im}(z) = \frac{z-\bar{z}}{2i}$

(i)z is real $\Rightarrow z = x + 0i$

$\bar{z} = \overline{x + 0i} = x - 0i = x \Rightarrow$ z is real if and only if $z = \bar{z}$

(ii) *Let* $z = x + iy$; $Re(z) = x \rightarrow (1)$

$\Rightarrow \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy = 2x \Rightarrow \frac{z+\bar{z}}{2} = x \rightarrow (2)$

From (1) & (2), $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$

(iii) *Let* $z = x + iy$; $Im(z) = y \rightarrow (1)$

$\Rightarrow \bar{z} = x - iy$

$z - \bar{z} = x + iy - x + iy = 2yi \Rightarrow \frac{z-\bar{z}}{2i} = y \rightarrow (2)$

From (1) & (2), $\mathrm{Im}(z) = \frac{z-\bar{z}}{2i}$

6.Find the least value of the positive integer n for which $\left(\sqrt{3} + i\right)^n$

$(i)real$ $(ii)purely\ imaginary$

$\left(\sqrt{3} + i\right)^3 = 3\sqrt{3} + 9i - 3\sqrt{3} - i = 8i$

$\left(\sqrt{3} + i\right)^6 = \left(\left(\sqrt{3} + i\right)^3\right)^2 = (8i)^2 = -64$

$(i)When\ n = 6, \left(\sqrt{3} + i\right)^n is\ real$

$(ii)When\ n = 3, \left(\sqrt{3} + i\right)^n is\ purely\ imaginary$

7.Show that $(i)\left(2 + i\sqrt{3}\right)^{10} - \left(2 - i\sqrt{3}\right)^{10}$ is purely imaginary.

$(ii)\left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$ is real.

$(i)\ z = \left(2 + i\sqrt{3}\right)^{10} - \left(2 - i\sqrt{3}\right)^{10}$

$\bar{z} = \overline{\left(2 + i\sqrt{3}\right)^{10} - \left(2 - i\sqrt{3}\right)^{10}} = \overline{\left(2 + i\sqrt{3}\right)^{10}} - \overline{\left(2 - i\sqrt{3}\right)^{10}}$

$= \overline{\left(2+\iota\sqrt{3}\right)}^{10} - \overline{\left(2-\iota\sqrt{3}\right)}^{10} = \left(2-i\sqrt{3}\right)^{10} - \left(2+i\sqrt{3}\right)^{10}$

$= -\left\{\left(2+i\sqrt{3}\right)^{10} - \left(2-i\sqrt{3}\right)^{10}\right\} = -z$

$\therefore \left(2+i\sqrt{3}\right)^{10} - \left(2-i\sqrt{3}\right)^{10}$ is purely imaginary.

$(ii)\ \frac{19-7i}{9+i} = \frac{19-7i}{9+i} \times \frac{9-i}{9-i} = \frac{171-19i-63i+7i^2}{9^2+1^2} = \frac{164-82i}{82} = \frac{82(2-i)}{82} = 2-i$

$\frac{20-5i}{7-6i} = \frac{20-5i}{7-6i} \times \frac{7+6i}{7+6i} = \frac{140+120i-35i-30i^2}{7^2+6^2} = \frac{170+85i}{85} = 2+i$

$z = \left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12} = (2-i)^{12} - (2+i)^{12}$

$\bar{z} = \overline{(2-\iota)^{12} - (2+\iota)^{12}} = \overline{(2-\iota)^{12}} - \overline{(2+\iota)^{12}} = (2+i)^{12} - (2-i)^{12} = z$

$\therefore \left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$ is real.

EXAMPLE 2.8 Show that $(i)\left(2+i\sqrt{3}\right)^{10} + \left(2-i\sqrt{3}\right)^{10}$ is real and

$(ii)\left(\frac{19+9ii}{5-3i}\right)^{15} - \left(\frac{8+i}{1+2i}\right)^{15}$ is purely imaginary.

EXAMPLE 2.3 Write $\frac{3+4i}{5-12i}$ in the x + iy form , hence find its real and imaginary parts.

$$\boxed{\frac{z_1}{z_2} = \frac{a+ib}{c+id} = \left[\frac{ac+bd}{c^2+d^2}\right] + i\left[\frac{bc-ad}{c^2+d^2}\right]}$$

$\frac{3+4i}{5-12i} = \left[\frac{15-48}{5^2+(-12)^2}\right] + i\left[\frac{20+36}{5^2+(-12)^2}\right] = \frac{-33+56i}{169} = -\frac{33}{169} + \frac{56}{169}i$

Real part $= -\frac{33}{169}$ and imaginary part$= \frac{56}{169}$

EXAMPLE 2.4 Simplify $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$

$\frac{1+i}{1-i} = \left[\frac{1-1}{1^2+(-1)^2}\right] + i\left[\frac{1+1}{1^2+(-1)^2}\right] == \frac{2i}{2} = i$

$\frac{1-i}{1+i} = \left(\frac{1+i}{1-i}\right)^{-1} = -i$

$\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3 = (i)^3 - (-i)^3 = -i - i = -2i$

EXAMPLE 2.5 If $\frac{z+3}{z-5i} = \frac{1+4i}{2}$, find the complex number z.

$\frac{z+3}{z-5i} = \frac{1+4i}{2}$

$\Rightarrow 2(z+3) = (1+4i)(z-5i)$

$\Rightarrow 2z+6 = z-5i+4zi-20i^2$

$\Rightarrow 2z-z-4zi = -5i+20-6$

$\Rightarrow z-4zi = 14-5i$

$\Rightarrow z(1-4i) = 14-5i$

$\Rightarrow z = \frac{14-5i}{1-4i}$

$$\boxed{\frac{z_1}{z_2} = \frac{a+ib}{c+id} = \left[\frac{ac+bd}{c^2+d^2}\right] + i\left[\frac{bc-ad}{c^2+d^2}\right]}$$

$= \left[\frac{14+20}{1^2+(-4)^2}\right] + i\left[\frac{-5+56}{1^2+(-4)^2}\right] = \frac{34}{17} + i\frac{51}{17} = 2+3i$

EXAMPLE 2.6 If $z_1 = 3-2i$ and $z_2 = 6+4i$, find $\frac{z_1}{z_2}$.

$\frac{z_1}{z_2} = \frac{3-2i}{6+4i} = \frac{3-2i}{6+4i} \times \frac{6-4i}{6-4i} = \frac{18-12i-12i-8}{6^2+4^2} = \frac{10-24i}{52} = \frac{2(5-12i)}{52} = \frac{5-12i}{26}$

EXAMPLE 2.7 Find z^{-1}, if $z = (2+3i)(1-i)$.

$z = (2+3i)(1-i) = 2-2i+3i-3i^2 = 2+i+3 = 5+i$

$z^{-1} = \left(\frac{x}{x^2+y^2}\right) + \boldsymbol{i}\left(-\frac{y}{x^2+y^2}\right) = \left(\frac{5}{5^2+1^2}\right) + \mathrm{i}\left(-\frac{1}{5^2+1^2}\right) = \frac{5}{26} - \frac{1}{26}\mathrm{i}$

MODULUS OF A COMPLEX NUMBER

- If $z = \mathrm{x} + \mathrm{iy}$, then the modulus of z is $|z| = \sqrt{x^2+y^2}$
- $z\bar{z} = |z|^2$
- **PROPERTIES OF MODULUS OF A COMPLEX NUMBER:**
  - ✓ $|Z| = |\bar{Z}|$
  - ✓ $|z_1+z_2| \le |z_1| + |z_2|$ (Triangle inequality)
  - ✓ $|z_1-z_2| \ge |z_1| - |z_2|$
  - ✓ $|z_1 z_2| = |z_1||z_2|$
  - ✓ $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, z_2 \neq 0$
  - ✓ $|z^n| = |z|^n$, n is an integer.
  - ✓ $Re(z) \le |z|$
  - ✓ $Im(z) \le |z|$
  - ✓ $\big||z_1| - |z_2|\big| \le |z_1+z_2| \le |z_1| + |z_2|$
- The distance between the two points z_1 and z_2 in the complex plane is $|z_1-z_2|$ (or) $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$
- Formula for finding the square root of the complex number $z = a+ib$ is

$$\sqrt{a+ib} = \pm\left(\sqrt{\frac{|z|+a}{2}} + i\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right),$$

- If b is negative $\frac{b}{|b|} = -1$, x and y have different signs.
- If b is positive $\frac{b}{|b|} = 1$, x and y have same signs.

Exercise 2.5

1.Find the modulus of the following complex numbers:

$(i)\frac{2i}{3+4i} \quad (ii)\frac{2-i}{1+i}+\frac{1-2i}{1-i} \quad (iii)(1-i)^{10} \quad (iv)2i(3-4i)(4-3i)$

$$\boxed{|z| = \sqrt{x^2+y^2}}$$

$(i)\left|\frac{2i}{3+4i}\right| = \frac{|2i|}{|3+4i|} = \frac{\sqrt{0^2+2^2}}{\sqrt{3^2+4^2}} = \frac{\sqrt{4}}{\sqrt{25}} = \frac{2}{5}$

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$(ii)\left|\frac{2-i}{1+i}+\frac{1-2i}{1-i}\right| = \left|\frac{(1-i)(2-i)+(1+i)(1-2i)}{(1+i)(1-i)}\right| = \left|\frac{2-i-2i+i^2+1-2i+i-2i^2}{1^2+1^2}\right|$

$= \left|\frac{2-i-2i-1+1-2i+2}{2}\right| = \left|\frac{4-4i}{2}\right| = |2-2i| = \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2}$

$(iii)|(1-i)^{10}| = |1-i|^{10} = \left(\sqrt{1^2+1^2}\right)^{10} = \sqrt{2}^{10} = 2^5 = 32$

$(iv)|2i(3-4i)(4-3i)| = |2i||3-4i||4-3i| = \sqrt{0^2+2^2}\sqrt{3^2+4^2}\sqrt{4^2+3^2}$

$= \sqrt{4}\sqrt{25}\sqrt{25} = 2.5.5 = 50$

EXAMPLE 2.9 If $z_1 = 3+4i$, $z_2 = 5-12i$ and $z_3 = 6+8i$, find $|z_1|, |z_2|, |z_3|, |z_1+z_2|, |z_2-z_3|$ and $|z_1+z_3|$.

EXAMPLE 2.10 Find the following : $(i)\left|\frac{2+i}{-1+2i}\right| \quad (ii)\left|\overline{(1+i)}(2+3i)(4i-3)\right|$

$(iii)\left|\frac{i(2+i)^3}{(1+i)^2}\right|$

2.For any two complex numbers z_1 and z_2, such that $|z_1| = |z_2| = 1$ and $z_1 z_2 \neq -1$ then show that $\frac{z_1+z_2}{1+z_1 z_2}$ is a real number.

Given $|z_1| = |z_2| = 1$

$|z_1| = 1 \Longrightarrow |z_1|^2 = 1 \Longrightarrow z_1\bar{z_1} = 1 \Longrightarrow \bar{z_1} = \frac{1}{z_1}$

$lll^{ly}\ \bar{z_2} = \frac{1}{z_2}$

Let $w = \frac{z_1+z_2}{1+z_1 z_2} \Longrightarrow \bar{w} = \overline{\left(\frac{z_1+z_2}{1+z_1 z_2}\right)} = \frac{\bar{z_1}+\bar{z_2}}{\bar{1}+\bar{z_1}\bar{z_2}} = \frac{\bar{z_1}+\bar{z_2}}{1+\bar{z_1}\bar{z_2}} = \frac{\frac{1}{z_1}+\frac{1}{z_2}}{1+\frac{1}{z_1}\frac{1}{z_2}} = \frac{\frac{z_1+z_2}{z_1 z_2}}{\frac{1+z_1 z_2}{z_1 z_2}} = \frac{z_1+z_2}{1+z_1 z_2} = w$

Since $\bar{w} = w$, w is purely real.

3.Which one of the points $10-8i, 11+6i$ is closest to $1+i$.

Distance between the two points z_1 and $z_2 = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

$z = 1+i, z_1 = 10-8i$

Distance between the two points z and $z_1 = \sqrt{(1-10)^2+(1+8)^2}$

$= \sqrt{81+81} = \sqrt{162} = \sqrt{2\times 81} = 9\sqrt{2}$

$z = 1+i, z_2 = 11+6i$

Distance between the two points z and $z_2 = \sqrt{(1-11)^2+(1-6)^2}$

$= \sqrt{100+125} = \sqrt{125} = \sqrt{5\times 25} = 5\sqrt{5}$

$\because 5\sqrt{5} < 9\sqrt{2}, 11+6i$ is the closest point to $1+i$

EXAMPLE 2.11 Which one of the points $i, -2+i$ *and* 3 is farthest from the origin?

4.If $|z| = 3$, show that $7 \leq |z+6-8i| \leq 13$.

Let $z_1 = z$ *and* $z_2 = 6-8i$

W.K.T, $\big||z_1|-|z_2|\big| \leq |z_1+z_2| \leq |z_1|+|z_2|$

$\Longrightarrow \left|3-\sqrt{6^2+8^2}\right| \leq |z+6-8i| \leq 3+\sqrt{6^2+8^2}$

$\Longrightarrow \left|3-\sqrt{100}\right| \leq |z+6-8i| \leq 3+\sqrt{100}$

$\Longrightarrow |3-10| \leq |z+6-8i| \leq 3+10$

$\Longrightarrow |-7| \leq |z+6-8i| \leq 13$

$\Longrightarrow 7 \leq |z+6-8i| \leq 13$

EXAMPLE 2.13 If $|z| = 2$, show that $3 \leq |z+3+4i| \leq 7$.

5. If $|z| = 1$, show that $2 \leq |z^2-3| \leq 4$.

Let $z_1 = z^2$ *and* $z_2 = -3$

W.K.T, $\big||z_1|-|z_2|\big| \leq |z_1+z_2| \leq |z_1|+|z_2|$

$\Longrightarrow \big||z|^2-|-3|\big| \leq |z^2-3| \leq |z|^2+|-3|$

$\Longrightarrow |1^2-3| \leq |z^2-3| \leq 1^2+3$

$\Longrightarrow |-2| \leq |z^2-3| \leq 4$

$\Longrightarrow 2 \leq |z^2-3| \leq 4$

6.If $\left|z-\frac{2}{z}\right| = 2$, show that the greatest and least value of $|z|$ are $\sqrt{3}+1$ and $\sqrt{3}-1$ respectively.

$\left|z-\frac{2}{z}\right| = 2 \Longrightarrow 2 = \left|z-\frac{2}{z}\right| \geq \left||z|-\left|\frac{2}{z}\right|\right|$

$\Rightarrow \left||z| - \left|\frac{2}{z}\right|\right| \le 2 \Rightarrow -2 \le |z| - \left|\frac{2}{z}\right| \le 2$ $\boxed{\because |x-a| \le r \Rightarrow -r \le x-a \le r}$

$\Rightarrow -2 \le |z| - \frac{2}{|z|} \le 2 \Rightarrow -2 \le \frac{|z|^2-2}{|z|} \le 2 \Rightarrow -2|z| \le |z|^2 - 2 \le 2|z| \rightarrow (1)$

From $(1) \Rightarrow -2|z| \le |z|^2 - 2$

$\Rightarrow 2 \le |z|^2 + 2|z|$

$\Rightarrow 3 \le |z|^2 + 2|z| + 1$ (Adding 1 on both sides)

$\Rightarrow 3 \le (|z|+1)^2$

$\Rightarrow \sqrt{3} \le |z| + 1$ (Taking square root on both sides)

$\Rightarrow \boxed{(\sqrt{3}-1) \le |z| \rightarrow (2)}$

From $(1) \Rightarrow |z|^2 - 2 \le 2|z|$

$\Rightarrow |z|^2 - 2|z| \le 2$

$\Rightarrow |z|^2 - 2|z| + 1 \le 3$ (Adding 1 on both sides)

$\Rightarrow (|z|-1)^2 \le 3$

$\Rightarrow |z| - 1 \le \sqrt{3}$ (Taking square root on both sides)

$\Rightarrow \boxed{|z| \le (\sqrt{3}+1) \rightarrow (3)}$

From (2) & (3), $\sqrt{3}-1 \le |z| \le \sqrt{3}+1$

Hence the greatest and least value of $|z|$ are $\sqrt{3}+1$ and $\sqrt{3}-1$ respectively.

7.If z_1, z_2 *and* z_3 are three complex numbers such that $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$.

$|z_1| = 1 \Rightarrow |z_1|^2 = 1 \Rightarrow z_1\bar{z_1} = 1 \Rightarrow z_1 = \frac{1}{\bar{z_1}}$ $\boxed{\because z\bar{z} = |z|^2}$

$|z_2| = 2 \Rightarrow |z_2|^2 = 4 \Rightarrow z_2\bar{z_2} = 4 \Rightarrow z_2 = \frac{4}{\bar{z_2}}$

$|z_3| = 3 \Rightarrow |z_3|^2 = 9 \Rightarrow z_3\bar{z_3} = 9 \Rightarrow z_3 = \frac{9}{\bar{z_3}}$

$z_1 + z_2 + z_3 = \frac{1}{\bar{z_1}} + \frac{4}{\bar{z_2}} + \frac{9}{\bar{z_3}}$

$\Rightarrow z_1 + z_2 + z_3 = \frac{\bar{z_2}\bar{z_3} + 4\bar{z_1}\bar{z_3} + 9\bar{z_1}\bar{z_2}}{\bar{z_1}\bar{z_2}\bar{z_3}}$

$\Rightarrow |z_1 + z_2 + z_3| = \left|\frac{\bar{z_2}\bar{z_3} + 4\bar{z_1}\bar{z_3} + 9\bar{z_1}\bar{z_2}}{\bar{z_1}\bar{z_2}\bar{z_3}}\right|$

$\Rightarrow |z_1 + z_2 + z_3| = \frac{|\bar{z_2}\bar{z_3} + 4\bar{z_1}\bar{z_3} + 9\bar{z_1}\bar{z_2}|}{|\bar{z_1}\bar{z_2}\bar{z_3}|}$

$\Rightarrow |z_1 + z_2 + z_3| = \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{|z_1z_2z_3|}$ $\because \boxed{\bar{\bar{z}} = z}$

$\Rightarrow |z_1 + z_2 + z_3| = \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{|z_1||z_2||z_3|}$ $\boxed{\because |z_1z_2| = |z_1||z_2|}$

$\Rightarrow 1 = \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{1.2.3}$

$\Rightarrow |9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$

(OR)

$|z_1| = 1 \Rightarrow |z_1|^2 = 1 \Rightarrow z_1\bar{z_1} = 1$ $\boxed{\because z\bar{z} = |z|^2}$

$|z_2| = 2 \Rightarrow |z_2|^2 = 4 \Rightarrow z_2\bar{z_2} = 4$

$|z_3| = 3 \Rightarrow |z_3|^2 = 9 \Rightarrow z_3\bar{z_3} = 9$

$|9z_1z_2 + 4z_1z_3 + z_2z_3| = |z_3\bar{z_3}z_1z_2 + z_2\bar{z_2}z_1z_3 + z_1\bar{z_1}z_2z_3|$

$= |(z_1z_2z_3)(\bar{z_1} + \bar{z_2} + \bar{z_3})|$

$= |z_1z_2z_3||\bar{z_1} + \bar{z_2} + \bar{z_3}|$

$= |z_1z_2z_3|\overline{|z_1 + z_2 + z_3|}$

$= |z_1||z_2||z_3||z_1 + z_2 + z_3|$

$= 1.2.3.1$

$= 6$

EXAMPLE 2.12 If z_1, z_2 *and* z_3 are three complex numbers such that $|z_1| = |z_2| = |z_3| = |z_1 + z_2 + z_3| = 1$, find the value of $\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right|$.

EXAMPLE 2.15 If z_1, z_2 *and* z_3 are three complex numbers such that $|z_1| = |z_2| = |z_3| = r > 0$ and $z_1 + z_2 + z_3 \neq 0$. Prove that $\left|\frac{z_1z_2 + z_2z_3 + z_3z_1}{z_1 + z_2 + z_3}\right| = r$

$|z_1| = r \Rightarrow |z_1|^2 = r^2 \Rightarrow z_1\bar{z_1} = r^2 \Rightarrow z_1 = \frac{r^2}{\bar{z_1}}$ $\boxed{\because z\bar{z} = |z|^2}$

$|z_2| = r \Rightarrow |z_2|^2 = r^2 \Rightarrow z_2\bar{z_2} = r^2 \Rightarrow z_2 = \frac{r^2}{\bar{z_2}}$

$|z_3| = r \Rightarrow |z_3|^2 = r^2 \Rightarrow z_3\bar{z_3} = r^2 \Rightarrow z_3 = \frac{r^2}{\bar{z_3}}$

$z_1 + z_2 + z_3 = \frac{r^2}{\bar{z_1}} + \frac{r^2}{\bar{z_2}} + \frac{r^2}{\bar{z_3}} = r^2\left(\frac{\bar{z_2}\bar{z_3} + \bar{z_1}\bar{z_3} + \bar{z_1}\bar{z_2}}{\bar{z_1}\bar{z_2}\bar{z_3}}\right)$

$\Rightarrow |z_1 + z_2 + z_3| = |r^2|\left|\frac{\bar{z_2}\bar{z_3} + \bar{z_1}\bar{z_3} + \bar{z_1}\bar{z_2}}{\bar{z_1}\bar{z_2}\bar{z_3}}\right| = r^2\frac{|\bar{z_2}\bar{z_3} + \bar{z_1}\bar{z_3} + \bar{z_1}\bar{z_2}|}{|\bar{z_1}\bar{z_2}\bar{z_3}|} = r^2\frac{|z_1z_2 + z_2z_3 + z_3z_1|}{|z_1||z_2||z_3|}$

$\Rightarrow |z_1 + z_2 + z_3| = r^2\frac{|z_1z_2 + z_2z_3 + z_3z_1|}{r.r.r} \Rightarrow |z_1 + z_2 + z_3| = \frac{|z_1z_2 + z_2z_3 + z_3z_1|}{r}$

$\Rightarrow \left|\frac{z_1z_2 + z_2z_3 + z_3z_1}{z_1 + z_2 + z_3}\right| = r$

8.If the area of the triangle formed by the vertices $z, iz \text{ and } z+iz$ is 50 square units, find the value of $|z|$.

Let $z=a+ib=(a,b) \Rightarrow |z|=\sqrt{a^2+b^2}$

$iz=i(a+ib)=ia+i^2b=-b+ia \Rightarrow |iz|=\sqrt{a^2+b^2}$

$z+iz=a+ib-b+ia=(a-b)+i(a+b)$

$|z+iz|=\sqrt{(a-b)^2+(a+b)^2}=\sqrt{2(a^2+b^2)}$

$|z|^2+|iz|^2=\mathbf{2}(a^2+b^2)=|z+iz|^2$

∴ The given vertices form an isosceles right triangle.

Area of the triangle = 50 sq. units

$\Rightarrow \frac{1}{2}\sqrt{a^2+b^2}.\sqrt{a^2+b^2}=50$

$\Rightarrow a^2+b^2=100$

$\Rightarrow \sqrt{a^2+b^2}=10$

$\Rightarrow |z|=10$

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EXAMPLE 2.14 Show that the points $1, -\frac{1}{2}+i\frac{\sqrt{3}}{2}$ and $-\frac{1}{2}-i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle.

Let $z_1=1, z_2=-\frac{1}{2}+i\frac{\sqrt{3}}{2}$ and $z_3=-\frac{1}{2}-i\frac{\sqrt{3}}{2}$

The length of the sides of the triangle are

$$|z_1-z_2|=\left|1-\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)\right|=\left|\frac{3}{2}-i\frac{\sqrt{3}}{2}\right|=\sqrt{\left(\frac{3}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}=\sqrt{\frac{9}{4}+\frac{3}{4}}=\sqrt{\frac{12}{4}}=\sqrt{3}$$

$$|z_2-z_3|=\left|\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)-\left(-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)\right|=\left|-\frac{1}{2}+i\frac{\sqrt{3}}{2}+\frac{1}{2}+i\frac{\sqrt{3}}{2}\right|=\sqrt{\left(\frac{2\sqrt{3}}{2}\right)^2}=\sqrt{3}$$

$$|z_3-z_1|=\left|\left(-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)-1\right|=\left|-\frac{3}{2}-i\frac{\sqrt{3}}{2}\right|=\sqrt{\left(\frac{3}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}=\sqrt{\frac{9}{4}+\frac{3}{4}}=\sqrt{3}$$

Since all the three sides are equal, the given points form an equilateral triangle.

EXAMPLE 2.18 Given the complex number $=3+2i$, represent the complex numbers z, iz and $z+iz$ in one Argand diagram. Show that these complex numbers form the vertices of an isosceles right triangle.

Given : $z=3+2i=A(3,2)$

$iz=i(3+2i)=-2+3i=B(-2,3)$

$z+iz=3+2i-2+3i=1+5i=C(1,5)$

The length of the sides of the triangle are

$AB=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}=\sqrt{(3+2)^2+(2-3)^2}=\sqrt{25+1}=\sqrt{26}$

$BC=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}=\sqrt{(-2-1)^2+(3-5)^2}=\sqrt{9+4}=\sqrt{13}$

$CA=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}=\sqrt{(1-3)^2+(5-2)^2}=\sqrt{4+9}=\sqrt{13}$

$AB^2=26, BC^2=13, CA^2=13$

Since $(i)BC=CA=\sqrt{13}$ and $(ii)BC^2+CA^2=13+13=26=AB^2$, So the given vertices form an isosceles right triangle.

9.Show that the equation $z^3+2\bar{z}=0$ has five solutions.

$z^3+2\bar{z}=0$

$\Rightarrow z^3=-2\bar{z}$

$\Rightarrow |z^3|=|-2||\bar{z}|$

$\Rightarrow |z^3|=2|z|$

$\Rightarrow |z|^3-2|z|=0$

$$\Rightarrow |z|(|z|^2-2)=0 \Rightarrow \begin{cases} |z|=0 \Rightarrow z=0 \; is\; one\; of\; the\; solution \\ |z|^2-2=0 \Rightarrow z\bar{z}-2=0 \Rightarrow \boxed{\bar{z}=\frac{2}{z}} \end{cases}$$

Put $\bar{z}=\frac{2}{z}$ in $z^3+2\bar{z}=0 \Rightarrow z^3+2\left(\frac{2}{z}\right)=0 \Rightarrow z^4+4=0$, which provides four solution.

Hence $z^3+2\bar{z}=0$ has five solutions.

EXAMPLE 2.16 Show that the equation $z^2=\bar{z}$ has four solutions.

10.Find the square root of $(i)4+3i \;\; (ii)-6+8i \;\; (iii)-5-12i$.

$$\boxed{\sqrt{a+ib}=\pm\left(\sqrt{\frac{|z|+a}{2}}+i\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right)}$$

$(i)4+3i$

$a=4, b=3, |z|=\sqrt{4^2+3^2}=\sqrt{25}=5$

$\sqrt{4+3i}=\pm\left(\sqrt{\frac{5+4}{2}}+i\frac{3}{3}\sqrt{\frac{5-4}{2}}\right)=\pm\left(\frac{3}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right)$

$(ii)-6+8i$

$a=-6, b=8, |z|=\sqrt{6^2+8^2}=\sqrt{100}=10$

$\sqrt{-6+8i}=\pm\left(\sqrt{\frac{10-6}{2}}+i\frac{8}{8}\sqrt{\frac{10+6}{2}}\right)=\pm\left(\sqrt{2}+i\sqrt{8}\right)=\pm\left(\sqrt{2}+i2\sqrt{2}\right)$

$(iii)\ -5-12i\ 4$

$a=-5, b=-12, |z|=\sqrt{5^2+12^2}=\sqrt{169}=13$

$\sqrt{-5-12i}=\pm\left(\sqrt{\frac{13-5}{2}}+i\frac{-12}{12}\sqrt{\frac{13+5}{2}}\right)=\pm(\sqrt{4}-i\sqrt{9})=\pm(2-3i)$

EXAMPLE 2.17 Find the square root of $6+8i$

GEOMETRY AND LOCUS OF COMPLEX NUMBERS

- A circle is defined as the locus of a point which moves in a plane such that its distance from a fixed point in that plane is always a constant. The fixed point is the centre and the constant distant is the radius of the circle.
- Equation of complex form of a circle is $|z-z_0|=r$.
- $|z-z_0|<r$ represents the points interior of the circle.
- $|z-z_0|>r$ represents the points exterior of the circle.
- $x^2+y^2=r^2$ represents a circle with centre at the origin and the radius r units.
- $r=\sqrt{x^2+y^2}$
- If $z=\frac{a+ib}{c+id}$ then $\mathrm{Re}(Z)=\frac{ac+bd}{c^2+d^2}$
- If $z=\frac{a+ib}{c+id}$ then $\mathrm{Im}(Z)=\frac{bc-ad}{c^2+d^2}$

EXERCISE 2.6

1.If $z=x+iy$ is a complex number such that $\left|\frac{z-4i}{z+4i}\right|=1$. Show that the locus of z is real axis.

Given: $z=x+iy$

$\left|\frac{z-4i}{z+4i}\right|=1\Rightarrow|z-4i|=|z+4i|$

$\Rightarrow|x+iy-4i|=|x+iy+4i|$

$\Rightarrow|x+i(y-4)|=|x+i(y+4)|$

$\Rightarrow\sqrt{x^2+(y-4)^2}=\sqrt{x^2+(y-4)^2}$

Squaring on both sides,

$\Rightarrow x^2+(y-4)^2=x^2+(y-4)^2$

$\Rightarrow x^2+y^2-8y+16=x^2+y^2+8y+16$

$\Rightarrow 8y=0$

$\Rightarrow y=0$

$\therefore$ The locus of z is real axis.

2. If $z=x+iy$ is a complex number such that $Im\left(\frac{2z+1}{iz+1}\right)=0$, show that the locus of z is $2x^2+2y^2+x-2y=0$.

Given: $z=x+iy$

$\frac{2z+1}{iz+1}=\frac{2(x+iy)+1}{i(x+iy)+1}=\frac{(2x+1)+i2y}{(1-y)+ix}$

$\boxed{\mathrm{Im}(Z)=\frac{bc-ad}{c^2+d^2}}$

Here $a=2x+1, b=2y, c=1-y, d=x$

$Im\left(\frac{2z+1}{iz+1}\right)=0\Rightarrow\frac{2y(1-y)-x(2x+1)}{(1-y)^2+x^2}=0\Rightarrow 2y-2y^2-2x^2-x=0$

$\Rightarrow 2x^2+2y^2+x-2y=0$

3.Obtain the Cartesian form of the locus of $z=x+iy$ in each of the following cases:

$(i)[Re(iz)]^2=3\ (ii)Im[(1-i)z+1]=0\ \ (iii)|z+i|=|z-1|\ (iv)\bar{z}=z^{-1}$

Given: $z=x+iy$

$(i)iz=i(x+iy)=y-ix$

$[Re(iz)]^2=3$

$\Rightarrow[Re(y-ix)]^2=3$

$\Rightarrow y^2=3$

$(ii)(1-i)z+1=(1-i)(x+iy)+1=x+iy-ix+y+1$

$=(x+y+1)+i(y-x)$

$Im[(1-i)z+1]=0$

$\Rightarrow Im[(x+y+1)+i(y-x)]=0$

$\Rightarrow y-x=0\ (or)x-y=0$

$(iii)\quad |z+i|=|z-1|$

$\Rightarrow|x+iy+i|=|x+iy-1|$

$\Rightarrow|x+i(y+1)|=|(x-1)+iy|$

$\Rightarrow\sqrt{(x)^2+(y+1)^2}=\sqrt{(x-1)^2+(y)^2}$

Squaring on both sides,

$\Rightarrow(x)^2+(y+1)^2=(x-1)^2+(y)^2$

$\Rightarrow x^2+y^2+2y+1=x^2-2x+1+y^2$

$\Rightarrow 2x+2y=0$

$\Rightarrow x+y=0$

$(iv)\bar{z}=z^{-1}\Rightarrow\bar{z}=\frac{1}{z}\Rightarrow z\bar{z}=1\Rightarrow(x+iy)(x-iy)=1\Rightarrow x^2+y^2=1$

4.Show that the following equations represent a circle, and find its centre and radius: $(i)|z-2-i|=3$ $(ii)|2z+2-4i|=2$ $(iii)|3z-6+12i|=8$

$\boxed{Equation\ of\ the\ circle\ is\ |z-z_0|=r}$

$(i)|z-2-i|=3 \implies |z-(2+i)|=3$

Centre $=2+i=(2,1)$; Radius $=3\ units$

$(ii)|2z+2-4i|=2 \implies 2|z+1-2i|=2 \implies |z-(-1+2i)|=1$

Centre $=-1+2i=(-1,2)$; Radius $=1\ unit$

$(iii)|3z-6+12i|=8 \implies 3|z-2+4i|=8 \implies |z-(2-4i)|=\frac{8}{3}$

Centre $=2-4i=(2,-4)$; Radius $=\frac{8}{3}\ units$

EXAMPLE 2.19 Show that $|3z-5+i|=4$ represent a circle, and find its centre and radius.

5. Obtain the Cartesian form of the locus of $z=x+iy$ in each of the following cases: $(i)|z-4|=16$ $(ii)|z-4|^2-|z-1|^2=16$

Given: $z=x+iy$

$(i)|z-4|=16 \implies |x+iy-4|=16$

$\implies |(x-4)+iy|=16$

$\implies \sqrt{(x-4)^2+y^2}=16$

Squaring on both sides,

$\implies (x-4)^2+y^2=16^2$

$\implies x^2-8x+16+y^2=256$

$\implies x^2+y^2-8x-240=0$ is the locus of z.

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$(ii)|z-4|^2-|z-1|^2=16 \implies |x+iy-4|^2-|x+iy-1|^2=16$

$\implies |(x-4)+iy|^2-|(x-1)+iy|^2=16$

$\implies \left(\sqrt{(x-4)^2+y^2}\right)^2-\sqrt{(x-1)^2+y^2}^2=16$

$\implies (x-4)^2+y^2-[(x-1)^2+y^2]=16$

$\implies x^2-8x+16+y^2-x^2+2x-1-y^2=16$

$\implies -6x-1=0$

$\implies 6x+1=0$ is the locus of z.

EXAMPLE 2.21 Obtain the Cartesian form of the locus of z in each of the following cases: $(i)|z|=|z-i|$ $(ii)|2z-3-i|=3$

EXAMPLE 2.20 Show that $|z+2-i|<2$ represents interior points of a circle. Find its centre and radius.

Consider $|z+2-i|=2 \implies |z-(-2+i)|=2$.

Centre $=-2+i=(-2,1)$; Radius $=2\ units$

$\therefore |z+2-i|<2$ represents all points inside the circle.

POLAR AND EULER FORM OF A COMPLEX NUMBER

- Polar form of $z=x+iy$ is $z=r(\cos\theta+i\sin\theta)\ (or)\ z=rcis\theta$
- $Modulus\ r=\sqrt{x^2+y^2}$
- $\alpha=tan^{-1}\left(\frac{y}{x}\right)$
- $\arg z=\theta$; $\begin{cases}\theta=\alpha\ ;if(+,+)\ \theta\ lies\ in\ the\ Ist\ quadrant\\ \theta=\pi-\alpha\ ;if(-,+)\ \theta\ lies\ in\ the\ 2nd\ quadrant\\ \theta=-\pi+\alpha\ ;if(-,-)\ \theta\ lies\ in\ the\ 3rd\ quadrant\\ \theta=-\alpha\ ;if(+,-)\ \theta\ lies\ in\ the\ 4th\ quadrant\end{cases}$
- In general $\arg z=Arg\ z+2n\pi\ , n\in z.$
- Properties of arguments:
  - ✓ $arg(z_1z_2)=arg(z_1)+arg(z_2)$
  - ✓ $arg\left(\frac{z_1}{z_2}\right)=arg(z_1)-arg(z_2)$
  - ✓ $arg(z^n)=n\ argz$
- Some of the principal argument and argument:

z	1	i	-1	$-i$
$Arg\ z$	0	$\frac{\pi}{2}$	π	$-\frac{\pi}{2}$
$argz$	$2n\pi$	$2n\pi+\frac{\pi}{2}$	$2n\pi+\pi$	$2n\pi-\frac{\pi}{2}$

- Euler's form of the complex number: $z=re^{i\theta}$; $e^{i\theta}=\cos\theta+i\sin\theta$
- Properties of polar form:
  - $z=r(\cos\theta+i\sin\theta) \implies z^{-1}=\frac{1}{r}(\cos\theta-i\sin\theta)$
  - If $z_1=r_1(\cos\theta_1+i\sin\theta_1), z_2=r_2(\cos\theta_2+i\sin\theta_2)\ then$ $z_1z_2=r_1r_2[\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2)]$
  - If $z_1=r_1(\cos\theta_1+i\sin\theta_1), z_2=r_2(\cos\theta_2+i\sin\theta_2)\ then$

$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]$

❖ Some of the useful polar forms :

$1 = cis(0)\ ;\ -1 = cis(\pi)\ ; i = cis\left(\frac{\pi}{2}\right);\ -i = cis\left(-\frac{\pi}{2}\right)$

❖ $tan^{-1}(m_1) - tan^{-1}(m_2) = \frac{m_1 - m_2}{1 + m_1 m_2}$

EXAMPLE 2.22 Find the modulus and principal argument of the following complex numbers: $(i)\sqrt{3} + i\ \ (ii) -\sqrt{3} + i\ \ (iii) -\sqrt{3} - i\ \ \ (iv)\sqrt{3} - i$

$(i)\sqrt{3} + i$

$Modulus\ r = \sqrt{x^2 + y^2} = \sqrt{\sqrt{3}^2 + 1^2} = \sqrt{4} = 2$

$\alpha = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{1}{\sqrt{3}}\right| = \frac{\pi}{6}$

Since the complex number $\sqrt{3} + i$ lies in the first quadrant, principal argument

$\theta = \alpha \Longrightarrow \theta = \frac{\pi}{6}$

$(ii) -\sqrt{3} + i$

$Modulus\ r = \sqrt{x^2 + y^2} = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = \sqrt{4} = 2$

$\alpha = tan^{-1}\left(\frac{y}{x}\right) = tan^{-1}\left|\frac{1}{-\sqrt{3}}\right| = \frac{\pi}{6}$

Since the complex number $-\sqrt{3} + i$ lies in the second quadrant, principal argument $\theta = \pi - \alpha \Longrightarrow \theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$(iii) -\sqrt{3} - i$

$Modulus\ r = \sqrt{x^2 + y^2} = \sqrt{\left(-\sqrt{3}\right)^2 + (-1)^2} = \sqrt{4} = 2$

$\alpha = tan^{-1}\left(\frac{y}{x}\right) = tan^{-1}\left|\frac{-1}{-\sqrt{3}}\right| = \frac{\pi}{6}$

Since the complex number $-\sqrt{3} - i$ lies in the third quadrant, principal argument

$\theta = -\pi + \alpha \Longrightarrow \theta = -\pi + \frac{\pi}{6} = -\frac{5\pi}{6}$

$(iv)\sqrt{3} - i$

$Modulus\ r = \sqrt{x^2 + y^2} = \sqrt{\sqrt{3}^2 + (-1)^2} = \sqrt{4} = 2$

$\alpha = tan^{-1}\left(\frac{y}{x}\right) = tan^{-1}\left|\frac{-1}{\sqrt{3}}\right| = \frac{\pi}{6}$

Since the complex number $\sqrt{3} - i$ lies in the fourth quadrant, principal argument

$\theta = -\alpha \Longrightarrow \theta = -\frac{\pi}{6}$

EXAMPLE 2.24 Find the principal argument $Arg\ z$, when $z = \frac{-2}{1 + i\sqrt{3}}$

$argz = arg\left(\frac{-2}{1 + i\sqrt{3}}\right) = arg(-2) - arg\left(1 + i\sqrt{3}\right) \rightarrow (1)$

$arg(-2) = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{0}{-2}\right| = 0$

Since the complex number -2 lies in the second quadrant, principal argument

$\theta = \pi - \alpha \Longrightarrow \theta = \pi - 0 = \pi$

$arg\left(1 + i\sqrt{3}\right) = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{\sqrt{3}}{1}\right| = \frac{\pi}{3}$

Since the complex number $1 + i\sqrt{3}$ lies in the first quadrant, principal argument

$\theta = \alpha \Longrightarrow \theta = \frac{\pi}{3}$

$(1) \Longrightarrow arg\left(\frac{-2}{1 + i\sqrt{3}}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

EXERCISE 2.7

1.Write in polar form of the following complex numbers:

$(i)2 + i2\sqrt{3}\ \ \ (ii)3 - i\sqrt{3}\ \ \ (iii) -2 - i2\ \ \ (iv)\frac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$

$(i)2 + i2\sqrt{3} = rcis(\theta) \rightarrow (1)$

$r = \sqrt{x^2 + y^2} = \sqrt{(2)^2 + \left(2\sqrt{3}\right)^2} = \sqrt{4 + 12} = \sqrt{16} = 4$

$\alpha = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{2\sqrt{3}}{2}\right| = \frac{\pi}{3}$

Principal argument lies in the first quadrant $\theta = \alpha = \frac{\pi}{3}$

$(1) \Longrightarrow 2 + i2\sqrt{3} = 4cis\left(\frac{\pi}{3}\right) = 4cis\left(2k\pi + \frac{\pi}{3}\right), k \in z$

$(ii)3 - i\sqrt{3} = rcis(\theta) \rightarrow (1)$

$r = \sqrt{x^2 + y^2} = \sqrt{(3)^2 + \left(-\sqrt{3}\right)^2} = \sqrt{9 + 3} = \sqrt{4 \times 3} = 2\sqrt{3}$

$\alpha = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{-\sqrt{3}}{3}\right| = tan^{-1}\left|\frac{1}{\sqrt{3}}\right| = \frac{\pi}{6}$

Principal argument lies in the 4th quadrant $\theta = -\alpha = -\frac{\pi}{6}$

$(1) \Longrightarrow 3 - i\sqrt{3} = 2\sqrt{3}cis\left(-\frac{\pi}{6}\right) = 2\sqrt{3}cis\left(2k\pi - \frac{\pi}{6}\right), k \in z$

$(iii) -2 - i2 = rcis(\theta) \rightarrow (1)$

$r = \sqrt{x^2 + y^2} = \sqrt{(-2)^2 + (-2)^2} = \sqrt{4+4} = \sqrt{2 \times 4} = 2\sqrt{2}$

$\alpha = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|-\frac{-2}{2}\right| = \frac{\pi}{4}$

Principal argument lies in the 3rd quadrant $\theta = -\pi + \alpha = -\pi + \frac{\pi}{4} = -\frac{3\pi}{4}$

$(1) \Longrightarrow -2 - i2 = 2\sqrt{2}cis\left(-\frac{3\pi}{4}\right) = 2\sqrt{2}cis\left(2k\pi - \frac{3\pi}{4}\right), k \in z$

$(iv) \frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$

$i - 1 = -1 + i = rcis(\theta) \rightarrow (1)$

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$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$

$\alpha = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{1}{-1}\right| = \frac{\pi}{4}$

Principal argument lies in the 2nd quadrant $\theta = \pi - \alpha \Longrightarrow \theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$(1) \Longrightarrow i - 1 = \sqrt{2}cis\left(\frac{3\pi}{4}\right)$

$\frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}} = \frac{\sqrt{2}cis\left(\frac{3\pi}{4}\right)}{cis\left(\frac{\pi}{3}\right)} = \sqrt{2}cis\left(\frac{3\pi}{4} - \frac{\pi}{3}\right) = \sqrt{2}cis\left(\frac{5\pi}{12}\right) = \sqrt{2}cis\left(2k\pi + \frac{5\pi}{12}\right), k \in z$

EXAMPLE 2.23 Represent the complex number $(i) -1 - i \;\; (ii) 1 + i\sqrt{3}$ in polar form.

2.Find the rectangular form of the complex numbers:

$(i)\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) \quad (ii) \frac{\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}}{2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)}$

$(i)\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) = \text{cis}\left(\frac{\pi}{6}\right) + \text{cis}\left(\frac{\pi}{12}\right)$

$= \text{cis}\left(\frac{\pi}{6} + \frac{\pi}{12}\right)$

$= \text{cis}\left(\frac{3\pi}{12}\right) = cis\left(\frac{\pi}{4}\right)$

$= \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$

$= \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$

$(ii) \frac{\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}}{2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)}$

$= \frac{1}{2}cis\left(-\frac{\pi}{6} - \frac{\pi}{3}\right) = \frac{1}{2}cis\left(-\frac{3\pi}{6}\right) = \frac{1}{2}cis\left(-\frac{\pi}{2}\right) = \frac{1}{2}\left[\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right]$

$= \frac{1}{2}\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right)$

$= \frac{1}{2}(0 - i)$

$= -\frac{1}{2}i$

EXAMPLE 2.25 Find the product $\frac{3}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right).6\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$ in rectangular form.

EXAMPLE 2.26 Find the quotient $\frac{2\left(\cos\frac{9\pi}{4}+i\sin\frac{9\pi}{4}\right)}{4\left(\cos\left(\frac{-3\pi}{2}\right)+i\sin\left(\frac{-3\pi}{2}\right)\right)}$ in rectangular form.

3.If $(x_1 + iy_1)(x_2 + iy_2) \ldots\ldots (x_n + iy_n) = a + ib$, show that (i) $(x_1^2 + y_1^2)(x_2^2 + y_2^2) \ldots\ldots (x_n^2 + y_n^2) = a^2 + b^2$ (ii) $\sum_{r=1}^{n} tan^{-1}\left(\frac{b_r}{a_r}\right) = tan^{-1}\left(\frac{b}{a}\right) + 2k\pi, k\epsilon Z.$

$(i)(x_1 + iy_1)(x_2 + iy_2) \ldots\ldots (x_n + iy_n) = a + ib$

Taking modulus on both sides,

$\Longrightarrow |(x_1 + iy_1)(x_2 + iy_2) \ldots\ldots (x_n + iy_n)| = |a + ib|$

$\Longrightarrow |x_1 + iy_1||x_2 + iy_2| \ldots\ldots |x_n + iy_n| = |a + ib|$

$\Longrightarrow \sqrt{\left(x_1^2 + y_1^2\right)}\sqrt{\left(x_2^2 + y_2^2\right)} \ldots\ldots \sqrt{\left(x_n^2 + y_n^2\right)} = \sqrt{a^2 + b^2}$

Squaring on both sides,

$\Longrightarrow (x_1^2 + y_1^2)(x_2^2 + y_2^2) \ldots\ldots (x_n^2 + y_n^2) = a^2 + b^2$

$(i)(x_1 + iy_1)(x_2 + iy_2) \ldots\ldots (x_n + iy_n) = a + ib$

Taking argument on both sides,

$\Longrightarrow arg[(x_1 + iy_1)(x_2 + iy_2) \ldots\ldots (x_n + iy_n)] = arg(a + ib)$

$\Longrightarrow tan^{-1}\left(\frac{y_1}{x_1}\right) + tan^{-1}\left(\frac{y_2}{x_2}\right) + \cdots + tan^{-1}\left(\frac{y_n}{x_n}\right) = tan^{-1}\left(\frac{b}{a}\right)$

$\Longrightarrow \sum_{r=1}^{n} tan^{-1}\left(\frac{b_r}{a_r}\right) = tan^{-1}\left(\frac{b}{a}\right) + 2k\pi, k\epsilon Z$

4.If $\frac{1+z}{1-z} = \cos 2\theta + i\sin 2\theta$, show that $z = i\tan\theta$.

$\frac{1+z}{1-z} = \cos 2\theta + i\sin 2\theta$

Add 1 on both sides, $\frac{1+z}{1-z} + 1 = (1 + \cos 2\theta) + i\sin 2\theta$

$\Longrightarrow \frac{1+z+1-z}{1-z} = 2cos^2\theta + i2\sin\theta\cos\theta$

$\Longrightarrow \frac{2}{1-z} = 2\cos\theta\,[\cos\theta + i\sin\theta]$

$\div by\ 2 \Longrightarrow \frac{1}{1-z} = \cos\theta\,[\cos\theta + i\sin\theta]$

$\Longrightarrow \frac{1}{\cos\theta[\cos\theta+i\sin\theta]} = 1 - z$

$\Longrightarrow z = 1 - \frac{1}{\cos\theta[\cos\theta+i\sin\theta]}$

$\Longrightarrow z = 1 - \frac{1}{\cos\theta[\cos\theta+i\sin\theta]} \times \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta}$

$\Longrightarrow z = 1 - \frac{\cos\theta - i\sin\theta}{\cos\theta(cos^2\theta + sin^2\theta)}$

$\Longrightarrow z = 1 - \left[\frac{\cos\theta}{\cos\theta} - i\frac{\sin\theta}{\cos\theta}\right]$

$\Longrightarrow z = 1 - 1 + i\tan\theta$

$\Longrightarrow z = i\tan\theta$

5.If $cos\alpha + cos\beta + cos\gamma = sin\alpha + sin\beta + sin\gamma = 0$, then show that

(i) $cos3\alpha + cos3\beta + cos3\gamma = 3\cos(\alpha + \beta + \gamma)$

(ii)$sin3\alpha + sin3\beta + sin3\gamma = 3\sin(\alpha + \beta + \gamma)$

Let $a = cis\alpha, b = cis\beta, c = cis\gamma$

W.K.T, If $a + b + c = 0\ then\ \ a^3 + b^3 + c^3 = 3abc$

$a + b + c = (cos\alpha + cos\beta + cos\gamma) + i(sin\alpha + sin\beta + sin\gamma) = 0$

$a^3 + b^3 + c^3 = 3abc$

$\Longrightarrow (cis\alpha)^3 + (cis\beta)^3 + (cis\gamma)^3 = 3cis\alpha.cis\beta.cis\gamma$

$\Longrightarrow cis(3\alpha) + cis(3\beta) + cis(3\gamma) = 3cis(\alpha + \beta + \gamma)$

$\Longrightarrow (cos3\alpha + cos3\beta + cos3\gamma) + i(sin3\alpha + sin3\beta + sin3\gamma) = 3(\cos(\alpha + \beta + \gamma)) + i3(\sin(\alpha + \beta + \gamma))$

Equating the real and imaginary parts on both sides,

$cos3\alpha + cos3\beta + cos3\gamma = 3\cos(\alpha + \beta + \gamma)$

$sin3\alpha + sin3\beta + sin3\gamma = 3\sin(\alpha + \beta + \gamma)$

6.If $z = x + iy$ and $arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$, then show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

Given: $Z = x + iy$

$arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$

$\Rightarrow arg(z - i) - arg(z + 2) = \frac{\pi}{4}$

$\Rightarrow arg(x + iy - i) - arg(x + iy + 2) = \frac{\pi}{4}$

$\Rightarrow arg\big(x + i(y-1)\big) - arg(x + 2 + iy) = \frac{\pi}{4}$

$\Rightarrow tan^{-1}\frac{y-1}{x} - tan^{-1}\frac{y}{x+2} = \frac{\pi}{4}$

$\Rightarrow tan^{-1}\frac{\frac{y-1}{x} - \frac{y}{x+2}}{1 + (\frac{y-1}{x} \times \frac{y}{x+2})} = \frac{\pi}{4}$

$\Rightarrow \frac{\big((y-1)\big)(x+2) - xy}{x(x+2)} = \tan\frac{\pi}{4}$

$\Longrightarrow \frac{xy + 2y - x - 2 - xy}{\frac{x^2 + 2x + y^2 - y}{(x+2)}} = 1$

$\Longrightarrow \frac{-x + 2y - 2}{x^2 + y^2 + 2x - y} = 1$

$\Longrightarrow x^2 + y^2 + 2x - y + x - 2y + 2 = 0$

$\Longrightarrow x^2 + y^2 + 3x - 3y + 2 = 0$

EXAMPLE 2.27 If $z = x + iy$ and $arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, then show that $x^2 + y^2 = 1$.

De MOIVRE'S THEOREM AND ITS APPLICATIONS

- De Moivre's theorem: Given any complex number $\cos\theta + i\sin\theta$ and any integer n, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$
- Some results:
  - ✓ $(\cos\theta - i\sin\theta)^n = \cos n\theta - i\sin n\theta$
  - ✓ $(\cos\theta + i\sin\theta)^{-n} = \cos n\theta - i\sin n\theta$
  - ✓ $(\cos\theta - i\sin\theta)^{-n} = \cos n\theta + i\sin n\theta$
  - ✓ $\sin\theta + i\cos\theta = i(\cos\theta - i\sin\theta)$
- Formula for finding n^{th} root of a complex number:
  $z^{1/n} = r^{1/n}cis\left(\frac{\theta+2k\pi}{n}\right), k = 0,1,\ldots,(n-1)$
- $\omega^3 = 1 \Longrightarrow 1 + \omega + \omega^2 = 0$
- The sum of all the nth roots of unity is $0.(ie., 1 + \omega + \omega^2 + \cdots . + \omega^{n-1} = 0)$

- The product of all the nth roots of unity is $(-1)^{n-1}$. $(ie., 1+\omega+\omega^2+\cdots.+\omega^{n-1}=(-1)^{n-1})$
- All the n roots of nth roots of unity are in geometrical progression.
- All the n roots of nth roots of unity lie on the circumference of a circle whose centre is at the origin and radius equal to 1 and these roots divide the circle into n equal parts and form a polygon of n sides.

EXERCISE 2.8

1.If $\omega \neq 1$ is a cube root of unity, then show that $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}+\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}=-1$.

$$\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}+\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}$$
$$=\left(\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}\times\frac{\omega}{\omega}\right)+\left(\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}\times\frac{\omega^2}{\omega^2}\right)$$
$$=\frac{\omega(a+b\omega+c\omega^2)}{a\omega^3+c\omega^2+b\omega}+\frac{\omega^2(a+b\omega+c\omega^2)}{a\omega^3+b\omega^4+c\omega^2}$$
$$=\frac{\omega(a+b\omega+c\omega^2)}{a+b\omega+c\omega^2}+\frac{\omega^2(a+b\omega+c\omega^2)}{a+b\omega+c\omega^2}$$
$$=\omega+\omega^2$$
$$=-1$$

$\omega^3=1 \;;\; 1+\omega+\omega^2=0$

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2.Show that $\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5=-\sqrt{3}$

$\frac{\sqrt{3}}{2}+\frac{i}{2}=rcis(\theta)\rightarrow(1)$

$r=\sqrt{x^2+y^2}=\sqrt{\left(\frac{\sqrt{3}}{2}\right)^2+\left(\frac{1}{2}\right)^2}=\sqrt{\frac{3}{4}+\frac{1}{4}}=\sqrt{1}=1$

$\alpha=tan^{-1}\left|\frac{y}{x}\right|=tan^{-1}\left|\frac{1/2}{\sqrt{3}/2}\right|=\frac{\pi}{6}$

Principal argument lies in the first quadrant $\theta=\alpha=\frac{\pi}{6}$

$(1)\Longrightarrow\frac{\sqrt{3}}{2}+\frac{i}{2}=cis\left(\frac{\pi}{6}\right)$

$\Longrightarrow\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5=\left[cis\left(\frac{\pi}{6}\right)\right]^5=cis\left(\frac{5\pi}{6}\right)=\cos\left(\frac{5\pi}{6}\right)+i\sin\left(\frac{5\pi}{6}\right)$

Similarly, $\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5=\cos\left(\frac{5\pi}{6}\right)-i\sin\left(\frac{5\pi}{6}\right)$

$\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5=\cos\left(\frac{5\pi}{6}\right)+i\sin\left(\frac{5\pi}{6}\right)+\cos\left(\frac{5\pi}{6}\right)-i\sin\left(\frac{5\pi}{6}\right)$

$=2\cos\left(\frac{5\pi}{6}\right)=2\cos\left(\frac{5\times180}{6}\right)=2\cos150=-2\cos30=-2\times\frac{\sqrt{3}}{2}=-\sqrt{3}$

EXAMPLE 2.31 Simplify $(i)(1+i)^{18}\quad(ii)\left(-\sqrt{3}+3i\right)^{31}$

3.Find the value of $\left(\frac{1+\sin\frac{\pi}{10}+i\cos\frac{\pi}{10}}{1+\sin\frac{\pi}{10}-i\cos\frac{\pi}{10}}\right)^{10}$

Let $z=\sin\frac{\pi}{10}+i\cos\frac{\pi}{10}\Longrightarrow\frac{1}{z}=\sin\frac{\pi}{10}-i\cos\frac{\pi}{10}$

$\left(\frac{1+\sin\frac{\pi}{10}+i\cos\frac{\pi}{10}}{1+\sin\frac{\pi}{10}-i\cos\frac{\pi}{10}}\right)^{10}=\left(\frac{1+z}{1+\frac{1}{z}}\right)^{10}=\left(\frac{1+z}{1+z}\times z\right)^{10}=z^{10}$

$z^{10}=\left(\sin\frac{\pi}{10}+i\cos\frac{\pi}{10}\right)^{10}=\left[i\left(\cos\frac{\pi}{10}-i\sin\frac{\pi}{10}\right)\right]^{10}$
$=i^{10}\left[\cos\left(10\times\frac{\pi}{10}\right)-i\sin\left(10\times\frac{\pi}{10}\right)\right]$
$=-1[\cos\pi-i\sin\pi]$
$=-1(-1-0)$
$=1$

EXAMPLE 2.29 Simplify $\left(\sin\frac{\pi}{6}+i\cos\frac{\pi}{6}\right)^{18}$

EXAMPLE 2.30 Simplify $\left(\frac{1+\cos2\theta+i\sin2\theta}{1+\cos2\theta-i\sin2\theta}\right)^{30}$

4.If $2\cos\alpha=x+\frac{1}{x}$ and $2\cos\beta=y+\frac{1}{y}$, show that

$(i)\frac{x}{y}+\frac{y}{x}=2\cos(\alpha-\beta)\quad(ii)xy-\frac{1}{xy}=2i\sin(\alpha+\beta)$

$(iii)\frac{x^m}{y^n}-\frac{y}{x^m}^n=2i\sin(m\alpha-n\beta)\quad(iv)x^my^n+\frac{1}{x^my^n}=2\cos(m\alpha+n\beta)$

$x+\frac{1}{x}=2\cos\alpha\Longrightarrow x=x=\cos\alpha+i\sin\alpha$

$y+\frac{1}{y}=2\cos\beta\Longrightarrow y=\cos\beta+i\sin\beta$

$(i)\frac{x}{y}=\frac{\cos\alpha+i\sin\alpha}{\cos\beta+i\sin\beta}=\cos(\alpha-\beta)+i\sin(\alpha-\beta)$
$\frac{y}{x}=\left(\frac{x}{y}\right)^{-1}=\cos(\alpha-\beta)-i\sin(\alpha-\beta)$
$\frac{x}{y}+\frac{y}{x}=2\cos(\alpha-\beta)$

$(ii)\;xy=(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta)=\cos(\alpha+\beta)+i\sin(\alpha+\beta)$
$\frac{1}{xy}=(xy)^{-1}=\cos(\alpha+\beta)-i\sin(\alpha+\beta)$

$$xy - \frac{1}{xy} = 2i\sin(\alpha+\beta)$$

$(iii)\ x^m = (\cos\alpha + i\sin\alpha)^m = \cos m\alpha + i\sin m\alpha$

$$y^n = (\cos\beta + i\sin\beta)^n = \cos n\beta + i\sin n\beta$$

$$\frac{x^m}{y^n} = \frac{\cos m\alpha + i\sin m\alpha}{\cos n\beta + i\sin n\beta} = \cos(m\alpha - n\beta) + i\sin(m\alpha - n\beta)$$

$$\frac{y}{x^m}^n = \left(\frac{x^m}{y^n}\right)^{-1} = \cos(m\alpha - n\beta) - i\sin(m\alpha - n\beta)$$

$$\frac{x^m}{y^n} - \frac{y}{x^m}^n = 2i\sin(m\alpha - n\beta)$$

$(iv) x^m = (\cos\alpha + i\sin\alpha)^m = \cos m\alpha + i\sin m\alpha$

$$y^n = (\cos\beta + i\sin\beta)^n = \cos n\beta + i\sin n\beta$$

$$x^m y^n = (\cos m\alpha + i\sin m\alpha)(\cos n\beta + i\sin n\beta)$$

$$= \cos(m\alpha + n\beta) + i\sin(m\alpha + n\beta)$$

$$\frac{1}{x^m y^n} = \cos(m\alpha + n\beta) - i\sin(m\alpha + n\beta)$$

$$x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\alpha + n\beta)$$

EXAMPLE 2.28 If $z = \cos\theta + i\sin\theta$, show that $z^n + \frac{1}{z^n} = 2\cos n\theta$ and $z^n - \frac{1}{z^n} = 2\sin n\theta$.

5.Solve the equation $z^3 + 27 = 0$

$z^3 + 27 = 0 \Rightarrow z^3 = -27 \Rightarrow z = (-27)^{1/3} = (27 \times -1)^{1/3}$

$z = 3(-1)^{1/3} = 3[cis(\pi)]^{1/3} = 3[cis(2k\pi + \pi)]^{1/3}, k = 0,1,2$

$= 3cis(2k+1)\frac{\pi}{3}, k = 0,1,2$

$= 3cis\left(\frac{\pi}{3}\right), 3cis(\pi), 3cis\left(\frac{5\pi}{3}\right)$

6. If $\omega \neq 1$ is a cube root of unity, show that the roots of the equation $(z-1)^3 + 8 = 0$ are $-1, 1-2\omega, 1-2\omega^2$.

$(z-1)^3 + 8 = 0$

$\Rightarrow (z-1)^3 = -8$

$\Rightarrow (z-1)^3 = (-2)^3 \Rightarrow \frac{(z-1)^3}{(-2)^3} = 1$

$\Rightarrow \left(\frac{z-1}{-2}\right)^3 = 1$

$\Rightarrow \frac{z-1}{-2} = (1)^{1/3}$

$\Rightarrow z - 1 = -2(1)^{1/3} \Rightarrow z - 1 = -2[1\ (or)\ \omega\ \ (or)\ \omega^2]$

$$\Rightarrow \begin{cases} z-1 = -2(1) \Rightarrow z - 1 = -2 \Rightarrow z = -1 \\ z-1 = -2\omega \Rightarrow z = 1 - 2\omega \\ z-1 = -2\omega^2 \Rightarrow z = 1 - 2\omega^2 \end{cases}$$

Thus the roots of the equation $(z-1)^3 + 8 = 0$ are $-1, 1-2\omega, 1-2\omega^2$.

EXAMPLE 2.34 Solve the equation $z^3 + 8i = 0$, where $z \in \mathbb{C}$.

Note: $z^3 + 8i = 0 \Rightarrow z = (-8i)^{1/3} \Rightarrow z = -2(i)^{1/3}$

find polar form for i and use de Moivre'theorem

EXAMPLE 2.35 Find all cube roots of $\sqrt{3} + i$.

Note: $\left(\sqrt{3} + i\right)^{1/3}$; *find polar form for* $\sqrt{3} + i$ *and use de Moivre'theorem*

7.Find the value of $\sum_{k=1}^{8}\left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9}\right)$

W.K.T,the sum of all the roots Is equal to zero.

$\sum_{k=0}^{8}\left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9}\right) = 0$

$\Rightarrow cis(0) + \sum_{k=1}^{8}\left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9}\right) = 0$

$\Rightarrow 1 + \sum_{k=1}^{8}\left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9}\right) = 0$

$\Rightarrow \sum_{k=1}^{8} cis\left(\frac{2k\pi}{9}\right) = -1$

$\Rightarrow \sum_{k=1}^{8}\left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9}\right) = -1$

8. If $\omega \neq 1$ is a cube root of unity, show that

$(i)(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 128$

$(ii)(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)\ldots.\left(1+\omega^{2^{11}}\right) = 1$

W.K.T, $\boxed{\omega^3 = 1\ \ ;\ \ 1+\omega+\omega^2 = 0}$

$(i)(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6$

$= (-\omega-\omega)^6 + (-\omega^2-\omega^2)^6$

$= (-2\omega)^6 + (-2\omega^2)^6 = 2^6\omega^6 + 2^6\omega^{12} = 64(\omega^3)^2 + 64(\omega^3)^4 = 64 + 64 = 128$

$(ii)(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)\ldots.\left(1+\omega^{2^{11}}\right)$

$= (1+\omega)(1+\omega^2)\left(1+\omega^{2^2}\right)\left(1+\omega^{2^3}\right)\ldots.\left(1+\omega^{2^{11}}\right)$

Clearly the above expression contains 12 terms,

$= (1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)\ldots..12\ terms$

$= [(1+\omega)(1+\omega^2)][(1+\omega)(1+\omega^2)] \ldots .6\ terms$
$= [(1+\omega)(1+\omega^2)]^6$
$= (1+\omega+\omega^2+\omega^3)^6$
$= (0+1)^6$
$= 1$

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9.If $z = 2-2i$, find the rotation of z by θ radians in the counter clockwise direction about the origin when $(i)\theta = \frac{\pi}{3}$ $(ii)\theta = \frac{2\pi}{3}$ $(iii)\theta = \frac{3\pi}{2}$

$z = 2-2i = rcis(\theta) \rightarrow (1)$

$r = \sqrt{x^2+y^2} = \sqrt{(2)^2+(-2)^2} = \sqrt{4+4} = \sqrt{4\times 2} = 2\sqrt{2}$

$\alpha = tan^{-1}\left|\frac{y}{x}\right| = tan^{-1}\left|\frac{-2}{2}\right| = \frac{\pi}{4}$

Principal argument lies in the 4th quadrant $\theta = -\alpha = -\frac{\pi}{4}$

$(1) \Longrightarrow z = 2-2i = 2\sqrt{2}cis\left(-\frac{\pi}{4}\right)$

$(i) when\ \theta = \frac{\pi}{3} \Longrightarrow z = 2\sqrt{2}cis\left(-\frac{\pi}{4}\right)cis\left(\frac{\pi}{3}\right)$
$= 2\sqrt{2}cis\left(-\frac{\pi}{4}+\frac{\pi}{3}\right)$
$= 2\sqrt{2}cis\left(\frac{-3\pi+4\pi}{12}\right)$
$= 2\sqrt{2}cis\left(\frac{\pi}{12}\right)$

$(ii) when\ \theta = \frac{2\pi}{3} \Longrightarrow z = 2\sqrt{2}cis\left(-\frac{\pi}{4}\right)cis\left(\frac{2\pi}{3}\right)$
$= 2\sqrt{2}cis\left(-\frac{\pi}{4}+\frac{2\pi}{3}\right)$
$= 2\sqrt{2}cis\left(\frac{-3\pi+8\pi}{12}\right)$
$= 2\sqrt{2}cis\left(\frac{5\pi}{12}\right)$

$(ii) when\ \theta = \frac{3\pi}{2} \Longrightarrow z = 2\sqrt{2}cis\left(-\frac{\pi}{4}\right)cis\left(\frac{3\pi}{2}\right)$
$= 2\sqrt{2}cis\left(-\frac{\pi}{4}+\frac{3\pi}{2}\right)$
$= 2\sqrt{2}cis\left(\frac{-\pi+6\pi}{4}\right)$
$= 2\sqrt{2}cis\left(\frac{5\pi}{4}\right)$

EXAMPLE 2.36 Suppose z_1, z_2 and z_3 are the vertices of an equilateral triangle inscribed in the circle $|z| = 2$. If $z_1 = 1+i\sqrt{3}$, then find z_2 and z_3.

$|z| = 2$ represents the circle with centre $(0,0)$ and radius 2.

Let $A, B\ and\ C$ be the vertices of the given triangle. Since the vertices z_1, z_2 and z_3 form an equilateral triangle inscribed in the circle $|z| = 2$, the sides of this triangle $AB, BC\ and\ CA$ subtend $\frac{2\pi}{3}$ radian at the origin.

We obtain z_2 and z_3 by the rotation of z_1 by $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$ respectively.

Given: $z_1 = 1+i\sqrt{3}$

$z_2 = z_1 cis\left(\frac{2\pi}{3}\right) = \left(1+i\sqrt{3}\right)\left[\cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right]$
$= \left(1+i\sqrt{3}\right)[\cos(180-60)+i\sin(180-60)]$
$= \left(1+i\sqrt{3}\right)(-\cos 60+i\sin 60)$
$= \left(1+i\sqrt{3}\right)\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)$
$= -2$

$z_3 = z_2 cis\left(\frac{2\pi}{3}\right) = -2\left[\cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right]$
$= -2[\cos(180-60)+i\sin(180-60)]$
$= -2(-\cos 60+i\sin 60)$
$= -2\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)$
$= 1-i\sqrt{3}$

10.Prove that the values of $\sqrt[4]{-1}$ are $\pm\frac{1}{\sqrt{2}}(1\pm i)$

Let $x = \sqrt[4]{-1}$

$= (-1)^{1/4}$

$= (cis\pi)^{\frac{1}{4}}$

$= [cis(2k\pi+\pi)]^{\frac{1}{4}}, k = 0,1,2,3$

$= cis(2k+1)\frac{\pi}{4}, k = 0,1,2,3$

$= cis\left(\frac{\pi}{4}\right), cis\left(\frac{3\pi}{4}\right), cis\left(\frac{5\pi}{4}\right), cis\left(\frac{7\pi}{4}\right)$

$= \cos 45+i\sin 45, \cos 135+i\sin 135, \cos 225+i\sin 225, \cos 315+i\sin 315$

$= \cos 45+i\sin 45, \cos(90+45)+i\sin(90+45), \cos(180+45)+i\sin(180+45), \cos(360-45)+i\sin(360-45)$

$= \cos 45 + i \sin 45 , - \cos 45 + i \sin 45 , - \cos 45 - i \sin 45 , \cos 45 - i \sin 45$

$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i , - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i , - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i , \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i$

$\therefore x = \pm \frac{1}{\sqrt{2}} (1 \pm i)$

EXAMPLE 2.32 Find the cube roots of unity.

Let $z^3 = 1 \Longrightarrow z = (1)^{1/3} = [cis(0)]^{1/3}$

$\Longrightarrow z = [cis(2k\pi + 0)]^{1/3} , k = 0,1,2$

$\Longrightarrow z = cis \left(\frac{2k\pi}{3} \right) , k = 0,1,2$

$\Longrightarrow z = cis(0) , cis \left(\frac{2\pi}{3} \right) , cis \left(\frac{4\pi}{3} \right)$

$\Longrightarrow z = cis(0) , cis \left(\pi - \frac{\pi}{3} \right) , cis \left(\pi + \frac{\pi}{3} \right)$

$\Longrightarrow z = \cos 0 + i \sin 0 , \cos \left(\pi - \frac{\pi}{3} \right) + i \sin \left(\pi - \frac{\pi}{3} \right) , \cos \left(\pi + \frac{\pi}{3} \right) + i \sin \left(\pi + \frac{\pi}{3} \right)$

$\Longrightarrow z = 1 , - \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} , - \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}$

$\Longrightarrow z = 1 , - \frac{1}{2} + i \frac{\sqrt{3}}{2} , - \frac{1}{2} + i \frac{\sqrt{3}}{2} \Longrightarrow z = 1 , \frac{-1+i\sqrt{3}}{2} , \frac{-1-i\sqrt{3}}{2}$

$\therefore$ The cube roots of unity $1, \omega, \omega^2$ are $1 , \frac{-1+i\sqrt{3}}{2} , \frac{-1-i\sqrt{3}}{2}$

EXAMPLE 2.33 Find the fourth roots of unity.

Let $z^4 = 1 \Longrightarrow z = (1)^{1/4} = [cis(0)]^{1/4}$

$\Longrightarrow z = [cis(2k\pi + 0)]^{1/4} , k = 0,1,2,3$

$\Longrightarrow z = cis \left(\frac{2k\pi}{4} \right) , k = 0,1,2,3$

$\Longrightarrow z = cis(0) , cis \left(\frac{2\pi}{4} \right) , cis \left(\frac{4\pi}{4} \right) , cis \left(\frac{6\pi}{4} \right)$

$\Longrightarrow z = cis(0) , cis \left(\frac{\pi}{2} \right) , cis(\pi) , cis \left(\frac{3\pi}{2} \right)$

$\Longrightarrow z = \cos 0 + i \sin 0 , \cos \left(\frac{\pi}{2} \right) + i \sin \left(\frac{\pi}{2} \right) , \cos(\pi) + i \sin(\pi) , \cos \left(\frac{3\pi}{2} \right) + i \sin \left(\frac{3\pi}{2} \right)$

$\Longrightarrow z = 1, i, -1, -i$

$\therefore$ The fourth roots of unity $1, \omega, \omega^2, \omega^3$ are $1, i, -1, -i$.

1.Prove that the multiplicative inverse of a non- zero complex number $z = x + iy$ is $z^{-1} = \left(\frac{x}{x^2+y^2} \right) + i \left(- \frac{y}{x^2+y^2} \right)$

PROOF:

Let $z^{-1} = u + iv$ be the inverse of $z = x + iy$

W.K.T, $zz^{-1} = 1 \Longrightarrow (x + iy)(u + iv) = 1$

$\Longrightarrow (xu - yv) + i(xv + yu) = 1 + i0$

Equating the Real and Imaginary parts on both sides,

$xu - yv = 1 \rightarrow (1) \; ; \; xv + yu = 0 \rightarrow (2)$

$\Delta = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2 ; \; \Delta_u = \begin{vmatrix} 1 & -y \\ 0 & x \end{vmatrix} = x ; \; \Delta_v = \begin{vmatrix} x & 1 \\ y & 0 \end{vmatrix} = -y$

$u = \frac{\Delta_u}{\Delta} = \frac{x}{x^2+y^2} \; ; v = \frac{\Delta_v}{\Delta} = \frac{-y}{x^2+y^2}$

$z^{-1} = u + iv \Longrightarrow z^{-1} = \left(\frac{x}{x^2+y^2} \right) + i \left(- \frac{y}{x^2+y^2} \right)$

2.For any two complex numbers z_1 and z_2 prove that $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

PROOF:

Let $z_1 = a + ib$ and $z_2 = c + id \; , a, b, c, d \in R$

$\overline{z_1 + z_2} = \overline{(a + \imath b) + (c + \imath d)}$

$= \overline{(a + c) + \imath (b + d)}$

$= (a + c) - i(b + d)$

$= (a - ib) + (c - id)$

$= \overline{z_1} + \overline{z_2}$

3.Prove that $\overline{z_1 . z_2} = \overline{z_1} . \overline{z_2}$ where $a, b, c, d \in R$.

PROOF:

Let $z_1 = a + ib$ and $z_2 = c + id \; , a, b, c, d \in R$

$\overline{z_1 . z_2} = \overline{(a + \imath b)(c + \imath d)}$

$= \overline{(ac - bd) + \imath (ad + bc)}$

$= (ac - bd) - i(ad + bc) \rightarrow (1)$

$\overline{z_1} . \overline{z_2} = \overline{a + \imath b} . \overline{c + \imath d}$

$= (a - ib)(c - id)$

$= (ac - bd) - i(ad + bc) \rightarrow (2)$

From (1) and (2), $\overline{z_1 . z_2} = \overline{z_1} . \overline{z_2}$

4.Prove that z is purely imaginary if and only if $z = -\bar{z}$

PROOF:

Let $z = x + iy \Longrightarrow \bar{z} = x - iy$

$z = -\bar{z} \Leftrightarrow x + iy = -(x - iy)$

$\Leftrightarrow x + iy = -x + iy$

$\Leftrightarrow 2x = 0$

$\Leftrightarrow x = 0$

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$\Leftrightarrow z\ is\ purely\ imaginary$

5.State and prove triangle inequality of two complex numbers (or)
For any two complex numbers z_1 and z_2 prove that $|z_1 + z_2| \leq |z_1| + |z_2|$

PROOF:

$$\begin{aligned}
|z_1 + z_2|^2 &= (z_1 + z_2)\overline{(z_1 + z_2)} && (\because z\bar{z} = |z|^2)\\
&= (z_1 + z_2)(\bar{z_1} + \bar{z_2}) && (\because \overline{z_1 + z_2} = \bar{z_1} + \bar{z_2})\\
&= z_1\bar{z_1} + (z_1\bar{z_2} + \bar{z_1}z_2) + z_2\bar{z_2}\\
&= z_1\bar{z_1} + (z_1\bar{z_2} + \overline{z_1\bar{z_2}}) + z_2\bar{z_2} && (\because \bar{\bar{z}} = z)\\
&= |z_1|^2 + 2Re(z_1\bar{z_2}) + |z_2|^2 && (\because z + \bar{z} = 2Re(z))\\
&\leq |z_1|^2 + 2|z_1\bar{z_2}| + |z_2|^2 && (\because Re(z) \leq |z|)\\
&= |z_1|^2 + 2|z_1||z_2| + |z_2|^2 && (\because \overline{z_1.z_2} = \bar{z_1}.\bar{z_2}\ \&|z| = |\bar{z}|)\\
&= (|z_1| + |z_2|)^2
\end{aligned}$$

Taking square root on both sides,

$|z_1 + z_2|^2 \leq |z_1| + |z_2|$

6.For any two complex numbers z_1 and z_2 prove that $|z_1z_2| = |z_1||z_2|$

PROOF:

$$\begin{aligned}
|z_1z_2|^2 &= (z_1z_2)\overline{(z_1z_2)} && (\because z\bar{z} = |z|^2)\\
&= (z_1)(z_2)(\bar{z_1})(\bar{z_2}) && (\because \overline{z_1.z_2} = \bar{z_1}.\bar{z_2})\\
&= (z_1\bar{z_1})(z_2\bar{z_2})\\
&= |z_1|^2|z_2|^2
\end{aligned}$$

Taking square root on both sides,

$|z_1z_2| = |z_1||z_2|$

7.If $z = r(\cos\theta + i\sin\theta)$ then prove that $z^{-1} = \frac{1}{r}(\cos\theta - i\sin\theta)$

PROOF:

$z = r(\cos\theta + i\sin\theta)$

$$\begin{aligned}
z^{-1} = \frac{1}{z} &= \frac{1}{r(\cos\theta + i\sin\theta)}\\
&= \frac{1}{r}\frac{1}{\cos\theta + i\sin\theta} \times \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta}\\
&= \frac{1}{r}.\frac{\cos\theta - i\sin\theta}{cos^2\theta + sin^2\theta}\\
&= \frac{1}{r}(\cos\theta - i\sin\theta)
\end{aligned}$$

8. If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1), z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ then
$z_1z_2 = r_1r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$

PROOF:

$$\begin{aligned}
z_1z_2 &= [r_1(\cos\theta_1 + i\sin\theta_1)][r_2(\cos\theta_2 + i\sin\theta_2)]\\
&= r_1r_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)]\\
&= r_1r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]
\end{aligned}$$

9. If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1), z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ then
$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]$

PROOF:

$$\begin{aligned}
\frac{z_1}{z_2} &= \frac{r_1(\cos\theta_1 + i\sin\theta_1)}{r_2(\cos\theta_2 + i\sin\theta_2)}\\
&= \frac{r_1}{r_2}.\frac{\cos\theta_1 + i\sin\theta_1}{\cos\theta_2 + i\sin\theta_2} \times \frac{\cos\theta_2 - i\sin\theta_2}{\cos\theta_2 - i\sin\theta_2}\\
&= \frac{r_1}{r_2}.\frac{(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)}{\cos^2(\theta_2) + \sin^2(\theta_2)}\\
&= \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]
\end{aligned}$$

3. THEORY OF EQUATIONS

BASICS OF POLYNOMIAL EQUATIONS

- $a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$ is called a **polynomial equation.**
- A polynomial with the leading coefficient 1 is called a **monic polynomial.**
- Polynomial functions are defined for all values of x.
- Every non-zero constant is a polynomial of degree 0.
- The constant 0 is also a polynomial called **zero polynomial.**
- The degree of a polynomial is a non- negative integer.
- The zero polynomial is the only polynomial with leading coefficient 0.
- Polynomials of degree two are called **quadratic polynomials.**
- Polynomials of degree three are called **cubic polynomials.**
- Polynomials of degree four are called **quartic polynomials.**
- Formula for finding roots of the quadratic equation $ax^2 + bx + c = 0$ is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- Discriminant $\Delta = b^2 - 4ac$
- Nature of the roots of the quadratic equation:
  - If $\Delta > 0$, then the roots are real and distinct.

- If $\Delta < 0$, then the roots are not real.
- If $\Delta = 0$, then the roots are real and equal.

❖ **Vieta's formula for polynomial equation of degree 2:**
$x^2 - (\alpha + \beta)x + \alpha\beta = 0$. (or)
$x^2 - (\text{Sum of the roots})x + (\text{Product of the roots}) = 0$

❖ Sum of the roots $\alpha + \beta = -\frac{b}{a}$

❖ Product of the roots $\alpha\beta = \frac{c}{a}$

❖ If $P(x) = 0$ is a quadratic equation whose roots are α and β, then $cP(x)$ is also a quadratic equation with roots α and β for any non-zero constant c.

❖ **Fundamental Theorem of Algebra:** A polynomial equation of degree n has exactly n root in $\mathbb{C}$ when the roots are counted with their multiplicities.

❖ **Vieta's formula for polynomial equation of degree 3:**
$x^3 - (\sum 1)x^2 + (\sum 2)x - (\sum 3) = 0$, where
$\sum_1 = \alpha + \beta + \gamma = -\frac{b}{a}$; $\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$; $\sum 3 = \alpha\beta\gamma = -\frac{d}{a}$

❖ **Vieta's formula for polynomial equation of degree 4:**
$x^4 + (\sum 1)x^3 + (\sum 2)x^2 + (\sum 3)x + (\sum 4) = 0$, where
$\sum 1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a}$
$\sum 2 = \alpha\beta + \beta\gamma + \gamma\delta + \delta\alpha = \frac{c}{a}$
$\sum 3 = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a}$
$\sum 4 = \alpha\beta\gamma\delta = \frac{e}{a}$

❖ **Vieta's formula for polynomial equation of degree $n > 3$:**
A polynomial equation of degree n with roots $\alpha_1, \alpha_2, \ldots, \alpha_n$ is
$x^n - (\sum \alpha_1)x^{n-1} + (\sum \alpha_1\alpha_2)x^{n-2} - (\sum \alpha_1\alpha_2\alpha_3)x^{n-3} + \cdots + (-1)^n\alpha_1\alpha_2 \ldots \alpha_n = 0$

EXERCISE 3.1

1. If the sides of a cubic box are increased by 1,2,3 units respectively to form a cuboid, then the volume is increased by 52 cubic units. Find the volume of the cuboid.

Let x be the length of the side of a cube.
From the given data,
$(x+1), (x+2)$ and $(x+3)$ are the length, breadth and height of the cuboid.
Volume of the cuboid = volume of the cube + 52 cubic units.
$\Rightarrow (x+1)(x+2)(x+3) = x^3 + 52$
$\Rightarrow (x^2 + 3x + 2)(x+3) = x^3 + 52 \quad [\because (x+a)(x+b) = x^2 + (a+b)x + ab]$
$\Rightarrow x^3 + 3x^2 + 3x^2 + 9x + 2x + 6 = x^3 + 52$
$\Rightarrow x^3 + 3x^2 + 3x^2 + 9x + 2x + 6 - x^3 - 52 = 0$
$\Rightarrow 6x^2 + 11x - 46 = 0$
$\Rightarrow (x-2)(6x+23) = 0$
$\Rightarrow x = 2 \quad (or) \quad x = -23$ ($Not\ admisible$)
Volume of the cuboid= $x^3 + 52 = 2^3 + 52 = 8 + 52 = 60$ cubic units.

2. Construct a cubic equation with roots $(i)1,2\ and\ 3 \quad ii)\ 1,1\ and -2$
$(iii)2, -2\ and 4$

$(i)1,2\ and\ 3$
Here $\alpha = 1, \beta = 2, \gamma = 3$
$\sum_1 = \alpha + \beta + \gamma = 1 + 2 + 3 = 6$
$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = (1 \times 2) + (2 \times 3) + (3 \times 1) = 2 + 6 + 3 = 11$
$\sum_3 = \alpha\beta\gamma = 1 \times 2 \times 3 = 6$
Vieta's formula for polynomial equation of degree 3:
$x^3 - (\sum \alpha)x^2 + (\sum \alpha\beta)x - (\sum \alpha\beta\gamma) = 0$
$\Rightarrow x^3 - 6x^2 + 11x - 6 = 0$

$ii)\ 1,1\ and -2$
Here $\alpha = 1, \beta = 1, \gamma = -2$
$\sum_1 = \alpha + \beta + \gamma = 1 + 1 - 2 = 0$
$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = (1 \times 1) + (1 \times -2) + (-2 \times 1) = 1 - 2 - 2 = -3$
$\sum_3 = \alpha\beta\gamma = 1 \times 1 \times (-2) = -2$
Vieta's formula for polynomial equation of degree 3:
$x^3 - (\sum 1)x^2 + (\sum 2)x - (\sum 3) = 0$
$\Rightarrow x^3 - 0x^2 - 3x - (-2) = 0$
$\Rightarrow x^3 - 3x + 2 = 0$

$(iii)2, -2\ and\ 4$
Here $\alpha = 2, \beta = -2, \gamma = 4$
$\sum_1 = \alpha + \beta + \gamma = 2 - 2 + 4 = 4$
$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = (2 \times -2) + (-2 \times 4) + (4 \times 2) = -4 - 8 + 8 = -4$
$\sum_3 = \alpha\beta\gamma = 2 \times (-2) \times 4 = -16$

Vieta's formula for polynomial equation of degree 3:

$x^3 - (\sum 1)x^2 + (\sum 2)x - (\sum 3) = 0$

$\Rightarrow x^3 - 4x^2 - 4x - (-16) = 0$

$\Rightarrow x^3 - 4x^2 - 4x + 16 = 0$

3. If $\alpha, \beta \text{ and } \gamma$ are the roots of the cubic equation $x^3 + 2x^2 + 3x + 4 = 0$, form a cubic equation whose roots are $(i) 2\alpha, 2\beta, 2\gamma \quad (ii) \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma} \quad (iii) -\alpha, -\beta, -\gamma$

Let $\alpha, \beta \text{ and } \gamma$ are the roots of the equation $x^3 + 2x^2 + 3x + 4 = 0$.

$\sum_1 = \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{2}{1} = -2$

$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = \frac{3}{1} = 3$

$\sum_3 = \alpha\beta\gamma = -\frac{d}{a} = -\frac{4}{1} = -4$

(i)Roots are $2\alpha, 2\beta, 2\gamma$

$\sum_1 = 2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) = 2(-2) = -4$

$\sum_2 = (2\alpha)(2\beta) + (2\beta)(2\gamma) + (2\gamma)(2\alpha) = 4(\alpha\beta + \beta\gamma + \gamma\alpha) = 4(3) = 12$

$\sum_3 = (2\alpha)(2\beta)(2\gamma) = 8(\alpha\beta\gamma) = 8 \times (-4) = -32$

∴ The required cubic polynomial equation is

$x^3 - (\sum 1)x^2 + (\sum 2)x - (\sum 3) = 0$

$\Rightarrow x^3 - (-4)x^2 + 12x - (-32) = 0$

$\Rightarrow x^3 + 4x^2 + 12x + 32 = 0$

(ii) Roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

$\sum_1 = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = -\frac{3}{4}$

$\sum_2 = \left(\frac{1}{\alpha} \times \frac{1}{\beta}\right) + \left(\frac{1}{\beta} \times \frac{1}{\gamma}\right) + \left(\frac{1}{\gamma} \times \frac{1}{\alpha}\right) = \frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{-2}{-2} = \frac{1}{2}$

$\sum_3 = \frac{1}{\alpha} \times \frac{1}{\beta} \times \frac{1}{\gamma} = \frac{1}{\alpha\beta\gamma} = -\frac{1}{4}$

∴ The required cubic polynomial equation is

$x^3 - (\sum 1)x^2 + (\sum 2)x - (\sum 3) = 0$

$\Rightarrow x^3 - \left(-\frac{3}{4}\right)x^2 + \frac{1}{2}x - \left(-\frac{1}{4}\right) = 0$

$\Rightarrow 4x^3 + 3x^2 + 2x + 1 = 0$

(iii) Roots are $-\alpha, -\beta, -\gamma$

$\sum_1 = -(\alpha + \beta + \gamma) = -(-2) = 2$

$\sum_2 = (-\alpha)(-\beta) + (-\beta)(-\gamma) + (-\gamma)(-\alpha) = \alpha\beta + \beta\gamma + \gamma\alpha = 3$

$\sum_3 = (-\alpha)(-\beta)(-\gamma) = -(\alpha\beta\gamma) = -(-4) = 4$

∴ The required cubic polynomial equation is

$x^3 - (\sum 1)x^2 + (\sum 2)x - \sum 3 = 0$

$\Rightarrow x^3 - 2x^2 + 3x - 4 = 0$

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4. Solve the equation $3x^3 - 16x^2 + 23x - 6 = 0$ if the product of two roots is 1.

Let $\alpha, \beta \text{ and } \gamma$ are the roots of the equation $3x^3 - 16x^2 + 23x - 6 = 0$.

The product of two roots is $1 \Rightarrow \alpha\beta = 1 \Rightarrow \beta = \frac{1}{\alpha} \rightarrow (1)$

$\sum_1 = \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{-16}{3} = \frac{16}{3} \Rightarrow \alpha + \beta + \gamma = \frac{16}{3} \rightarrow (2)$

$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = \frac{23}{3} \Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = \frac{23}{3} \rightarrow (3)$

$\sum_3 = \alpha\beta\gamma = -\frac{d}{a} = -\frac{-6}{3} = 2 \Rightarrow \alpha\beta\gamma = 2 \rightarrow (4)$

From (1) & $(4) \Rightarrow (1)\gamma = 2 \Rightarrow \boxed{\gamma = 2}$

$(2) \Rightarrow \alpha + \beta + 2 = \frac{16}{3} \Rightarrow \alpha + \beta = \frac{10}{3}$

$\Rightarrow \alpha + \frac{1}{\alpha} = \frac{10}{3} \qquad (From\ (1))$

$\Rightarrow \frac{\alpha^2+1}{\alpha} = \frac{10}{3} \Rightarrow 3\alpha^2 - 10\alpha + 3 = 0 \Rightarrow (\alpha - 3)(3\alpha - 1) = 0 \Rightarrow \boxed{\alpha = 3\ (or)\ \frac{1}{3}}$

If $\alpha = 3\ (or)\ \frac{1}{3}$, $(1) \Rightarrow \boxed{\beta = \frac{1}{3}\ (or) 3}$

Thus the roots are $3, \frac{1}{3}, 2$

5.Find the sum of the squares of root of the equation $2x^4 - 8x^3 + 6x^2 - 3 = 0$.

Let $\alpha, \beta, \gamma, \delta$ are the roots of the equation $2x^4 - 8x^3 + 6x^2 - 3 = 0$

$a = 2, b = -8, c = 6, d = 0, e = -3$

$\sum_1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a} = -\frac{-8}{2} = 4$

$\sum_2 = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\alpha = \frac{c}{a} = \frac{6}{2} = 3$

W.K.T,

$(\alpha + \beta + \gamma + \delta)^2 = \alpha^2 + \beta^2 + \gamma^2 + \delta^2 - 2\sum \alpha\beta$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2\sum \alpha\beta$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (4)^2 - 2(3) = 16 - 6 = 10$

$\boxed{EXAMPLE\ 3.4}$ Find the sum of the squares of roots of $ax^4 + bx^3 + cx^2 + dx + e = 0$.

6. Solve the equation $x^3 - 9x^2 + 14x + 24 = 0$ if it is given that two of its roots are in the ratio 3: 2.

Let $\alpha, \beta \; and \; \gamma$ are the roots of the equation $x^3 - 9x^2 + 14x + 24 = 0$.

Two of its roots are in the ratio 3: 2. So, the roots are $\alpha, \beta = 3p, \gamma = 2p$

$$\sum_1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a} = -\frac{-9}{1} = 9 \Longrightarrow \alpha + 3p + 2p = 9$$
$$\Longrightarrow \alpha = 9 - 5p \rightarrow (1)$$
$$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = \frac{14}{1} \Longrightarrow \alpha\beta + \beta\gamma + \gamma\alpha = 14$$
$$\Longrightarrow (9 - 5p)3p + (3p)(2p) + (2p)(9 - 5p) = 14$$
$$\Longrightarrow 27p - 15p^2 + 6p^2 + 18p - 10p^2 - 14 = 0$$
$$\Longrightarrow -19p^2 + 45p - 14 = 0$$
$$\Longrightarrow 19p^2 - 45p + 14 = 0$$
$$\Longrightarrow 19p^2 - 38p - 7p + 14 = 0$$
$$\Longrightarrow 19p(p - 2) - 7(p - 2) = 0$$
$$\Longrightarrow (p - 2)(19p - 7) = 0$$
$$\Longrightarrow \boxed{p = 2} \; (or) \frac{7}{19} \; (Not \; admisible)$$

Thus the roots are $9 - 5p, 3p, 2p = -1,6,4$.

7. If $\alpha, \beta \; and \; \gamma$ are the roots of the polynomial equation $ax^3 + bx^2 + cx + d = 0$, find the value of $\sum \frac{\alpha}{\beta\gamma}$ in terms of the coefficients.

$\alpha, \beta \; and \; \gamma$ are the roots of the equation $ax^3 + bx^2 + cx + d = 0$

$$\sum_1 = \alpha + \beta + \gamma = -\frac{b}{a}$$
$$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$
$$\sum_3 = \alpha\beta\gamma = -\frac{d}{a}$$
$$\sum 1 = \sum \frac{\alpha}{\beta\gamma} = \frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta} = \frac{\alpha^2+\beta^2+\gamma^2}{\alpha\beta\gamma} = \frac{(\alpha+\beta+\gamma)^2 - 2\sum\alpha\beta}{\alpha\beta\gamma}$$
$$= \frac{\left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{-\frac{d}{a}}$$
$$= \left(\frac{b^2}{a^2} - \frac{2c}{a}\right)\left(-\frac{a}{d}\right)$$
$$= \left(\frac{b^2 - 2ac}{a^2}\right)\left(-\frac{a}{d}\right)$$
$$= \frac{2ac - b^2}{ad}$$

EXAMPLE 3.3 If $\alpha, \beta \; and \; \gamma$ are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \frac{1}{\beta\gamma}$ in terms of the coefficients.

8. . If $\alpha, \beta, \gamma \; and \; \delta$ are the roots of the polynomial equation $2x^4 + 5x^3 - 7x^2 + +8 = 0$, find a quadratic equation with integer coefficient whose root are $\alpha + \beta + \gamma + \delta$ and $\alpha\beta\gamma\delta$

If $\alpha, \beta, \gamma \; and \; \delta$ are the roots of the equation $2x^4 + 5x^3 - 7x^2 + +8 = 0$

$$a = 2, b = 5, c = -7, d = 0, e = 8$$
$$\sum_1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a} = -\frac{5}{2}$$
$$\sum_4 = \alpha\beta\gamma\delta = \frac{e}{a} = \frac{8}{2} = 4$$

Sum of the roots $= -\frac{5}{2} + 4 = \frac{3}{2}$

Product of the roots $= \left(-\frac{5}{2}\right)4 = -10$

The required quadratic equation is

$$x^2 - (\text{Sum of the roots })x + (\text{Product of the roots}) = 0$$
$$\Longrightarrow x^2 - \left(\frac{3}{2}\right)x + (-10) = 0$$
$$\Longrightarrow 2x^2 - 3x - 20 = 0$$

9.If p and are roots of the equation $lx^2 + nx + n = 0$, show that $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \frac{\sqrt{n}}{\sqrt{l}} = 0$.

$$p + q = -\frac{n}{l} \quad ; \quad pq = \frac{n}{l}$$
$$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = \frac{\sqrt{p}}{\sqrt{q}} + \frac{\sqrt{q}}{\sqrt{p}} = \frac{p+q}{\sqrt{pq}} = \frac{-n/l}{\sqrt{n/l}} = -\sqrt{n/l} = -\frac{\sqrt{n}}{\sqrt{l}}$$
$$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \frac{\sqrt{n}}{\sqrt{l}} = -\frac{\sqrt{n}}{\sqrt{l}} + \frac{\sqrt{n}}{\sqrt{l}} = 0$$

10.If the equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root, show that it must be equal to $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$

$x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root. Let α be the common root of the above equations. So the equations become $\alpha^2 + p\alpha + q = 0 \rightarrow (1)$ and $\alpha^2 + p'\alpha + q' = 0 \rightarrow (2)$

Solving (1) and (2) by cross multiplication method,

$$\begin{matrix} \alpha^2 & \alpha & 1 \\ p & q & 1 \\ p' & q' & 1 \end{matrix} \quad \begin{matrix} p \\ p \\ p' \end{matrix}$$

$$\Rightarrow \frac{\alpha^2}{pq'-p'q} = \frac{\alpha}{q-q'} = \frac{1}{p'-p}$$

$$\Rightarrow \frac{\alpha^2}{pq'-p'q} = \frac{\alpha}{q-q'} \text{ (or) } \frac{\alpha}{q-q'} = \frac{1}{p'-p}$$

$$\Rightarrow \alpha = \frac{pq'-p'q}{q-q'} \text{ (or) } \alpha = \frac{q-q'}{p'-p}$$

11. Formulate into a mathematical problem to find a number such that when it cube root is added to it, the result is 6.

Let x be the required number.

Given: $x + \sqrt[3]{x} = 6 \Rightarrow \sqrt[3]{x} = 6 - x$

Raising the power to 3 on both sides, $x = (6-x)^3$

$$\Rightarrow x = (6)^3 - 3(6)^2x + 3(6)x^2 - x^3$$
$$\Rightarrow x = 216 - 108x + 18x^2 - x^3$$
$$\Rightarrow x^3 - 18x^2 + 109x - 216 = 0$$

12. A 12 metre tall tree was broken into two parts. It was found that the height of the part which was left standing was the cube root of the length of the part that was cut away. Formulate this into a mathematical problem to find the height of the part which was cut away.

Let x^3 be the length of the part of the tree standing and x be the length of the part of the tree cut away.

From the given data, $x^3 + x = 12 \Rightarrow x^3 + x - 12 = 0$

EXAMPLE 3.1 If α and β are the roots of the quadratic equation $17x^2 + 43x - 73 = 0$, construct a quadratic equation whose roots are $\alpha + 2$ and $\beta + 2$.

$\alpha + \beta = -\frac{43}{17}$; $\alpha\beta = -\frac{73}{17}$

Sum of the roots $= \alpha + 2 + \beta + 2 = \alpha + \beta + 4 = -\frac{43}{17} + 4 = \frac{25}{17}$

Product of the roots $= (\alpha + 2)(\beta + 2)$

$$= \alpha\beta + 2\alpha + 2\beta + 4$$
$$= \alpha\beta + 2(\alpha + \beta) + 4$$
$$= -\frac{73}{17} + 2\left(-\frac{43}{17}\right) + 4$$
$$= \frac{-73-86}{17} + 4$$
$$= \frac{-159+68}{17}$$
$$= -\frac{91}{17}$$

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The required quadratic equation is

$$x^2 - (\text{Sum of the roots})x + (\text{Product of the roots}) = 0$$
$$\Rightarrow x^2 - \left(\frac{25}{17}\right)x + \left(-\frac{91}{17}\right) = 0 \Rightarrow 17x^2 - 25x - 91 = 0$$

EXAMPLE 3.2 If α and β are the roots of the quadratic equation $2x^2 - 7x + 13 = 0$, construct a quadratic equation whose roots are α^2 and β^2.

$\alpha + \beta = \frac{7}{2}$; $\alpha\beta = \frac{13}{2}$

Sum of the roots $= \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{7}{2}\right)^2 - 2\left(\frac{13}{2}\right) = \frac{49}{4} - 13 = -\frac{3}{4}$

Product of the roots $= \alpha^2\beta^2 = (\alpha\beta)^2 = \left(\frac{13}{2}\right)^2 = \frac{169}{4}$

The required quadratic equation is

$$x^2 - (\text{Sum of the roots})x + (\text{Product of the roots}) = 0$$
$$\Rightarrow x^2 - \left(-\frac{3}{4}\right)x + \left(\frac{169}{4}\right) = 0 \Rightarrow 4x^2 + 3x + 169 = 0$$

EXAMPLE 3.5 Find the condition that the roots of $x^3 + ax^2 + bx + c = 0$ are in the ratio $p:q:r$.

Since the roots are in the ratio $p:q:r$, then the roots are $p\lambda, q\lambda, r\boldsymbol{\lambda}$.

$$\sum_1 = p\lambda + q\lambda + r\boldsymbol{\lambda} = -\frac{a}{1} \Rightarrow \lambda(\mathrm{p+q+r}) = -\mathrm{a} \Rightarrow \lambda = -\frac{\mathrm{a}}{\mathrm{p+q+r}} \rightarrow (1)$$

$$\sum_3 = (p\lambda)(q\lambda)(r\boldsymbol{\lambda}) = -\frac{c}{1} \Rightarrow \boldsymbol{\lambda}^3 pqr = -c \Rightarrow \boldsymbol{\lambda}^3 = -\frac{c}{pqr} \rightarrow (2)$$

Sub. (1) in (2) $\Rightarrow \left(-\frac{\mathrm{a}}{\mathrm{p+q+r}}\right)^3 = -\frac{c}{pqr}$

$$\Rightarrow -\frac{a^3}{(\mathrm{p+q+r})^3} = -\frac{c}{pqr} \Rightarrow pqra^3 = c(\mathrm{p+q+r})^3$$

EXAMPLE 3.6 Form the equation whose roots are the squares of the roots of the cubic equation $x^3 + ax^2 + bx + c = 0$.

Let α, β and γ are the roots of the equation $x^3 + ax^2 + bx + c = 0$.

$$\sum_1 = \alpha + \beta + \gamma = -\frac{a}{1} = -a$$

$$\sum_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{b}{1} = b$$

$\sum_3 = \alpha\beta\gamma = -\frac{c}{1} = -c$

Now the roots are $\alpha^2, \beta^2, \gamma^2$

$\sum_1 = \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = a^2 - 2b$

$\sum_2 = \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$= b^2 - 2(-c)(-a) = b^2 - 2ca$

$\sum_3 = \alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = (-c)^2 = c^2$

Vieta's formula for polynomial equation of degree 3:

$x^3 - (\sum 1)x^2 + (\sum 2)x - (\sum 3) = 0$

$x^3 - (a^2 - 2b\ )x^2 + (b^2 - 2ca)x - c^2 = 0$

$EXAMPLE\ 3.7$ If p is real, discuss the nature of the roots of the equation $4x^2 + 4px + p + 2 = 0$, in terms of p.

Discriminant $\Delta = b^2 - 4ac$

$= (4p)^2 - 4(4)(p + 2)$

$= 16p^2 - 16p - 32$

$= 16(p^2 - p - 2)$

$= 16(p + 1)(p - 2)$

$\Delta < 0\ if\ -1 < p < 2$

$\Delta = 0\ if\ \ p = -1\ or\ p = 2$

$\Delta > 0\ if\ -\infty < p < -1\ \ or\ \ 2 < p < \infty$

Thus the given polynomial has

Imaginary roots $if\ -1 < p < 2$;

Equal and real roots $if\ \ p = -1\ or\ p = 2$;

Distinct real roots $if\ -\infty < p < -1\ \ or\ \ 2 < p < \infty$.

NATURE OF ROOT AND NATURE OF COEFFICIENTS OF POLYNOMIAL EQUATIONS

- ❖ COMPLEX CONJUGATE ROOT THEOREM: If a complex number z_0 is a root of a polynomial equation with real coefficients, then its complex conjugate $\bar{z_0}$ is also a root.
- ❖ Non-real complex roots or imaginary roots occur as conjugate pairs.
- ❖ IRRATIONAL ROOTS:
  - If $\sqrt{\Delta}$ is rational, then both the roots are rational.
  - If $\sqrt{\Delta}$ is irrational, then both the roots are irrational.
  - If $\Delta = \frac{m}{n}$ for some positive integers m and n with G.C.D of m, n=1.
  - $\sqrt{\Delta}$ is rational $\Leftrightarrow$ m and n are perfect squares.
  - $\sqrt{\Delta}$ is irrational $\Leftrightarrow$ at least one of m or n is not a perfect square
- ❖ If p and q are rational numbers such that $\sqrt{q}$ is irrational. If $p + \sqrt{q}$ is a root of a quadratic equation with rational coefficients, then $p - \sqrt{q}$ is also a root of the same equation.
- ❖ If p and q be rational numbers so that $\sqrt{p}$ and $\sqrt{q}$ are irrational numbers; further let one of $\sqrt{p}$ and $\sqrt{q}$ be not a rational multiple of the other. If $\sqrt{p} + \sqrt{q}$ is a root of a polynomial equation with rational coefficients, then $\sqrt{p} - \sqrt{q}, -\sqrt{p} + \sqrt{q}$ and $-\sqrt{p} - \sqrt{q}$ are also roots of the same polynomial equation.

EXERCISE 3.2

1.If k is real, discuss the nature of the roots of the polynomial equation $2x^2 + kx + k = 0$, in terms of k.

Here $a = 2, b = k, c = k$

$\Delta = b^2 - 4ac = k^2 - 8k = k(k - 8)$

If $\Delta = 0$, then $k(k - 8) = 0 \Rightarrow k = 0\ or\ k = 8$

When $k < 0 \Rightarrow\ \Delta > 0$, then the equation has real roots.

When $k = 0\ , k = 8 \Rightarrow \Delta = 0$, then the equation has equal roots.

When $0 < k < 8 \Rightarrow\ \Delta < 0$, then the equation has imaginary roots.

When $k > 8 \Rightarrow\ \Delta > 0$, then the equation has real and distinct roots.

2.Find the polynomial equation of minimum degree with rational coefficients, having $2 + \sqrt{3}i$ as a root.

Given: One root $\alpha = 2 + \sqrt{3}i$

∴ Another root $\beta = 2 - \sqrt{3}i$

$\alpha + \beta = 2 + \sqrt{3}i + 2 - \sqrt{3}i = 4$

$\alpha\beta = \left(2 + \sqrt{3}i\right)\left(2 - \sqrt{3}i\right) = 4 + 3 = 7$

∴ The required polynomial is $x^2 - (\alpha + \beta\ )x + (\alpha\beta) = 0\ \ \Rightarrow x^2 - 4x + 7 = 0$

$EXAMPLE\ 3.8$ Find the monic polynomial equation of minimum degree with real coefficients having $2 - \sqrt{3}i$ as a root.

$EXAMPLE\ 3.9$ Find the polynomial equation of minimum degree with rational

coefficients, having $2-\sqrt{3}$ as a root.

3. Find the polynomial equation of minimum degree with rational coefficients, having $2i+3$ as a root.

One root $\alpha = 3+2i$

$\therefore$ Another root $\beta = 3-2i$ www.nammakalvi.in

$\alpha+\beta = 3+2i+3-2i = 6$

$\alpha\beta = (3+2i)(3-2i) = 9+4 = 13$

$\therefore$ The required polynomial is $x^2-(\alpha+\beta)x+(\alpha\beta)=0 \Longrightarrow x^2-6x+13=0$

4. Find the polynomial equation of minimum degree with rational coefficients, having $\sqrt{5}-\sqrt{3}$ as a root.

Since $\sqrt{5}-\sqrt{3}$ is a root. $x = \sqrt{5}-\sqrt{3}$

$\Longrightarrow x+\sqrt{3} = \sqrt{5}$

Squaring on both sides, $\left(x+\sqrt{3}\right)^2 = \sqrt{5}^2$

$\Longrightarrow x^2+2\sqrt{3}x+3 = 5$

$\Longrightarrow x^2-2 = 2\sqrt{3}x$

Again squaring on both sides, $(x^2-2)^2 = \left(2\sqrt{3}x\right)^2$

$\Longrightarrow x^4-4x^2+4 = 12x^2$

$\Longrightarrow x^4-16x^2+4 = 0$

5.Prove that a straight line and parabola cannot intersect at more than two points.

Equation of the straight line is $y = mx+c \rightarrow (1)$

Equation of the parabola is $y^2 = 4ax \rightarrow (2)$

Sub. (1) in $(2) \Longrightarrow (mx+c)^2 = 4ax$

$\Longrightarrow m^2x^2+2mcx+c^2 = 4ax$

$\Longrightarrow m^2x^2+2mcx+c^2-4ax = 0$

$\Longrightarrow m^2x^2+(2mc-4a)x+c^2 = 0$

This is a quadratic equation which cannot have more than two solutions. Hence a line and a parabola cannot intersect at more than two points.

$EXAMPLE\ 3.14$ Prove that a line cannot intersect a circle at more than two points.

Equation of the circle is $x^2+y^2 = r^2 \rightarrow (1)$

Equation of the straight line is $y = mx+c \rightarrow (2)$

Sub. (2) in $(1) \Longrightarrow x^2+(mx+c)^2 = r^2$

$\Longrightarrow x^2+m^2x^2+2mcx+c^2 = r^2$

$\Longrightarrow (1+m^2)x^2+2mcx+(c^2-r^2) = 0$

This is a quadratic equation which cannot have more than two solutions. Hence a circle and a line cannot intersect at more than two points.

$EXAMPLE\ 3.10$ Form the polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

Since $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ is a root, $\left(x-\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right)$ is a factor. So $\left(x+\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right)$ is another factor.

Product of the factors $= \left(x-\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right)\left(x+\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right) = x^2-\left(\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right)^2 = x^2-\frac{\sqrt{2}}{\sqrt{3}}$

Now $\left(x^2-\frac{\sqrt{2}}{\sqrt{3}}\right)$ is one factor. So $\left(x^2+\frac{\sqrt{2}}{\sqrt{3}}\right)$ is another factor.

Again product of the factors$= \left(x^2-\frac{\sqrt{2}}{\sqrt{3}}\right)\left(x^2+\frac{\sqrt{2}}{\sqrt{3}}\right) = (x^2)^2-\left(\frac{\sqrt{2}}{\sqrt{3}}\right)^2 = x^4-\frac{2}{3}$

$\Longrightarrow 3x^4-2=0$ is the required polynomial equation with integer coefficients.

$EXAMPLE\ 3.11$ Show that the equation $2x^2-6x+7=0$ cannot be satisfied by any real values of x.

Here $a=2, b=-6, c=7$

$\Delta = b^2-4ac = (-6)^2-4(2)(7) = 36-56 = -20 < 0$

$\therefore$ The roots are imaginary numbers.

$EXAMPLE\ 3.12$ If $x^2+2(k+2)x+9k=0$ has equal roots, find k.

Here $a=1, b=2k+4, c=9k$

$\Delta = 0 \Longrightarrow b^2-4ac = 0$

$\Longrightarrow (2k+4)^2-4(1)(9k) = 0$

$\Longrightarrow 4k^2+16k+16-36k = 0$

$\Longrightarrow 4k^2-20k+16 = 0$

$\Longrightarrow k^2-5k+4 = 0 \Longrightarrow (k-1)(k-4) = 0 \Longrightarrow k = 1\ or\ 4$

$EXAMPLE\ 3.13$ Show that if p, q, r are rational, the roots of the equation $x^2-2px+p^2-q^2+2qr-r^2=0$ are rational.

Here $a=1, b=-2p, c=p^2-q^2+2qr-r^2$

$\Delta = b^2-4ac = (-2p)^2-4(1)(p^2-q^2+2qr-r^2)$

$= 4p^2 - 4p^2 + 4q^2 - 8qr + 4r^2$
$= 4q^2 - 8qr + 4r^2$
$= 4(q^2 - 2qr + r^2)$
$= [2(q-r)]^2$ is a perfect square.

Hence the roots are rational.

ROOTS OF HIGHER DEGREE POLYNOMIAL EQUATIONS

- ❖ IMAGINARY OR SURDS ROOTS:
  - i. If $\alpha + i\beta$ is an imaginary root of a quartic polynomial with real coefficients, then $\alpha - i\beta$ is also a root.
  - ii. Thus$[x - (\alpha + i\beta)]$ and $[x - (\alpha - i\beta)]$ are the factors of the polynomial.
  - iii. Divide the polynomial with the product of the factors $x^2 - 2\alpha\beta x + \alpha^2 + \beta^2$ and reduce the polynomial to quadratic form.
  - iv. solving the quadratic equation find the other roots.
- ❖ POLYNOMIAL EQUATIONS WITH EVEN POWERS ONLY:

  If $P(x)$ is a polynomial equation of degree 2n, having only even powers
  - i. Replace x^2 by y $(ie., x^2 = y)$ and solve the equation to find y.
  - ii. Substituting the values of y in $x^2 = y$ and find the roots.
- ❖ If the sum of all coefficients is zero, then 1 is a root of $P(x)$.
- ❖ If the sum of the coefficients of the even degree is equal to the sum of the coefficients of the odd degree, then -1 is a root of $P(x)$.
- ❖ If the roots of the polynomial equation are in A.P , then the roots are $\ldots, (a-d), a, (a+d), \ldots$
- ❖ If the roots of the polynomial equation are in G.P , then the roots are $\ldots, \left(\frac{a}{r}\right), a, (ar), \ldots$
- ❖ If the roots of the polynomial equation are in H.P , then the roots are $\ldots, \left(\frac{1}{a-d}\right), \frac{1}{a}, \left(\frac{1}{a+d}\right), \ldots$

EXERCISE 3.3

1.Solve the cubic equation: $2x^3 - x^2 - 18x + 9 = 0$ if sum of two of its roots vanishes.

Let $\alpha, -\alpha, \beta$ are the root of $2x^3 - x^2 - 18x + 9 = 0$. (∵sum of two of its roots vanishes.)

$\sum_1 = \alpha - \alpha + \beta = -\frac{b}{a} = -\frac{-1}{2} = \frac{1}{2} \Rightarrow \beta = \frac{1}{2}$

$\sum_2 = \alpha(-\alpha) + (-\alpha)(\beta) + (\beta)(\alpha) = \frac{c}{a} = -\frac{18}{2}$

$\Rightarrow -\alpha^2 - \alpha\beta + \alpha\beta = -9$

$\Rightarrow \alpha^2 = 9$

$\Rightarrow \alpha = \pm 3$

Thus the roots are $3, -3, \frac{1}{2}$

2. Solve the equation: $9x^3 - 36x^2 + 44x - 16 = 0$ if the roots form an arithmetic progression.

Let $(a-d), a, (a+d)$ are the roots of $9x^3 - 36x^2 + 44x - 16 = 0$. $(\because A.P)$

$\sum_1 = a - d + a + a + d = -\frac{b}{a} = -\frac{-36}{9} = 4 \Rightarrow 3a = 4 \Rightarrow a = \frac{4}{3}$

	9	−36	44	−16
$\frac{4}{3}$	0	12	−32	16
	9	−24	12	0

Quotient $= \frac{1}{3}(9x^2 - 24x + 12)$

$= 3x^2 - 8x + 4$

$= 3x^2 - 6x - 2x + 4$

$= 3x(x-2) - 2(x-2)$

$= (x-2)(3x-2)$

$(x-2)(3x-2) = 0 \Rightarrow x = 2, \frac{2}{3}$

Thus the roots $= \frac{2}{3}, \frac{4}{3}, 2.$ (or) $2, \frac{4}{3}, \frac{2}{3}$.

EXAMPLE 3.22 It is known that the roots of the equation $x^3 - 6x^2 - 4x + 24 = 0$ are in arithmetic progression. Find its roots.

3. Solve the equation: $3x^3 - 26x^2 + 52x - 24 = 0$ if the roots form a geometric progression.

Let $\left(\frac{a}{r}\right), a, (ar)$ are the roots of $3x^3 - 26x^2 + 52x - 24 = 0$. $(\because G.P)$

$\sum_3 = \left(\frac{a}{r}\right) a(ar) = -\frac{d}{a} = -\frac{-24}{3} = 8 \Rightarrow a^3 = 2^3 \Rightarrow a = 2$

	3	−26	52	−24
2	0	6	−40	24
	3	−20	12	0

Quotient $= 3x^2 - 20x + 12$

$= 3x^2 - 18x - 2x + 12$
$= 3x(x-6) - 2(x-6)$
$= (x-6)(3x-2)$

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$(x-6)(3x-2) = 0 \Rightarrow x = 6, \frac{2}{3}$.

Thus the roots $= \frac{2}{3}, 2, 6.$ (or) $6,2,,\frac{2}{3}$

4. Determine k and Solve the equation $2x^3 - 6x^2 + 3x + k = 0$ if one of its roots is twice the sum of the other two roots.

The polynomial equation is $2x^3 - 6x^2 + 3x + k = 0$

Let $\alpha, \beta \; and \; \gamma$ are the roots.

Given: $\alpha = 2(\beta + \gamma) \rightarrow (1)$

$\sum_1 = \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{-6}{2} = 3$

$\Rightarrow \alpha + \beta + \gamma = 3$

$\Rightarrow 2\alpha + 2(\beta + \gamma) = 6$

$\Rightarrow 2\alpha + \alpha = 6 \Rightarrow 3\alpha = 6 \Rightarrow \boxed{\alpha = 2}$

	2	−6	3	k
2	0	4	−4	−2
	2	−2	−1	0

$\because \alpha = 2$ is one of the root, so $k - 2 = 0 \Rightarrow \boxed{k = 2}$

Quotient $= 2x^2 - 2x - 1$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-(-2) \pm \sqrt{(-2)^2-4(2)(-1)}}{2(2)} = \frac{2 \pm \sqrt{4+8}}{4} = \frac{2 \pm 2\sqrt{3}}{4} = \frac{1 \pm \sqrt{3}}{2} = \frac{1+\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}$

Thus the roots are $2, \frac{1+\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}$.

5. Find all the zeros of the polynomial $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$, if it is known that $1 + 2i$ and $\sqrt{3}$ are two of its zeros.

Given roots are $1 + 2i$ and $\sqrt{3}$. Therefore $1 - 2i$ and $-\sqrt{3}$ are also roots.

$x = 1 + 2i \Rightarrow x - 1 = 2i$

$\Rightarrow (x-1)^2 = (2i)^2$

$\Rightarrow x^2 - 2x + 1 = -4$

$\Rightarrow x^2 - 2x + 5 = 0$

$x = \sqrt{3} \Rightarrow x^2 = 3 \Rightarrow x^2 - 3 = 0$

$x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135 \equiv (x^2 - 2x + 5)(x^2 - 3)(x^2 + px - 9)$

Equating the coefficient of x on both side, $-39 = -54 - 15p \Rightarrow p = -1$

So, $x^2 + px - 9 = x^2 - x - 9$, solving $x^2 - x - 9 = 0$.

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-(-1) \pm \sqrt{(-1)^2-4(1)(-9)}}{2(1)} = \frac{1 \pm \sqrt{1+36}}{2} = \frac{1 \pm \sqrt{37}}{2} = \frac{1+\sqrt{37}}{2}, \frac{1-\sqrt{37}}{2}$

Thus the roots are $1 + 2i, 1 - 2i, \sqrt{3}, -\sqrt{3}, \frac{1+\sqrt{37}}{2}, \frac{1-\sqrt{37}}{2}$

EXAMPLE 3.15 If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation $x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x - 140 = 0$, find all roots.

6. Solve the cubic equations:

$(i)\; 2x^3 - 9x^2 + 10x = 3 \quad (ii) 8x^3 - 2x^2 - 7x + 3 = 0$

$(i)\; 2x^3 - 9x^2 + 10x = 3 \Rightarrow 2x^3 - 9x^2 + 10x - 3 \; = 0$

Sum of all coefficients $= 2 - 9 + 10 - 3 = 0$

$\therefore (x-1)$ is a factor.

	2	−9	10	−3
1	0	2	−7	3
	2	−7	3	0

Quotient $= 2x^2 - 7x + 3$

$= 2x^2 - x - 6x + 3$

$= x(2x-1) - 3(2x-1)$

$= (x-3)(2x-1)$

$x = 3, \frac{1}{2}$

Thus the roots are $1,3, \frac{1}{2}$.

$(ii) 8x^3 - 2x^2 - 7x + 3 = 0$

Sum of the coefficient of odd degree $= 8 - 7 = 1$

Sum of the coefficient of even degree $= -2 + 3 = 1$

$\therefore (x+1)$ is a factor.

	8	−2	−7	3
−1	0	−8	10	−3
	8	−10	3	0

Quotient $= 8x^2 - 10x + 3$

$= 8x^2 - 4x - 6x + 3$

$= 4x(2x-1) - 3(2x-1)$

$= (4x-3)(2x-1)$

$x = \frac{3}{4}, \frac{1}{2}$

Thus the roots are $-1, \frac{1}{2}, \frac{3}{4}$.

EXAMPLE 3.17 Solve the equation: $x^3 - 3x^2 - 33x + 35 = 0$

EXAMPLE 3.18 Solve the equation: $2x^3 + 11x^2 - 9x - 18 = 0$

7. Solve the equation: $x^4 - 14x^2 + 45 = 0$

Let $x^2 = y \rightarrow (1)$

$$x^4 - 14x^2 + 45 = 0 \Rightarrow y^2 - 14y + 45 = 0$$
$$\Rightarrow y^2 - 9y - 5y + 45 = 0$$
$$\Rightarrow y(y-9) - 5(y-9) = 0$$
$$\Rightarrow (y-5)(y-9) = 0$$
$$\Rightarrow y = 5, 9$$

Sub. $y = 5,9\ in\ (1) \Rightarrow x^2 = 5\ ;\ x^2 = 9 \Rightarrow x = \pm\sqrt{5}\ ; x = \pm 3$

Thus the roots are $-3, 3, -\sqrt{5}, \sqrt{5}$.

EXAMPLE 3.16 Solve the equation: $x^4 - 9x^2 + 20 = 0$

EXAMPLE 3.19 Obtain the condition that the roots of $x^3 + px^2 + qx + r = 0$ are in A.P.

Let $(a-d), a, (a+d)$ are the roots of $x^3 + px^2 + qx + r = 0$. $(\because A.P)$

Applying the Vieta's formula, $a - d + a + a + d = -\frac{b}{a} = -p$

$$\Rightarrow 3a = -p \Rightarrow a = -\frac{p}{3}$$

Since $a = -\frac{p}{3}$ is the root of the equation $x^3 + px^2 + qx + r = 0$

$$\Rightarrow \left(-\frac{p}{3}\right)^3 + p\left(-\frac{p}{3}\right)^2 + q\left(-\frac{p}{3}\right) + r = 0$$
$$\Rightarrow -\frac{p^3}{27} + p\left(\frac{p^2}{9}\right) - \frac{pq}{3} + r = 0$$
$$\Rightarrow -\frac{p^3}{27} + \frac{p^3}{9} + r = \frac{pq}{3}$$
$$(\times)\ 27 \Rightarrow -p^3 + 3p^3 + 27r = 9pq$$
$$\Rightarrow 9pq = 2p^3 + 27r$$

EXAMPLE 3.20 Find the condition that the roots of $ax^3 + bx^2 + cx + d = 0$ are in geometric progression. Assume $a, b, c, d \neq 0$.

Let $\left(\frac{a}{r}\right), a, (ar)$ are the roots of $ax^3 + bx^2 + cx + d = 0$. $(\because G.P)$

Applying the Vieta's formula,

$$\sum_1 = \frac{a}{r} + a + ar = -\frac{b}{a} \Rightarrow a\left(\frac{1}{r} + 1 + r\right) = -\frac{b}{a} \rightarrow (1)$$
$$\sum_2 = \left(\frac{a}{r}\right)(a) + a(ar) + (ar)\left(\frac{a}{r}\right) = \frac{c}{a} \Rightarrow a^2\left(\frac{1}{r} + 1 + r\right) = \frac{c}{a} \rightarrow (2)$$
$$\sum_3 = \left(\frac{a}{r}\right)a(ar) = -\frac{d}{a} \Rightarrow a^3 = -\frac{d}{a} \rightarrow (3)$$

Dividing (2) by $(2) \Rightarrow a = \left(\frac{c}{a}\right)\left(-\frac{a}{b}\right) \Rightarrow a = -\frac{c}{b} \rightarrow (4)$

Sub. (4) in $(3) \Rightarrow \left(-\frac{c}{b}\right)^3 = -\frac{d}{a} \Rightarrow -\frac{c^3}{b^3} = -\frac{d}{a} \Rightarrow ac^3 = db^3$

EXAMPLE 3.21 If the roots of $x^3 + px^2 + qx + r = 0$ are in H.P., prove that $9pqr = 27r^3 + 2p$.

Let the roots are in H.P. then their reciprocals are in A.P. and roots of the equation

$$\left(\frac{1}{x}\right)^3 + p\left(\frac{1}{x}\right)^2 + q\left(\frac{1}{x}\right) + r = 0 \Rightarrow \frac{1}{x^3} + \frac{p}{x^2} + \frac{q}{x} + r = 0$$
$$(\times)x^3 \Rightarrow rx^3 + qx^2 + px + 1 = 0$$

Let $(a-d), a, (a+d)$ are the roots of $rx^3 + qx^2 + px + 1 = 0$. $(\because A.P)$

$$\sum_\alpha = a - d + a + a + d = -\frac{b}{a} = -\frac{-q}{r} \Rightarrow 3a = \frac{-q}{r} \Rightarrow a = \frac{-q}{3r}$$

Since $a = \frac{-q}{3r}$ is the root of the equation $rx^3 + qx^2 + px + 1 = 0$

$$\Rightarrow r\left(\frac{-q}{3r}\right)^3 + q\left(\frac{-q}{3r}\right)^2 + p\left(\frac{-q}{3r}\right) + 1 = 0$$
$$\Rightarrow -\frac{rq^3}{27r^3} + \frac{q^3}{9r^2} - \frac{pq}{3r} + 1 = 0$$
$$\Rightarrow -\frac{q^3}{27r^2} + \frac{q^3}{9r^2} - \frac{pq}{3r} + 1$$
$$(\times)27r^2 \Rightarrow -q^3 + 3q^3 - 9pqr + 27r^2 = 0$$
$$\Rightarrow 9pqr = 2q^3 + 27r^2$$

PARTLY FACTORED POLYNOMIALS

- Quadratic polynomial equations of the form $(ax+b)(cx+d)(px+q)(rx+s) + k = 0, k \neq 0$ can be rewritten in the form $(\alpha x^2 + \beta x + \lambda)(\alpha x^2 + \beta x + \mu) + k = 0$
- $(x+a)(x+b) = x^2 + (a+b)x + ab$

EXERCISE 3.4

1.Solve : $(i)(x-5)(x-7)(x+6)(x+4) = 504$

$(ii)(x-4)(x-7)(x-2)(x+1) = 16$

$(i)(x-5)(x-7)(x+6)(x+4)=504$

$\Rightarrow [(x-5)(x+4)][(x-7)(x+6)]=504$

$\Rightarrow [x^2-x-20][x^2-x-42]=504$

Let $x^2-x=y \rightarrow (1)$

$\Rightarrow [y-20][y-42]=504$

$\Rightarrow y^2-62y+840-504=0$

$\Rightarrow y^2-62y+336=0$

$\Rightarrow y^2-6y-56y+336=0$

$\Rightarrow y(y-6)-56(y-6)=0$

$\Rightarrow (y-6)(y-56)=0$

$\Rightarrow y=6,56$

Put $y=6,56$ in (1),

$\Rightarrow x^2-x=6$	$\Rightarrow x^2-x=56$
$\Rightarrow x^2-x-6=0$	$\Rightarrow x^2-x-56=0$
$\Rightarrow (x+2)(x-3)=0$	$\Rightarrow (x+7)(x-8)=0$
$\Rightarrow x=-2,3$	$\Rightarrow x=-7,8$

Thus the roots are $-7,-2,3,8$.

$(ii)(x-4)(x-7)(x-2)(x+1)=16$

$\Rightarrow [(x-4)(x-2)][(x-7)(x+1)]=16$

$\Rightarrow [x^2-6x+8][x^2-6x-7]=16$

Let $x^2-6x=y \rightarrow (1)$

$\Rightarrow [y+8][y-7]=16$

$\Rightarrow y^2+y-56-16=0$

$\Rightarrow y^2+y-72=0$

$\Rightarrow y^2-8y+9y-72=0$

$\Rightarrow y(y-8)+9(y-8)=0$

$\Rightarrow (y-8)(y+)=0$

$\Rightarrow y=8,-9$

Put $y=8,-9$ in (1),

$\Rightarrow x^2-6x=8$	$\Rightarrow x^2-6x=-9$
$\Rightarrow x^2-6x-8=0$	$\Rightarrow x^2-6x+9=0$
$\Rightarrow x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$	$\Rightarrow (x-3)(x-3)=0$
	$\Rightarrow x=3,3$

$\Rightarrow x=\frac{-(-6)\pm\sqrt{(-6)^2-4(1)(-8)}}{2(1)}$	www.nammakalvi.in
$\Rightarrow x=\frac{6\pm\sqrt{36+32}}{2}$	
$\Rightarrow x=\frac{6\pm\sqrt{78}}{2}$	
$\Rightarrow x=\frac{6\pm\sqrt{4\times17}}{2}$	
$\Rightarrow x=\frac{6\pm2\sqrt{17}}{2}$	
$\Rightarrow x=3\pm\sqrt{17}$	

Thus the roots are $3,3,3+\sqrt{17},3-\sqrt{17}$.

2.Solve: $(2x-1)(x+3)(x-2)(2x+3)+20=0$

$\Rightarrow [(2x-1)(2x+3)][(x+3)(x-2)]+20=0$

$\Rightarrow [4x^2+4x-3][x^2+x-6]+20=0$

$\Rightarrow [4(x^2+x)-3][x^2+x-6]+20=0$

Let $x^2+x=y \rightarrow (1)$

$\Rightarrow [4y-3][y-6]+20=0$

$\Rightarrow 4y^2-24y-3y+18+20=0$

$\Rightarrow 4y^2-27y+38=0$

$\Rightarrow 4y^2-8y-19y+38=0$

$\Rightarrow 4y(y-2)-19(y-2)=0$

$\Rightarrow (y-2)(4y-19)=0$

$\Rightarrow y=2,\frac{19}{4}$

Put $y=2,\frac{19}{4}$ in (1),

$\Rightarrow x^2+x=2$	$\Rightarrow x^2+x=\frac{19}{4}$
$\Rightarrow x^2+x-2=0$	$\Rightarrow 4x^2+4x-19=0$
$\Rightarrow (x+2)(x-1)=0$	$\Rightarrow x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$
$\Rightarrow x=-2,1$	$\Rightarrow x=\frac{-(4)\pm\sqrt{(4)^2-4(4)(-19)}}{2(4)}$
	$\Rightarrow x=\frac{-4\pm\sqrt{320}}{8}$
	$\Rightarrow x=\frac{-4\pm4\sqrt{20}}{8}$

	$\Rightarrow x = \frac{-1\pm\sqrt{4\times5}}{2}$ $\Rightarrow x = \frac{-1\pm2\sqrt{5}}{2}$

Thus the roots are $-2,1,\frac{-1+2\sqrt{5}}{2},\frac{-1-2\sqrt{5}}{2}$.

$EXAMPLE\ 3.23$ Solve the equation $(x-2)(x-7)(x-3)(x+2)+19=0$

HINT: $[(x-2)(x-3)][(x-7)(x+2)]+19=0$

$EXAMPLE\ 3.24$ Solve the equation $(2x-3)(6x-1)(3x-2)(x-12)-7=0$

HINT: $[(2x-3)(3x-2)][(6x-1)(x-12)]-7=0$

POLYNOMIAL EQUATION WITH NO ADDITIONAL INFORMATION

- **RATIONAL ROOT THEOREM:** Let $a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$ with $a_n\neq0$, and $a_0\neq0$, be a polynomial with integer coefficients. If $\frac{p}{q}$ with $(p,q)=1$, is a root of the polynomial, then p is a factor of a_0 and q is a factor of a_n.
- **RECIPROCAL POLYNOMIAL:** A polynomial $P(x)$ of degree n is said to be a reciprocal polynomial if one of the following conditions is true: $(i)P(x)=x^nP\left(\frac{1}{x}\right)\ \ (ii)P(x)=-x^nP\left(\frac{1}{x}\right)$
- **RECIPROCAL EQUATION:** A polynomial equation $a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0=0,(a_n\neq0)$ is a reciprocal equation if and only if one of the following two statements is true: $(i)a_n=a_0,\ a_{n-1}=a_1,a_{n-2}=a_2,\ldots\ldots(ii)a_n=-a_0,\ a_{n-1}=-a_1,a_{n-2}=-a_2,\ldots\ldots$
  - A reciprocal equation cannot have 0 as a solution.
  - The coefficients and the solutions are not restricted to be real.
  - If a polynomial equation $P(x)=0$ has α as a root and if $\frac{1}{\alpha}$ is also a root then the polynomial $P(x)=0$ must be a reciprocal equation is not true.
  - For an odd degree reciprocal equation satisfying $a_n=a_0,\ a_{n-1}=a_1,a_{n-2}=a_2,\ldots\ldots,x=-1$ must be a solution.
  - For an even degree reciprocal equation satisfying $a_n=-a_0,\ a_{n-1}=-a_1,a_{n-2}=-a_2,\ldots\ldots,x=1$ must be a solution.
  - For an even degree reciprocal equation satisfying $a_n=-a_0,\ a_{n-1}=-a_1,a_{n-2}=-a_2,\ldots\ldots,$and the middle term must be 0, then $x=1$ and $x=-1$ are solutions.
  - For an even degree reciprocal equation by taking $x+\frac{1}{x}$ or $x-\frac{1}{x}$ as y, we can obtain a polynomial equation of degree one half of the degree of the given equation.

EXERCISE 3.5

1.Solve the following equations:

$(i)sin^2x-5\sin x+4=0\ \ (ii)12x^3+8x=29x^2-4$

$i)sin^2x-5\sin x+4=0$

Let $\sin x=y\rightarrow(1)$

$sin^2x-5\sin x+4=0\Rightarrow y^2-5y+4=0$

$\Rightarrow(y-1)(y-4)=0$

$\Rightarrow y=1,4$

Sub. $y=1,4$ in (1),

$\sin x=1$ $\Rightarrow\sin x=\sin\frac{\pi}{4}$ $\Rightarrow x=2n\pi+\frac{\pi}{2},n\in z$	$\sin x=4$ is not possible.

$(ii)12x^3+8x=29x^2-4\Rightarrow12x^3-29x^2+8x+4=0$

Let $P(x)=12x^3-29x^2+8x+4$

$P(2)=12(2)^3-29(2)^2+8(2)+4=96-116+16+4=116-116=0$

$\therefore(x-2)$ is a factor.

	12	−29	8	4
2	0	24	−10	−4
	12	−5	−2	0

Quotient $=12x^2-5x-2$

$=12x^2+3x-8x-2$

$=3x(4x+1)-2(4x+1)$

$=(4x+1)(3x-2)$

$(4x+1)(3x-2)=0\Rightarrow x=-\frac{1}{4},\frac{2}{3}$

Thus the roots are $2,-\frac{1}{4},\frac{2}{3}$.

EXAMPLE 3.25 Solve the equation $x^3 - 5x^2 - 4x + 20 = 0$.

2.Examine for the rational roots of $(i)2x^3 - x^2 - 1 = 0$ $(ii)x^8 - 3x + 1 = 0$

$(i)2x^3 - x^2 - 1 = 0$

Sum of all coefficients $= 2 - 1 - 1 = 0$

$\therefore (x - 1)$ is a factor.

	2	−1	0	−1
1	0	2	1	1
	2	1	1	0

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Quotient $= 2x^2 + x + 1$

$\Rightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-(1) \pm \sqrt{(1)^2-4(2)(1)}}{2(2)} = \frac{-1 \pm \sqrt{-7}}{4} = \frac{-1 \pm \sqrt{7}i}{4}$, which are imaginary.

$\therefore x = 1$ is a rational root.

$(ii)x^8 - 3x + 1 = 0$

Let $P(x) = x^8 - 3x + 1$.

$a_n = 1\ ;\ a_0 = 1 \Rightarrow p = \pm 1, q = \pm 1 \Rightarrow \frac{p}{q} = \pm 1$

Sum of all co.eff.$= 1 - 3 + 1 \neq 0 \Rightarrow 1$ is not a root.

Sum of even term co.eff.$\neq$ Sum of odd term co.eff.$\Rightarrow -1$ is not a root.

$\therefore P(x)$ has no rational roots.

3.Solve: $8x^{\frac{3}{2n}} - 8x^{\frac{-3}{2n}} = 63$

Let $y = x^{\frac{3}{2n}}$.

$8x^{\frac{3}{2n}} - 8x^{\frac{-3}{2n}} = 63 \Rightarrow 8x^{\frac{3}{2n}} - \frac{8}{x^{\frac{3}{2n}}} = 63$

$\Rightarrow 8y - \frac{8}{y} = 63$

$\Rightarrow \frac{8y^2-8}{y} = 63$

$\Rightarrow 8y^2 - 8 = 63y$

$\Rightarrow 8y^2 - 63y - 8 = 0$

$\Rightarrow 8y^2 - 64y + y - 8 = 0$

$\Rightarrow 8y(y - 8) + 1(y - 8) = 0$

$\Rightarrow (y - 8)(8y + 1) = 0$

$\Rightarrow \boxed{y = 8}\ (or)\ y = -\frac{1}{8}$ (Not a solution)

$y = x^{\frac{3}{2n}} \Rightarrow x^{\frac{3}{2n}} = 8 \Rightarrow x^{\frac{3}{2n}} = 2^3 \Rightarrow x^{\frac{1}{2n}} = 2 \Rightarrow x = 2^{2n} \Rightarrow \boxed{x = 4^n}$

4.Solve: $2\sqrt{\frac{x}{a}} + 3\sqrt{\frac{a}{x}} = \frac{b}{a} + \frac{6a}{b}$.

$2\sqrt{\frac{x}{a}} + 3\sqrt{\frac{a}{x}} = \frac{b}{a} + \frac{6a}{b}$

Squaring on both sides, $\left(2\sqrt{\frac{x}{a}} + 3\sqrt{\frac{a}{x}}\right)^2 = \left(\frac{b}{a} + \frac{6a}{b}\right)^2$

$\Rightarrow 4\frac{x}{a} + 9\frac{a}{x} + 12\sqrt{\frac{x}{a}}.\sqrt{\frac{a}{x}} = \frac{b^2}{a^2} + \frac{36a^2}{b^2} + 2.\frac{b}{a}.\frac{6a}{b}$

$\Rightarrow 4\frac{x}{a} + 9\frac{a}{x} + 12 = \frac{b^2}{a^2} + \frac{36a^2}{b^2} + 12$

$\Rightarrow 4\frac{x}{a} + 9\frac{a}{x} = \frac{b^2}{a^2} + \frac{36a^2}{b^2}$

$\Rightarrow \frac{4x^2+9a^2}{ax} = \frac{b^4+36a^4}{a^2b^2}$

$\Rightarrow (4x^2 + 9a^2)ab^2 = (b^4 + 36a^4)x$

$\Rightarrow 4ab^2x^2 + 9a^3b^2 = (b^4 + 36a^4)x$

$\Rightarrow 4ab^2x^2 - (b^4 + 36a^4)x + 9a^3b^2 = 0$

$\Rightarrow 4ab^2x^2 - b^4x - 36a^4x + 9a^3b^2 = 0$

$\Rightarrow b^2x(4ax - b^2) - 9a^3(4ax - b^2) = 0$

$\Rightarrow (4ax - b^2)(b^2x - 9a^3) = 0$

$\Rightarrow x = \frac{b^2}{4a}, \frac{9a^3}{b^2}$

5.Solve the equations: $(i)6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

$(ii)x^4 + 3x^3 - 3x - 1 = 0$

$(i)6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

The given polynomial equation is an even degree reciprocal equation of Type-I.

Dividing the given equation by x^2, we get

$6x^2 - 35x + 62 - \frac{35}{x} + \frac{6}{x^2} = 0 \Rightarrow 6\left(x^2 + \frac{1}{x^2}\right) - 35\left(x + \frac{1}{x}\right) + 62 = 0 \rightarrow (1)$

Let $\boxed{y = x + \frac{1}{x}} \Rightarrow y^2 = \left(x + \frac{1}{x}\right)^2 \Rightarrow y^2 = x^2 + \frac{1}{x^2} + 2 \Rightarrow x^2 + \frac{1}{x^2} = y^2 - 2$

$(1) \Rightarrow 6(y^2 - 2) - 35y + 62 = 0$

$\Rightarrow 6y^2 - 12 - 35y + 62 = 0$

$\Rightarrow 6y^2 - 35y + 50 = 0$

$\Rightarrow 6y^2 - 15y - 20y + 50 = 0$
$\Rightarrow 3y(2y-5) - 10(2y-5) = 0$
$\Rightarrow (2y-5)(3y-10) = 0 \Rightarrow \boxed{y = \frac{5}{2}\ ; y = \frac{10}{3}}$

Substituting the values of y in $y = x + \frac{1}{x}$,

When $y = \frac{5}{2} \Rightarrow x + \frac{1}{x} = \frac{5}{2}$
$\Rightarrow \frac{x^2+1}{x} = \frac{5}{2}$
$\Rightarrow 2x^2 + 2 = 5x$
$\Rightarrow 2x^2 - 5x + 2 = 0$
$\Rightarrow 2x^2 - 4x - x + 2 = 0$
$\Rightarrow 2x(x-2) - 1(x-2) = 0$
$\Rightarrow (x-2)(2x-1) = 0$
$\Rightarrow x = 2, \frac{1}{2}$

When $y = \frac{10}{3} \Rightarrow x + \frac{1}{x} = \frac{10}{3}$
$\Rightarrow \frac{x^2+1}{x} = \frac{10}{3}$
$\Rightarrow 3x^2 + 3 = 10x$
$\Rightarrow 3x^2 - 10x + 3 = 0$
$\Rightarrow 3x^2 - 9x - x + 3 = 0$
$\Rightarrow 3x(x-3) - 1(x-3) = 0$
$\Rightarrow (x-3)(3x-1) = 0$
$\Rightarrow x = 3, \frac{1}{3}$

Thus the roots are $2, \frac{1}{2}, 3, \frac{1}{3}$.

$(ii) x^4 + 3x^3 - 3x - 1 = 0$

The given polynomial equation is an even degree reciprocal equation of Type-II.

So 1 and -1 are the roots of the equation. Hence the other two roots are α and $\frac{1}{\alpha}$.

$\sum 1 = -1 + 1 + \alpha + \frac{1}{\alpha} = -3 \Rightarrow \frac{\alpha^2+1}{\alpha} = -3 \Rightarrow \alpha^2 + 1 = -3\alpha$
$\Rightarrow \alpha^2 + 3\alpha + 1 = 0$
$\Rightarrow \alpha = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-(3) \pm \sqrt{(3)^2-4(1)(1)}}{2(1)} = \frac{-3 \pm \sqrt{9-4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$

Thus the roots are $-1, 1, \frac{-3+\sqrt{5}}{2}, \frac{-3-\sqrt{5}}{2}$

6.Find all the real numbers satisfying $4^x - 3(2^{x+2}) + 2^5 = 0$.

$4^x - 3(2^{x+2}) + 2^5 = 0$
$\Rightarrow (2^x)^2 - 3(2^x 2^2) + 32 = 0$
$\Rightarrow (2^x)^2 - 12(2^x) + 32 = 0$

Let $2^x = y$

$\Rightarrow (y)^2 - 12(y) + 32 = 0$
$\Rightarrow y^2 - 4y - 8y + 32 = 0$
$\Rightarrow y(y-4) - 8(y-4) = 0$
$\Rightarrow (y-4)(y-8) = 0$
$\Rightarrow y = 4,8$

Substituting the values of y in $2^x = y$

When $y = 4 \Rightarrow 2^x = 4 \Rightarrow 2^x = 2^2 \Rightarrow \boxed{x = 2}$

When $y = 8 \Rightarrow 2^x = 8 \Rightarrow 2^x = 2^3 \Rightarrow \boxed{x = 3}$

Thus the roots are 2 and 3.

7. Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

The given polynomial equation is an even degree reciprocal equation of Type-I.

Since $\frac{1}{3}$ is a root, then 3 is also a root. Let the other roots are α and $\frac{1}{\alpha}$.

$\sum 1 = 3 + \frac{1}{3} + \alpha + \frac{1}{\alpha} = \frac{5}{6} \Rightarrow \frac{10}{3} + \frac{\alpha^2+1}{\alpha} = \frac{5}{6}$
$\Rightarrow \frac{\alpha^2+1}{\alpha} = \frac{5}{6} - \frac{10}{3}$
$\Rightarrow \frac{\alpha^2+1}{\alpha} = \frac{5-20}{6}$
$\Rightarrow \frac{\alpha^2+1}{\alpha} = -\frac{5}{2}$
$\Rightarrow 2\alpha^2 + 2 = -5\alpha$
$\Rightarrow 2\alpha^2 + 5\alpha + 2 = 0$
$\Rightarrow 2\alpha^2 + \alpha + 4\alpha + 2 = 0$
$\Rightarrow 2\alpha(\alpha+2) + 1(\alpha+2) = 0$
$\Rightarrow (\alpha+2)(2\alpha+1) = 0$
$\Rightarrow \alpha = -2, -\frac{1}{2}$

Thus the roots are $-2, -\frac{1}{2}, 3, \frac{1}{3}$.

$\boxed{EXAMPLE\ 3.28}$ Solve the equation $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$

$\boxed{EXAMPLE\ 3.26}$ Find the roots of $2x^3 + 3x^2 + 2x + 3$

Let $P(x) = 2x^3 + 3x^2 + 2x + 3$.

$P\left(-3/2\right) = 2\left(-3/2\right)^3 + 3\left(-3/2\right)^2 + 2\left(-3/2\right) + 3$
$= 2\left(-\frac{27}{8}\right) + 3\left(\frac{9}{4}\right) - 3 + 3$
$= -\frac{27}{4} + \frac{27}{4}$

$= 0$

$\therefore -\frac{3}{2}$ is one of the root.

To find other roots divide $2x^3 + 3x^2 + 2x + 3$ by $-\frac{3}{2}$

$$-\frac{3}{2}\Big|\begin{array}{cccc} 2 & 3 & 2 & 3 \\ 0 & -3 & 0 & -3 \\ \hline 2 & 0 & 2 & 0 \end{array}$$

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Quotient $= \frac{1}{2}[2x^2 + 2] = x^2 + 1$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(0) \pm \sqrt{(0)^2 - 4(1)(1)}}{2(1)} = \frac{\pm\sqrt{-4}}{2} = \frac{\pm 2i}{2} = \pm i$$

Thus the roots are $-\frac{3}{2}, i, -i$.

EXAMPLE 3.27 Solve the equation $7x^3 - 43x^2 - 43x + 7 = 0$

The given polynomial equation is an odd degree reciprocal equation of Type-I.

Thus -1 is one root.

To find the other roots divide $7x^3 - 43x^2 - 43x + 7$ by -1.

$$-1\Big|\begin{array}{cccc} 7 & -43 & -43 & 7 \\ 0 & -7 & 50 & -7 \\ \hline 7 & -50 & 7 & 0 \end{array}$$

Quotient $= 7x^2 - 50x + 7$

$= 7x^2 - 49x - x + 7$

$= 7x(x - 7) - 1(x - 7)$

$= (x - 7)(7x - 1)$

$\Rightarrow x = 7, \frac{1}{7}$

Thus the roots are $-1, 7, \frac{1}{7}$.

EXAMPLE 3.29 Find the solution, if any of the equation $2cos^2 x - 9\cos x + 4 = 0$

Let $\cos x = y \rightarrow (1)$

$2cos^2 x - 9\cos x + 4 = 0 \Rightarrow 2y^2 - 9y + 4 = 0$

$\Rightarrow 2y^2 - y - 8y + 4 = 0$

$\Rightarrow y(2y - 1) - 4(2y - 1) = 0$

$\Rightarrow (y - 4)(2y - 1) = 0$

$\Rightarrow y = 4, \frac{1}{2}$

When $y = \frac{1}{2} \Rightarrow \cos x = \frac{1}{2}$

$\Rightarrow \cos x = \cos\frac{\pi}{3}$

$\Rightarrow x = 2n\pi \pm \frac{\pi}{3}, n \in z$

When $y = 4$,

$\Rightarrow \cos x = 4$ is not possible

DESCARTES RULE

Descartes Rule: If p is the number of positive zero of a polynomial $P(x)$ with real coefficients and s is the number of sign changes in coefficients of $P(x)$, then $s - p$ is a non-negative even integer.

The number negative zeros of the polynomial $P(x)$ cannot be more than the number of sign changes in coefficients of $P(-x)$ and the difference between the number of sign changes in coefficients of $P(-x)$ and the number of negative zeros of the polynomial $P(x)$ is even.

EXERCISE 3.6

1.Discuss the maximum possible number of positive and negative roots of the polynomial equation $9x^9 - 4x^8 + 4x^7 - 3x^6 + 2x^5 + x^3 + 7x^2 + 7x + 2 = 0$

Let $P(x) = 9x^9 - 4x^8 + 4x^7 - 3x^6 + 2x^5 + x^3 + 7x^2 + 7x + 2$

$P(-x) = 9(-x)^9 - 4(-x)^8 + 4(-x)^7 - 3(-x)^6 + 2(-x)^5 + (-x)^3 + 7(-x)^2 + 7(-x) + 2$

$= -9x^9 - 4x^8 - 4x^7 - 3x^6 - 2x^5 - x^3 + 7x^2 - 7x + 2$

The number of sign changes for $P(x)$ and $P(-x)$ are 4 and 2 respectively. Hence it has at most 4 positive real roots and 2 negative real roots.

2. Discuss the maximum possible number of positive and negative roots of the polynomial equation $x^2 - 5x + 6$ and $x^2 - 5x + 16$. Also draw rough sketch of the graph.

$P(x) = (x^2 - 5x + 6)$; $P(-x) = (x^2 + 5x + 6)$

The number of sign changes for $P(x)$ and $P(-x)$ are 2 and 0 respectively. Hence it has at most 2 positive real roots and no negative real roots.

$Q(x) = x^2 - 5x + 16$; $Q(-x) = x^2 + 5x + 16$

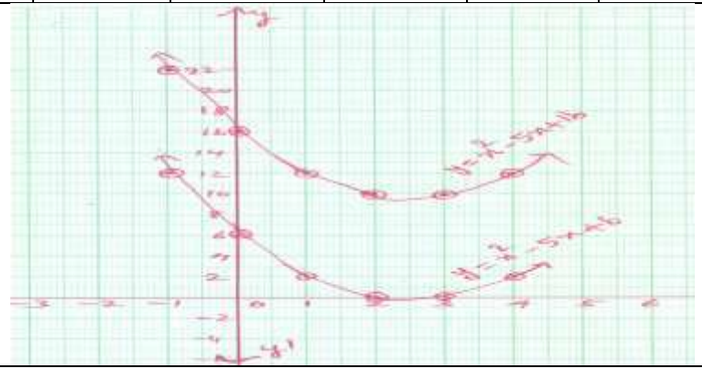
The number of sign changes for $Q(x)$ and $Q(-x)$ are 2 and 0 respectively. Hence it has at most 2 positive real roots and no negative real roots.

$y = x^2 - 5x + 6$

x	-1	0	1	2	3	4
y	12	6	2	0	0	2

$y = x^2 - 5x + 16$

x	-1	0	1	2	3	4
y	22	16	12	10	10	12



3. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has at least 6 imaginary solutions.

$P(x) = x^9 - 5x^5 + 4x^4 + 2x^2 + 1$

$P(-x) = (-x)^9 - 5(-x)^5 + 4(-x)^4 + 2(-x)^2 + 1$

$= -x^9 + 5x^5 + 4x^4 + 2x^2 + 1$

The number of sign changes for $P(x)$ and $P(-x)$ are 2 and 1 respectively. Hence it has at most 2 positive real root and 1 negative real root.

Number of imaginary roots $= 9 - (2 + 1) = 9 - 3 = 6.$

Hence it has at least 6 imaginary solutions.

$EXAMPLE\ 3.30$ Show that the polynomial $9x^9 + 2x^5 - x^4 - 7x^2 + 2$ has at least 6 imaginary roots.

4. Determine the number of positive and negative roots of the equation $x^9 - 5x^8 - 14x^7 = 0$

$P(x) = x^9 - 5x^8 - 14x^7$

$P(-x) = (-x)^9 - 5(-x)^8 - 14(-x)^7 = -x^9 + 5x^8 + 14x^7$

The number of sign changes for $P(x)$ and $P(-x)$ are 1 and 1 respectively. Hence it has at most 1 positive real root and 1 negative real root.

5. Find the exact number of real and imaginary roots of the equation $x^9 + 9x^7 + 7x^5 + 5x^3 + 3x = 0.$

$P(x) = x^9 + 9x^7 + 7x^5 + 5x^3 + 3x.$

$P(-x) = (-x)^9 + 9(-x)^7 + 7(-x)^5 + 5(-x)^3 + 3(-x)$

$= -x^9 - 9x^7 - 7x^5 - 5x^3 - 3x$

The number of sign changes for $P(x)$ and $P(-x)$ are 0 and 0 respectively. Hence it has no positive real roots and no negative real roots. 0 is one root of the equation. Hence the given equation has only one real root and 8 imaginary roots.

$EXAMPLE\ 3.31$ Discuss the nature of the roots of the following polynomials:

$(i) x^{2018} + 1947x^{1950} + 15x^8 + 26x^6 + 2019$ $(ii) x^5 - 19x^4 + 2x^3 + 5x^2 + 11$

4.INVERSE TRIGONOMETRIC FUNCTIONS

DOMAIN AND RANGE OF TRIGONOMETRIC FUNCTIONS:

	DOMAIN	RANGE
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\text{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\cot^{-1} x$	R	$(0, \pi)$

PROPERTIES OF TRIGONOMETRIC FUNCTIONS:

PROPERTY I :–

1) $sin^{-1}(\sin\theta) = \theta,\ if\ \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
2) $cos^{-1}(\cos\theta) = \theta, if\ \theta \in [0, \pi]$
3) $tan^{-1}(\tan\theta) = \theta, if\ \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
4) $cosec^{-1}(\text{cosec}\,\theta) = \theta, if\ \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
5) $sec^{-1}(\sec\theta) = \theta, if\ \theta \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$
6) $cot^{-1}(\cot\theta) = \theta, if\ \theta \in (0, \pi)$

PROPERTY II :–

1) $\sin(sin^{-1}x) = x, if\ x \in [-1, 1]$
2) $\cos(cos^{-1}x) = x, if\ x \in [-1, 1]$
3) $\tan(tan^{-1}x) = x, if\ x \in R$
4) $\text{cosec}(cosec^{-1}x) = x, if\ x \in R \backslash (-1, 1)$

5) $\sec(sec^{-1}x) = x, if\ x \in R\backslash(-1,1)$
6) $\cot(cot^{-1}x) = x, if\ x \in R$

PROPERTY III : −RECIPROCAL INVERSE IDENTITIES www.nammakalvi.in

1) $sin^{-1}\left(\frac{1}{x}\right) = cosecx, if\ x \in R\backslash(-1,1)$
2) $cos^{-1}\left(\frac{1}{x}\right) = secx^{-1}x, if\ x \in R\backslash(-1,1)$
3) $tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} cot^{-1}x\ , x > 0 \\ -\pi + cot^{-1}x\ , x < 0 \end{cases}$

PROPERTY IV : −REFLECTION IDENTITIES

1) $sin^{-1}(-x) = -sin^{-1}x, if\ x \in [-1,1]$
2) $tan^{-1}(-x) = -tan^{-1}x, if\ x \in R$
3) $cosec^{-1}(-x) = -cosec^{-1}x, if\ |x| \geq 1\ \ or\ \ x \in R\backslash(-1,1)$
4) $cos^{-1}(-x) = \pi - cos^{-1}x, if\ x \in [-1,1]$
5) $sec^{-1}(-x) = \pi - sec^{-1}x|x| \geq 1\ \ or\ \ x \in R\backslash(-1,1)$
6) $cot^{-1}(-x) = -cot^{-1}x, if\ x \in R$

PROPERTY V : −COFUNCTION INVERSE IDENTITIES

1) $sin^{-1}x + cos^{-1}x = \frac{\pi}{2}, x \in [-1,1]$
2) $tan^{-1}x + cot^{-1}x = \frac{\pi}{2}, x \in R$
3) $cosec^{-1}x + sec^{-1}x = \frac{\pi}{2}, x \in R\backslash(-1,1)\ \ or\ \ |x| \geq 1$

PROPERTY VI : −

1) $sin^{-1}x + sin^{-1}y = sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$, where either $x^2 + y^2 \leq 1$ or $xy < 0$
2) $sin^{-1}x - sin^{-1}y = sin^{-1}\left(x\sqrt{1-y^2} - y\sqrt{1-x^2}\right)$, where either $x^2 + y^2 \leq 1$ or $xy > 0$
3) $cos^{-1}x + cos^{-1}y = cos^{-1}\left[xy - \sqrt{1-x^2}\sqrt{1-y^2}\right], if\ x + y \geq 0$
4) $cos^{-1}x - cos^{-1}y = cos^{-1}\left[xy + \sqrt{1-x^2}\sqrt{1-y^2}\right], if\ x \leq y$
5) $tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1$
6) $tan^{-1}x - tan^{-1}y = tan^{-1}\left(\frac{x-y}{1+xy}\right), xy > -1$

PROPERTY VII : −

1) $2tan^{-1}x = tan^{-1}\left(\frac{2x}{1-x^2}\right), |x| < 1$
2) $2tan^{-1}x = co^{-1}\left(\frac{1-x^2}{1+x^2}\right), x \geq 0$
3) $2tan^{-1}x = sin^{-1}\left(\frac{2x}{1+x^2}\right), |x| \leq 1$

PROPERTY VIII : −

1) $sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2sin^{-1}x, if|x| \leq \frac{1}{\sqrt{2}}\ or\ -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$
2) $sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2cos^{-1}x, if\ \frac{1}{\sqrt{2}} \leq x \leq 1$

PROPERTY IX : −

1) $sin^{-1}(x) = cos^{-1}\sqrt{1-x^2}, if\ 0 \leq x \leq 1$
2) $sin^{-1}(x) = -cos^{-1}\sqrt{1-x^2}, if\ -1 \leq x < 0$
3) $sin^{-1}(x) = tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right), if\ -1 < x < 1$
4) $cos^{-1}(x) = sin^{-1}\sqrt{1-x^2}, if\ 0 \leq x \leq 1$
5) $cos^{-1}(x) = \pi - sin^{-1}\sqrt{1-x^2}, if\ -1 \leq x < 0$
6) $tan^{-1}x = tan^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right), if\ x > 0$

PROPERTY X : −

1) $3sin^{-1}x = sin^{-1}(3x - 4x^3), x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
2) $3cos^{-1}x = cos^{-1}(4x^3 - 3x), x \in \left[\frac{1}{2}, 1\right]$

EXERCISE 4.1

1.Find all the values of x such that

$(i) -10\pi \leq x \leq 10\pi$ and $\sin x = 0$ $(ii) -8\pi \leq x \leq 8\pi$ and $\sin x = -1$

$(i)\ \sin x = 0 \Longrightarrow x = n\pi, n \in z.$

$-10\pi \leq x \leq 10\pi \Longrightarrow\ x = n\pi, n = 0, \pm1, \pm2, ...$

$(ii)\ \sin x = -1$

$\Longrightarrow sinx = -sin\left(\frac{\pi}{2}\right)$

$\Longrightarrow x = n\pi + (-1)^n\left(-\frac{\pi}{2}\right), n \in z$

$\Longrightarrow x = n\pi - (-1)^{n+1}\left(\frac{\pi}{2}\right), n \in I$

$\Longrightarrow x = -\frac{\pi}{2}, \pi + \frac{\pi}{2}, -\pi + \frac{\pi}{2}, 2\pi - \frac{\pi}{2}, -2\pi - \frac{\pi}{2}, 3\pi + \frac{\pi}{2}, -3\pi + \frac{\pi}{2}, 4\pi - \frac{\pi}{2},$

$-4\pi - \frac{\pi}{2}, 5\pi + \frac{\pi}{2}, -5\pi + \frac{\pi}{2}, 6\pi - \frac{\pi}{2}, -6\pi - \frac{\pi}{2}, 7\pi + \frac{\pi}{2}, -7\pi + \frac{\pi}{2},$

$8\pi - \frac{\pi}{2}, -8\pi - \frac{\pi}{2}$.

$\Rightarrow x = -\frac{\pi}{2}, \frac{3\pi}{2}, -\frac{5\pi}{2}, \frac{7\pi}{2}, -\frac{9\pi}{2}, \frac{11\pi}{2}, -\frac{13\pi}{2}, \frac{15\pi}{2} \quad \left(-\frac{17\pi}{2} \notin [-8\pi, 8\pi]\right)$

$\Rightarrow x = -\frac{13\pi}{2}, -\frac{9\pi}{2}, -\frac{5\pi}{2}, -\frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}, \frac{15\pi}{2}$

$\Rightarrow (4n-1)\frac{\pi}{2}, n = 0, \pm1, \pm2, \pm3, 4$

2.Find the period and amplitude of $(i) y = \sin 7x \quad (ii) y = -\sin\left(\frac{1}{3}x\right) \quad (iii) y = 4\sin(-2x)$

Period and amplitude of $y = a\sin bx$ are $\frac{2\pi}{|b|}$ and $|a|$

$(i) y = \sin 7x$

$a = 1, b = 7$

Period $= \frac{2\pi}{|7|} = \frac{2\pi}{7}$; Amplitude $= |a| = |1| = 1$

$(ii) y = -\sin\left(\frac{1}{3}x\right)$

$a = -1, b = \frac{1}{3}$

Period $= \frac{2\pi}{\left|\frac{1}{3}\right|} = \frac{2\pi}{\frac{1}{3}} = 6\pi$; Amplitude $= |a| = |-1| = 1$

$(iii) y = 4\sin(-2x)$

$a = 4, b = -2$

Period $= \frac{2\pi}{|-2|} = \frac{2\pi}{2} = \pi$; Amplitude $= |a| = |4| = 4$

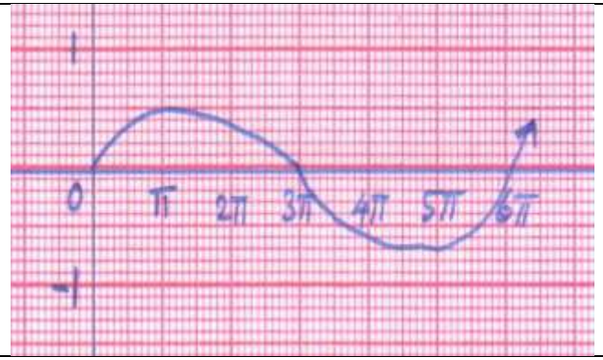
3.Sketch the graph of $y = \sin\left(\frac{1}{3}x\right)$ $for\ 0 \le x \le 6\pi$.

$y = \sin\left(\frac{1}{3}x\right)$

$a = 1, b = \frac{1}{3}$

Period $= \frac{2\pi}{\left|\frac{1}{3}\right|} = \frac{2\pi}{\frac{1}{3}} = 6\pi$; Amplitude= $|a| = |1| = 1$

x	0	π	2π	3π	4π	5π	6π
y	0	$^1/_2$	$^{\sqrt{3}}/_2$	0	$-^{\sqrt{3}}/_2$	$-^{\sqrt{3}}/_2$	0



4.Find the value of $(i) sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right) \quad (ii)\ sin^{-1}\left(\sin\left(\frac{5\pi}{4}\right)\right)$

$(i) sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$

$= sin^{-1}\left(\sin\left(\pi - \frac{\pi}{3}\right)\right) = sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$(ii)\ sin^{-1}\left(\sin\left(\frac{5\pi}{4}\right)\right)$

$= sin^{-1}\left(\sin\left(\pi + \frac{\pi}{4}\right)\right) = sin^{-1}\left(-\sin\left(\frac{\pi}{4}\right)\right) = sin^{-1}\left(\sin\left(-\frac{\pi}{4}\right)\right) = -\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

5.For what value of x does $\sin x = sin^{-1}x$?

$\sin x = 0$ and $sin^{-1}x = 0$

$\Rightarrow x = 0$ is the solution.

6.Find the domain of the following.

$(i) f(x) = sin^{-1}\left(\frac{x^2+1}{2x}\right) \quad (ii) g(x) = 2sin^{-1}(2x-1) - \frac{\pi}{4}$

Domain of $sin^{-1}\ is\ [-1,1]$

So, $-1 \le \frac{x^2+1}{2x} \le 1$

$\Rightarrow -2x \le x^2 + 1 \le 2x$

$\Rightarrow -2x \le x^2 + 1$	$\Rightarrow x^2 + 1 \le 2x$
$\Rightarrow 0 \le x^2 + 1 + 2x$	$\Rightarrow x^2 + 1 - 2x \le 0$
$\Rightarrow 0 \le (x+1)^2$	$\Rightarrow (x-1)^2 \le 0$
$\Rightarrow (x+1)^2 \ge 0$	$\Rightarrow x - 1 \le 0$
$\Rightarrow x + 1 \ge 0$	$\Rightarrow x \le 1 \rightarrow (2)$
$\Rightarrow x \ge -1 \rightarrow (1)$	

From (1) & (2) $\Rightarrow$ $-1 \le x \le 1$

$\therefore Domain\ of f(x) = sin^{-1}\left(\frac{x^2+1}{2x}\right)\ is\ [-1,1].$

(ii) Domain of $sin^{-1}\ is\ [-1,1]$

So, $-1 \le 2x-1 \le 1$

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$\Rightarrow -1 \le 2x-1$	$\Rightarrow 2x-1 \le 1$
$\Rightarrow 0 \le 2x-1+1$	$\Rightarrow 2x \le 2$
$\Rightarrow 0 \le 2x$	$\Rightarrow x \le 1 \rightarrow (2)$
$\Rightarrow x \ge 0 \rightarrow (1)$	

From (1) & (2) $\Rightarrow 0 \le x \le 1$

$\therefore Domain\ of g(x) = 2sin^{-1}(2x-1) - \frac{\pi}{4}\ is\ [0,1]$

7.Find the value of $sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9} + \cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$

$$sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9} + \cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$$
$$= sin^{-1}\left(\sin\left(\frac{5\pi}{9}+\frac{\pi}{9}\right)\right) \qquad (\because \sin(A+B) = sinAcosB + cosAsinB)$$
$$= sin^{-1}\left(\sin\left(\frac{6\pi}{9}\right)\right)$$
$$= sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$$
$$= sin^{-1}\left(\sin\left(\pi-\frac{\pi}{3}\right)\right)$$
$$= sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$$
$$= \frac{\pi}{3} \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$$

EXAMPLE 4.1 Find the principal value of $sin^{-1}\left(-\frac{1}{2}\right)$ (in radians and degrees).

$y = sin^{-1}\left(-\frac{1}{2}\right) = sin^{-1}\left(-sin\left(\frac{\pi}{6}\right)\right) = sin^{-1}\left(sin\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6}$

So, $y = -\frac{\pi}{6} \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]. y = -\frac{\pi}{6} = -30^0.$

EXAMPLE 4.2 Find the principal value of $sin^{-1}(2)$, if it exists.

Domain of $sin^{-1}\ is\ [-1,1]$. But $2 \notin [-1,1]$. So $sin^{-1}(2)$ does not exists.

EXAMPLE 4.3 Find the principal value of

$(i) sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\ (ii) sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right)\ (iii) sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$

$(i) y = sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = sin^{-1}\left(sin\left(\frac{\pi}{4}\right)\right) = \frac{\pi}{4} \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$(ii) sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3} \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$(iii) y = sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right) = sin^{-1}\left(\sin\left(\pi-\frac{\pi}{6}\right)\right)$
$$= sin^{-1}\left(\sin\left(\frac{\pi}{6}\right)\right)$$
$$= \frac{\pi}{6} \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$$

EXAMPLE 4.4 Find the domain of $sin^{-1}(2-3x^2)$.

Domain of $sin^{-1}\ is\ [-1,1]$. So, $-1 \le 2-3x^2 \le 1$.

$-1 \le 2-3x^2$	$2-3x^2 \le 1$
$\Rightarrow -3 \le -3x^2$	$\Rightarrow -3x^2 \le -1$
$\Rightarrow 3x^2 \le 3$	$\Rightarrow 3x^2 \ge 1$
$\Rightarrow x^2 \le 1 \rightarrow (1)$	$\Rightarrow x^2 \ge \frac{1}{3} \rightarrow (2)$

From(1) & (2), $\frac{1}{3} \le x^2 \le 1 \Rightarrow \frac{1}{\sqrt{3}} \le |x| \le 1 \Rightarrow x \in \left[-1,-\frac{1}{\sqrt{3}}\right] \cup \left[\frac{1}{\sqrt{3}},1\right].$

$(\because If\ a \le |x| \le b, then\ \ x \in [-b,-a] \cup [a,b])$

EXERCISE 4.2

1.Find all value of x such that

$(i) -6\pi \le x \le 6\pi$ and $\cos x = 0$ $(ii) -5\pi \le x \le 5\pi$ and $\cos x = 1$

$(i) \cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}, n \in z$

$\cos x = (2n+1)\frac{\pi}{2}, n = 0, \pm1, \pm2, \pm3, \pm4, \pm5.$

$(ii) \cos x = 1 \Rightarrow \cos x = \cos 0 \Rightarrow \alpha = 0$

$\Rightarrow x = 2n\pi \pm \alpha = 2n\pi, n = 0, \pm1, \pm2$

2.State the reason for $cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \ne -\frac{\pi}{6}$.

$cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6} \notin [0,\pi] \qquad (\because cos^{-1}(\cos(x)) = x)$

So, $cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \ne -\frac{\pi}{6}.$

3.Is $cos^{-1}(-x) = \pi - cos^{-1}(x)$ true? Justify your answer.

$cos^{-1}(-x) = \pi - cos^{-1}(x)$ is true.

$cos^{-1}(-x) = \alpha$

$\Rightarrow -x = \cos\alpha$

$\Rightarrow x = -\cos\alpha$

$\Rightarrow x = \cos(\pi - \alpha)$

$\Rightarrow cos^{-1}x = \pi - \alpha$

$\Rightarrow \alpha = \pi - cos^{-1}x$

$\Rightarrow cos^{-1}(-x) = \pi - cos^{-1}(x)$

4.Find the principal value of $cos^{-1}\left(\frac{1}{2}\right)$.

$cos^{-1}\left(\frac{1}{2}\right) = cos^{-1}\left(cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3} \in [0,\pi]$

$\therefore$ The principal value of $cos^{-1}\left(\frac{1}{2}\right)$ is $\frac{\pi}{3}$.

5.Find the value of $(i)2cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}\left(\frac{1}{2}\right)$

$(ii)cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}(-1)$ $(iii)cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17} - \sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$

$(i)2cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}\left(\frac{1}{2}\right)$

$= 2\left(\frac{\pi}{3}\right) + \left(\frac{\pi}{6}\right)$

$= \frac{4\pi+\pi}{6} = \frac{5}{6}$

$(ii)cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}(-1)$

$= cos^{-1}\left(\frac{1}{2}\right) - sin^{-1}(1)$

$= \frac{\pi}{3} - \frac{\pi}{2}$

$= \frac{2\pi-3\pi}{6} = -\frac{\pi}{6}$

$(iii)cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17} - \sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$

$= cos^{-1}\left(\cos\left(\frac{\pi}{7} + \frac{\pi}{17}\right)\right)$

$= cos^{-1}\left(\cos\left(\frac{17\pi+7\pi}{119}\right)\right)$

$= cos^{-1}\left(\cos\left(\frac{24\pi}{119}\right)\right)$

$= \frac{24\pi}{119}$

6.Find the domain of $(i)f(x) = sin^{-1}\left(\frac{|x|-2}{3}\right) + cos^{-1}\left(\frac{1-|x|}{4}\right)$

$(ii)g(x) = sin^{-1}x + cos^{-1}(x)$

The domain of $sin^{-1}x$ is $[-1,1]$ and the domain of $cos^{-1}x$ is $[-1,1]$.

So, $-1 \le \frac{|x|-2}{3} \le 1$; $-1 \le \frac{1-|x|}{4} \le 1$

$-1 \le \frac{\lvert x\rvert-2}{3} \le 1$	$-1 \le \frac{1-\lvert x\rvert}{4} \le 1$
$\Rightarrow -3 \le \lvert x\rvert - 2 \le 3$	$\Rightarrow -4 \le 1 - \lvert x\rvert \le 4$
$\Rightarrow -1 \le \lvert x\rvert \le 5$	$\Rightarrow -5 \le -\lvert x\rvert \le 3$
$\Rightarrow \lvert x\rvert \le 5$	$\Rightarrow -3 \le \lvert x\rvert \le 5$
	$\Rightarrow \lvert x\rvert \le 5$

$\therefore x \in [-5,5]$

(ii) The domain of $sin^{-1}x$ and $cos^{-1}x$ is$[-1,1]$.

Hence the domain of the given function is $[-1,1]$.

7.For what value of x, the inequality $\frac{\pi}{2} < cos^{-1}(3x-1) < \pi$ holds?

$\frac{\pi}{2} < cos^{-1}(3x-1) < \pi$

$cos^{-1}x$ lies in the second quadrant, if $-1 < x < 0$.

$-1 < 3x - 1 < 0 \Rightarrow 0 < 3x < 1 \Rightarrow 0 < x < \frac{1}{3}$

$\therefore x \in \left(0, \frac{1}{3}\right)$

8. Find the value of $(i)\cos\left(cos^{-1}\left(\frac{4}{5}\right) + sin^{-1}\left(\frac{4}{5}\right)\right)$

$(ii)cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right) + cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)\right)$

$(i)\cos\left(cos^{-1}\left(\frac{4}{5}\right) + sin^{-1}\left(\frac{4}{5}\right)\right)$

Let $sin^{-1}\left(\frac{4}{5}\right) = \alpha$.

$\Rightarrow \sin\alpha = \frac{4}{5}$

$\Rightarrow \cos\left(\frac{\pi}{2} - \alpha\right) = \frac{4}{5}$

$\Rightarrow \frac{\pi}{2} - \alpha = cos^{-1}\left(\frac{4}{5}\right)$

$\Rightarrow \frac{\pi}{2} = cos^{-1}\left(\frac{4}{5}\right) + \alpha$

$\Rightarrow cos^{-1}\left(\frac{4}{5}\right) + sin^{-1}\left(\frac{4}{5}\right) = \frac{\pi}{2}$

$\therefore \cos\left(cos^{-1}\left(\frac{4}{5}\right) + sin^{-1}\left(\frac{4}{5}\right)\right) = \cos\frac{\pi}{2} = 0$

$(ii)cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right) + cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)\right)$

$= cos^{-1}\left(\cos\left(\pi + \frac{\pi}{3}\right) + cos^{-1}\left(\cos\left(\pi + \frac{\pi}{4}\right)\right)\right)$

$= cos^{-1}\left(-\cos\left(\frac{\pi}{3}\right) + cos^{-1}\left(-\cos\left(\frac{\pi}{4}\right)\right)\right)$

$= cos^{-1}\left(\cos\left(\pi - \frac{\pi}{3}\right) + cos^{-1}\left(\cos\left(\pi - \frac{\pi}{4}\right)\right)\right)$

$= cos^{-1}\left(\cos\left(\frac{2\pi}{3}\right) + cos^{-1}\left(\cos\left(\frac{3\pi}{4}\right)\right)\right)$

$= \frac{2\pi}{3} + \frac{3\pi}{4}$

$= \frac{8\pi+9\pi}{12}$

$= \frac{17\pi}{12}$

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EXAMPLE 4.5 Find the principal value of $cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

The range of the principal values of $cos^{-1}(x)\ is [0, \pi]$.

$y = cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = cos^{-1}\left(cos\left(\frac{\pi}{6}\right)\right) = \frac{\pi}{6} \in [0, \pi]$

$\therefore$ The principal value of $cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is $\frac{\pi}{6}$.

EXAMPLE 4.6 Find $(i) cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)\ (ii) cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right)$

$(iii) cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$

Inverse cosine function $cos^{-1}: [-1,1] \to [0, \pi]$.

$(i) cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = cos^{-1}\left(-cos\left(\frac{\pi}{4}\right)\right)$

$= cos^{-1}\left(\cos\left(\pi - \frac{\pi}{4}\right)\right)$

$= \pi - \frac{\pi}{4}$

$= \frac{3\pi}{4} \in [0, \pi]$

$(ii) cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right) = cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right)$

$= \frac{\pi}{3} \in [0, \pi]$

$(iii) cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right) = cos^{-1}\left(\cos\left(\pi + \frac{\pi}{6}\right)\right)$

$= cos^{-1}\left(-\cos\left(\frac{\pi}{6}\right)\right)$

$= cos^{-1}\left(\cos\left(\pi - \frac{\pi}{6}\right)\right)$

$= \pi - \frac{\pi}{6}$

$= \frac{5\pi}{6} \in [0, \pi]$

EXAMPLE 4.7 Find the domain of $cos^{-1}\left(\frac{2+\sin x}{3}\right)$

The domain of $cos^{-1}x$ is $[-1,1]$. So $-1 \le \frac{2+\sin x}{3} \le 1$

$\Rightarrow -3 \le 2 + \sin x \le 3$

$\Rightarrow -5 \le sinx \le 1$

$\Rightarrow -1 \le sinx \le 1$

$\Rightarrow -sin^{-1}(1) \le x \le sin^{-1}(1)$

$\Rightarrow -\frac{\pi}{2} \le x \le \frac{\pi}{2}$

$\therefore$ The domain of $cos^{-1}\left(\frac{2+\sin x}{3}\right)\ is\ \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

EXERCISE 4.3

1. Find the domain of the following functions :

$(i) tan^{-1}\left(\sqrt{9-x^2}\right)\ (ii) \frac{1}{2} tan^{-1}(1-x^2) - \frac{\pi}{4}$

(i)The domain of $tan^{-1}x\ is\ (-\infty, \infty)$.

If $\sqrt{9-x^2}$ is real, then $\sqrt{9-x^2} \ge 0$

$\Rightarrow 9 - x^2 \ge 0$

$\Rightarrow 9 \ge x^2$

$\Rightarrow x^2 \le 9$

$\Rightarrow -3 \le x \le 3$

So, The domain of $tan^{-1}\left(\sqrt{9-x^2}\right)\ is\ [-3,3]$.

(ii)The domain of $tan^{-1}x\ is\ (-\infty, \infty)$.

For all values of $x, 1-x^2$ is real.

So,the domain of $\frac{1}{2} tan^{-1}(1-x^2) - \frac{\pi}{4}\ is\ R$.

2.Find the value of $(i) tan^{-1}\left(\tan\frac{5\pi}{4}\right)\ (ii) tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right)$

$(i) tan^{-1}\left(\tan\frac{5\pi}{4}\right) = tan^{-1}\left(\tan\left(\pi + \frac{\pi}{4}\right)\right) = tan^{-1}\left(\tan\left(\frac{\pi}{4}\right)\right) = \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$(ii) tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

3. Find the value of $(i) \tan\left(tan^{-1}\left(\frac{7\pi}{4}\right)\right)\ (ii) \tan\left(tan^{-1}(1947)\right)$

$(iii) \tan\left(tan^{-1}(-0.2021)\right)$

$(i)\ \tan\left(tan^{-1}\left(\frac{7\pi}{4}\right)\right) = \frac{7\pi}{4}$

$(ii)\ \tan\left(tan^{-1}(1947)\right) = 1947$

$(iii)\ \tan\left(tan^{-1}(-0.2021)\right) = -0.2021$

4. Find the value of $(i)\ \tan\left(cos^{-1}\left(\frac{1}{2}\right) - sin^{-1}\left(-\frac{1}{2}\right)\right)$

$(ii)\ \sin\left(tan^{-1}\left(\frac{1}{2}\right) - cos^{-1}\left(\frac{4}{5}\right)\right)$ $(iii)\ \cos\left(sin^{-1}\left(\frac{4}{5}\right) - tan^{-1}\left(\frac{3}{4}\right)\right)$

$(i)\ \tan\left(cos^{-1}\left(\frac{1}{2}\right) - sin^{-1}\left(-\frac{1}{2}\right)\right)$

$= \tan\left(\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right)$

$= \tan\left(\frac{\pi}{3} + \frac{\pi}{6}\right)$

$= \tan\left(\frac{\pi}{2}\right)$

$= \infty$

$(ii)\ \sin\left(tan^{-1}\left(\frac{1}{2}\right) - cos^{-1}\left(\frac{4}{5}\right)\right)$

$sin(x-y) = sinxcosy - cosxsiny$. Let $tan^{-1}\left(\frac{1}{2}\right) = x\ ; cos^{-1}\left(\frac{4}{5}\right) = y$.

$tan^{-1}\left(\frac{1}{2}\right) = x \Longrightarrow \tan x = \frac{1}{2}$

$sec^2x = 1 + tan^2x = 1 + \frac{1}{4} = \frac{5}{4} \Longrightarrow secx = \frac{\sqrt{5}}{2} \Longrightarrow cosx = \frac{2}{\sqrt{5}}$

$sinx = \sqrt{1 - cos^2x} = \sqrt{1 - \frac{4}{5}} = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}}$

$cos^{-1}\left(\frac{4}{5}\right) = y \ \Longrightarrow cosy = \frac{4}{5}$

$siny = \sqrt{1 - cos^2y} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

$\sin\left(tan^{-1}\left(\frac{1}{2}\right) - cos^{-1}\left(\frac{4}{5}\right)\right) = sin(x-y)$

$= sinxcosy - cosxsiny$

$= \frac{1}{\sqrt{5}} \cdot \frac{4}{5} - \frac{2}{\sqrt{5}} \cdot \frac{3}{5}$

$= \frac{4-6}{5\sqrt{5}}$

$= -\frac{2}{5\sqrt{5}}$

$= -\frac{2}{5\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}}$

$= -\frac{2\sqrt{5}}{25}$

$(iii)\ \cos\left(sin^{-1}\left(\frac{4}{5}\right) - tan^{-1}\left(\frac{3}{4}\right)\right)$

$cos(x-y) = cosxcosy + sinxsiny$. Let $sin^{-1}\left(\frac{4}{5}\right) = x\ ; tan^{-1}\left(\frac{3}{4}\right) = y$.

$sin^{-1}\left(\frac{4}{5}\right) = x \Longrightarrow sinx = \frac{4}{5}$

$cosx = \sqrt{1 - sin^2x} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

$tan^{-1}\left(\frac{3}{4}\right) = y \Longrightarrow tany = \frac{3}{4}$

$sec^2y = 1 + tan^2y = 1 + \frac{9}{16} = \frac{25}{16} \Longrightarrow secy = \frac{5}{4} \Longrightarrow cosy = \frac{4}{5}$

$siny = \sqrt{1 - cos^2y} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

$\cos\left(sin^{-1}\left(\frac{4}{5}\right) - tan^{-1}\left(\frac{3}{4}\right)\right) = cos(x-y)$

$= cosxcosy + sinxsiny$

$= \frac{3}{5} \cdot \frac{4}{5} + \frac{4}{5} \cdot \frac{3}{5}$

$= \frac{12+12}{25}$

$= \frac{25}{25}$

EXAMPLE 4.8 Find the principal value of $tan^{-1}\left(\sqrt{3}\right)$.

$y = tan^{-1}\left(\sqrt{3}\right) = tan^{-1}\left(tan\left(\frac{\pi}{3}\right)\right) \Longrightarrow y = \frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

EXAMPLE 4.9 Find $(i) tan^{-1}\left(-\sqrt{3}\right)$ $(ii) tan^{-1}\left(\tan\frac{3\pi}{5}\right)$

$(iii)\ \tan\left(tan^{-1}(2019)\right)$

$(i) tan^{-1}\left(-\sqrt{3}\right) = tan^{-1}\left(tan\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$(ii) tan^{-1}\left(\tan\left(\frac{3\pi}{5}\right)\right) = tan^{-1}\left(\tan\left(\frac{3\pi}{5} - \pi\right)\right)$

$= tan^{-1}\left(\tan\left(-\frac{2\pi}{5}\right)\right)$

$= -\frac{2\pi}{5} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

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$(iii)\ \tan\left(tan^{-1}(2019)\right) = 2019$

EXAMPLE 4.10 Find the value of $tan^{-1}(-1) + cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}\left(-\frac{1}{2}\right)$

$tan^{-1}(-1) = tan^{-1}\left(-tan\frac{\pi}{4}\right) = tan^{-1}\left(tan\left(-\frac{\pi}{4}\right)\right) = -\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$cos^{-1}\left(\frac{1}{2}\right) = cos^{-1}\left(cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3} \in [0, \pi]$

$sin^{-1}\left(-\frac{1}{2}\right) = sin^{-1}\left(-sin\frac{\pi}{6}\right) = sin^{-1}\left(sin\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$tan^{-1}(-1) + cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{4} + \frac{\pi}{3} - \frac{\pi}{6} = -\frac{\pi}{12}$

EXAMPLE 4.11 Prove that $\tan\left(sin^{-1}(x)\right) = \frac{x}{\sqrt{1-x^2}}, -1 < x < 1.$

If $x = 0$ then both sides are equal to $0 \rightarrow (1)$

Assume that $0 < x < 1$. Let $\theta = sin^{-1}x$. Then $0 < \theta < \frac{\pi}{2}$. Now

$\theta = sin^{-1}x \Rightarrow \sin\theta = \frac{x}{1} \Rightarrow$ Opposie side $= x$, Hypotenuse $= 1$

Adjacent side $= \sqrt{1-x^2}$.

$\tan\theta = \frac{x}{\sqrt{1-x^2}} \Rightarrow \tan\left(sin^{-1}(x)\right) = \frac{x}{\sqrt{1-x^2}} \rightarrow (2)$

Let $-1 < x < 0$. *Then* $\theta = sin^{-1}x$ gives $-\frac{\pi}{2} < \theta < 0$.

$\theta = sin^{-1}x \Rightarrow \sin\theta = \frac{x}{1} \Rightarrow \tan\theta = \frac{x}{\sqrt{1-x^2}} \Rightarrow \tan\left(sin^{-1}(x)\right) = \frac{x}{\sqrt{1-x^2}} \rightarrow (3)$

From $(1), (2), (3)$, $\tan\left(sin^{-1}(x)\right) = \frac{x}{\sqrt{1-x^2}}, -1 < x < 1$

EXERCISE 4.4

1.Find the principal value of $(i)sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ $(ii)cot^{-1}\left(\sqrt{3}\right)$ $(iii)cosec^{-1}\left(-\sqrt{2}\right)$

$(i)y = sec^{-1}\left(\frac{2}{\sqrt{3}}\right) \Rightarrow secy = \frac{2}{\sqrt{3}}$

$\Rightarrow \frac{1}{cosy} = \frac{2}{\sqrt{3}}$

$\Rightarrow cosy = \left(\frac{\sqrt{3}}{2}\right)$

$\Rightarrow y = cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$= cos^{-1}\left(cos\frac{\pi}{6}\right)$

$= \frac{\pi}{6} \in [0, \pi]$

$(ii)y = cot^{-1}\left(\sqrt{3}\right) \Rightarrow coty = \sqrt{3}$

$\Rightarrow \frac{1}{tany} = \sqrt{3}$

$\Rightarrow tany = \left(\frac{1}{\sqrt{3}}\right)$

$\Rightarrow y = tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$= tan^{-1}\left(tan\left(\frac{\pi}{6}\right)\right)$

$= \frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$(iii)y = cosec^{-1}\left(-\sqrt{2}\right) \Rightarrow cosecy = -\sqrt{2}$

$\Rightarrow \frac{1}{siny} = -\sqrt{2}$

$\Rightarrow siny = -\frac{1}{\sqrt{2}}$

$\Rightarrow y = sin\left(-\frac{1}{\sqrt{2}}\right)$

$= sin\left(-sin\left(\frac{\pi}{4}\right)\right)$

$= sin\left(sin\left(-\frac{\pi}{4}\right)\right)$

$= -\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2.Find the value of $(i)tan^{-1}\left(\sqrt{3}\right) - sec^{-1}(-2)$

$(ii)sin^{-1}(-1) + cos^{-1}\left(\frac{1}{2}\right) + cot^{-1}(2)$

$(iii)cot^{-1}(1) + sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - sec^{-1}\left(-\sqrt{2}\right)$

$(i)\ \ tan^{-1}\left(\sqrt{3}\right) - sec^{-1}(-2)$

$= tan^{-1}\left(\tan\left(\frac{\pi}{3}\right)\right) - cos\left(-\frac{1}{2}\right)$

$= tan^{-1}\left(\tan\left(\frac{\pi}{3}\right)\right) - \cos\left(-cos\left(\frac{\pi}{3}\right)\right)$

$= tan^{-1}\left(\tan\left(\frac{\pi}{3}\right)\right) - \cos\left(cos\left(\pi - \frac{\pi}{3}\right)\right)$

$= \frac{\pi}{3} - \left(\pi - \frac{\pi}{3}\right)$

$= \frac{\pi}{3} - \frac{2\pi}{3}$

$= -\frac{\pi}{3}$

$(ii) sin^{-1}(-1) + cos^{-1}\left(\frac{1}{2}\right) + cot^{-1}(2)$

$= sin^{-1}\left(sin\left(-\frac{\pi}{2}\right)\right) + cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right) + cot^{-1}(2)$

$= -\frac{\pi}{2} + \frac{\pi}{3} + cot^{-1}(2)$

$= -\frac{\pi}{6} + cot^{-1}(2)$

$(iii) cot^{-1}(1) + sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - sec^{-1}(-\sqrt{2})$

$= \tan(1) + sin^{-1}\left(-\sin\left(\frac{\sqrt{3}}{2}\right)\right) - \cos\left(-\frac{1}{\sqrt{2}}\right)$

$= \tan\left(\frac{\pi}{4}\right) + sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) - \cos\left(\pi - \frac{\pi}{4}\right)$

$= \frac{\pi}{4} - \frac{\pi}{3} - \frac{3\pi}{4}$

$= -\frac{2\pi}{4} - \frac{\pi}{3}$

$= -\frac{\pi}{2} - \frac{\pi}{3}$

$= -\frac{5\pi}{6}$

EXAMPLE 4.12 Find the principal value of $(i) cosec^{-1}(-1)$ $(ii) sec^{-1}(-2)$

$(i) cosec^{-1}(-1) = sin(-1) = -sin(1) \Rightarrow -\frac{\pi}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$(ii) sec^{-1}(-2) = cos\left(-\frac{1}{2}\right) \Rightarrow \pi - \frac{\pi}{3} = \frac{2\pi}{3} \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$

EXAMPLE 4.13 Find the value of $sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right)$

$sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right) = sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = cos\left(-\frac{\sqrt{3}}{2}\right) \Rightarrow \pi - \frac{\pi}{6} = \frac{5\pi}{6} \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$

EXAMPLE 4.14 If $cot^{-1}\left(\frac{1}{7}\right) = \theta$, Find the value of $\cos\theta$.

$cot^{-1}\left(\frac{1}{7}\right) = \theta$

$\Rightarrow \cot\theta = \frac{1}{7} \Rightarrow \tan\theta = \frac{7}{1} \Rightarrow Opposite\ side = 7\ ; adjacent\ side = 1$

$Hypotenuse = \sqrt{7^2 + 1^2} = \sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$

So, $\cos\theta = \frac{1}{5\sqrt{2}}$

EXAMPLE 4.15 Show that $cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = sec^{-1}x, |x| > 1$

$Let\ cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = y \rightarrow (1)$

$\Rightarrow coty = \frac{1}{\sqrt{x^2-1}}$

$\Rightarrow tany = \frac{\sqrt{x^2-1}}{1};\quad Opposite\ side\ = \sqrt{x^2-1}\ ;\ adjacent\ side\ = 1$

$Hypotenuse\ = \sqrt{\left(\sqrt{x^2-1}\right)^2 + (1)^2} = \sqrt{x^2-1+1} = \sqrt{x^2} = x$

$secy = \frac{x}{1} \Rightarrow sec^{-1}x = y \rightarrow (2)$

$(1), (2) \Rightarrow cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = sec^{-1}x$

EXERCISE 4.5

1.Find the value, if it exists. If not, give the reason for non-existence

$(i) sin^{-1}(\cos\pi)$ $(ii) tan^{-1}\left(\sin\left(-\frac{5\pi}{2}\right)\right)$ $(iii) sin^{-1}(\sin 5)$

$(i) sin^{-1}(\cos\pi) = sin^{-1}(-1) = sin^{-1}\left(-sin\left(\frac{\pi}{2}\right)\right) = sin^{-1}\left(sin\left(-\frac{\pi}{2}\right)\right) = -\frac{\pi}{2}$

$(ii) tan^{-1}\left(\sin\left(-\frac{5\pi}{2}\right)\right) = tan^{-1}\left(-\sin\left(\frac{5\pi}{2}\right)\right)$

$= tan^{-1}\left(-\sin\left(2\pi + \frac{\pi}{2}\right)\right)$

$= tan^{-1}\left(-\sin\left(\frac{\pi}{2}\right)\right)$

$= tan^{-1}(-1)$

$= tan^{-1}\left(-tan\left(\frac{\pi}{4}\right)\right)$

$= tan^{-1}\left(tan\left(-\frac{\pi}{4}\right)\right)$

$= -\frac{\pi}{4}$

$(iii) sin^{-1}(\sin 5) = sin^{-1}\left[sin(2\pi - (2\pi - 5))\right]$

$= sin^{-1}[-sin(2\pi - 5)]$

$= -sin^{-1}[sin(2\pi - 5)]$

$= -(2\pi - 5)$

$= 5 - 2\pi$

2.Find the value of the expression in terms of x, with the help of a reference triangle.

$(i)\sin\left(cos^{-1}(1-x)\right)\quad (ii)\cos\left(tan^{-1}(3x-1)\right)\quad (iii)\tan\left(sin^{-1}\left(x+\frac{1}{2}\right)\right)$

$(i)\sin\left(cos^{-1}(1-x)\right)$

$Let\ y = cos^{-1}(1-x) \rightarrow (1)$

$\Rightarrow cosy = 1-x$

$\Rightarrow siny = \sqrt{1-cos^2y}$

$= \sqrt{1-(1-x)^2}$

$= \sqrt{1-(1-2x+x^2)}$

$= \sqrt{1-1+2x-x^2}$

$= \sqrt{2x-x^2}$

$\Rightarrow \sin\left(cos^{-1}(1-x)\right) = \sqrt{2x-x^2}$ $\quad (From\ (1)\ )$

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$(ii)\cos\left(tan^{-1}(3x-1)\right)$

$Let\ y = tan^{-1}(3x-1) \rightarrow (1).$

$\Rightarrow tany = 3x-1$

$\Rightarrow secy = \sqrt{1+tan^2y}$

$= \sqrt{1+(3x-1)^2}$

$= \sqrt{1+(9x^2-6x+1)}$

$= \sqrt{1+9x^2-6x+1}$

$= \sqrt{9x^2-6x+2}$

$\Rightarrow cosy = \frac{1}{\sqrt{9x^2-6x+2}}$

$\Rightarrow \cos\left(tan^{-1}(3x-1)\right) = \frac{1}{\sqrt{9x^2-6x+2}}$ $\quad (\ From\ (1)\ )$

$(iii)\tan\left(sin^{-1}\left(x+\frac{1}{2}\right)\right)$

$Let\ y = sin^{-1}\left(x+\frac{1}{2}\right) \rightarrow (1).$

$\Rightarrow siny = x+\frac{1}{2} \Rightarrow cosy = \sqrt{1-sin^2y}$

$= \sqrt{1-\left(x+\frac{1}{2}\right)^2}$

$= \sqrt{1-x^2-x-\frac{1}{4}}$

$= \sqrt{\frac{4-4x^2-4x-1}{4}}$

$= \frac{\sqrt{3-4x^2-4x}}{2}$

$tany = \frac{siny}{cosy} = \frac{x+\frac{1}{2}}{\frac{\sqrt{3-4x^2-4x}}{2}} = \frac{(2x+1)/2}{(\sqrt{3-4x^2-4x})/2} = \frac{2x+1}{\sqrt{3-4x^2-4x}}$

$\Rightarrow \tan\left(sin^{-1}\left(x+\frac{1}{2}\right)\right) = \frac{2x+1}{\sqrt{3-4x^2-4x}}$ $\quad (\ From\ (1)\ )$

3.Find the value of $(i)sin^{-1}\left(\cos\left(sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right)$

$(ii)\cot\left(sin^{-1}\frac{3}{5}+sin^{-1}\frac{4}{5}\right)\quad (iii)\tan\left(sin^{-1}\frac{3}{5}+cot^{-1}\frac{3}{2}\right)$

$(i)sin^{-1}\left(\cos\left(sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right) = sin^{-1}\left(\cos\left(sin^{-1}\left(sin\frac{\pi}{3}\right)\right)\right)$

$= sin^{-1}\left(cos\frac{\pi}{3}\right)$

$= sin^{-1}\left(\frac{1}{2}\right)$

$= sin^{-1}\left(sin\frac{\pi}{6}\right)$

$= \frac{\pi}{6}$

$(ii)\cot\left(sin^{-1}\frac{3}{5}+sin^{-1}\frac{4}{5}\right)$

$sin^{-1}x+sin^{-1}y = sin^{-1}\left[x\sqrt{1-y^2}+y\sqrt{1-x^2}\right]$

$sin^{-1}\frac{3}{5}+sin^{-1}\frac{4}{5} = sin^{-1}\left[\frac{3}{5}\sqrt{1-\left(\frac{4}{5}\right)^2}+\frac{4}{5}\sqrt{1-\left(\frac{3}{5}\right)^2}\right]$

$= sin^{-1}\left[\frac{3}{5}\sqrt{\frac{9}{25}}+\frac{4}{5}\sqrt{\frac{16}{25}}\right]$

$= sin^{-1}\left[\frac{3}{5}.\frac{3}{5}+\frac{4}{5}.\frac{4}{5}\right]$

$= sin^{-1}\left[\frac{9}{25}+\frac{16}{25}\right]$

$= sin^{-1}\left[\frac{25}{25}\right]$

$= sin^{-1}[1]$

$$= sin^{-1}\left(sin\frac{\pi}{2}\right)$$
$$= \frac{\pi}{2}$$
$$\therefore \cot\left(sin^{-1}\frac{3}{5} + sin^{-1}\frac{4}{5}\right) = \cot\left(\frac{\pi}{2}\right) = 0$$

$(iii)\tan\left(sin^{-1}\frac{3}{5} + cot^{-1}\frac{3}{2}\right)$

$sin^{-1}\frac{3}{5} = x \Longrightarrow \sin x = \frac{3}{5} \Longrightarrow cosx = \sqrt{1 - sin^2x} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$

$\tan x = \frac{\sin x}{\cos x} = \frac{3/5}{4/5} = \frac{3}{4}$

$cot^{-1}\frac{3}{2} = y \Longrightarrow \cot y = \frac{3}{2} \Longrightarrow \tan y = \frac{2}{3}$

$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y} = \frac{\frac{3}{4}+\frac{2}{3}}{1-\left(\frac{3}{4}\right)\left(\frac{2}{3}\right)} = \frac{17/12}{6/12} = \frac{17}{6}$

4.Prove that $(i)tan^{-1}\frac{2}{11} + tan^{-1}\frac{7}{24} = tan^{-1}\frac{1}{2}$ $(ii)sin^{-1}\frac{3}{5} - cos^{-1}\frac{12}{13} = sin^{-1}\frac{16}{65}$

$(i)tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$$tan^{-1}\frac{2}{11} + tan^{-1}\frac{7}{24} = tan^{-1}\left(\frac{2/11 + 7/24}{1-\left(\frac{2}{11}\right)\left(\frac{7}{24}\right)}\right)$$
$$= tan^{-1}\left(\frac{\frac{48+77}{264}}{1-\frac{14}{264}}\right)$$
$$= tan^{-1}\left(\frac{125/264}{250/264}\right)$$
$$= tan^{-1}\left(\frac{125}{264}\times\frac{264}{250}\right)$$
$$= tan^{-1}\left(\frac{1}{2}\right)$$

$(ii)sin^{-1}x - sin^{-1}y = sin^{-1}\left(x\sqrt{1-y^2} - y\sqrt{1-x^2}\right)$

$y = cos^{-1}\frac{12}{13} \Longrightarrow cosy = \frac{12}{13}$

$siny = \sqrt{1-cos^2y} = \sqrt{1-\left(\frac{12}{13}\right)^2} = \sqrt{\frac{169-144}{169}} = \sqrt{\frac{25}{169}} = \frac{5}{13} \Longrightarrow y = sin^{-1}\left(\frac{5}{13}\right)$

$\therefore cos^{-1}\frac{12}{13} = sin^{-1}\left(\frac{5}{13}\right)$

$sin^{-1}\frac{3}{5} - cos^{-1}\frac{12}{13} = sin^{-1}\frac{3}{5} - sin^{-1}\left(\frac{5}{13}\right)$

$$= sin^{-1}\left(\frac{3}{5}\sqrt{1-\left(\frac{5}{13}\right)^2} - \frac{5}{13}\sqrt{1-\left(\frac{3}{5}\right)^2}\right)$$
$$= sin^{-1}\left(\frac{3}{5}\sqrt{\frac{169-25}{169}} - \frac{5}{13}\sqrt{\frac{25-9}{25}}\right)$$
$$= sin^{-1}\left(\frac{3}{5}\sqrt{\frac{144}{169}} - \frac{5}{13}\sqrt{\frac{16}{25}}\right)$$
$$= sin^{-1}\left(\frac{3}{5}.\frac{12}{13} - \frac{5}{13}.\frac{4}{5}\right)$$
$$= sin^{-1}\left(\frac{36-20}{65}\right)$$
$$= sin^{-1}\frac{16}{65}$$

5.Prove that $tan^{-1}x + tan^{-1}y + tan^{-1}z = tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$

$tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$$tan^{-1}x + tan^{-1}y + tan^{-1}z = tan^{-1}\left(\frac{x+y}{1-xy}\right) + tan^{-1}z$$
$$= tan^{-1}\left(\frac{\frac{x+y}{1-xy}+z}{1-\left(\frac{x+y}{1-xy}\right)z}\right)$$
$$= tan^{-1}\left(\frac{x+y+z-xy/1-xy}{1-xy-xz-yz/1-xy}\right)$$
$$= tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$$

6.*If* $tan^{-1}x + tan^{-1}y + tan^{-1}z = \pi$, *show that* $x + y + z = xyz$.

$tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$$tan^{-1}x + tan^{-1}y + tan^{-1}z = tan^{-1}\left(\frac{x+y}{1-xy}\right) + tan^{-1}z$$
$$= tan^{-1}\left(\frac{\frac{x+y}{1-xy}+z}{1-\left(\frac{x+y}{1-xy}\right)z}\right)$$
$$= tan^{-1}\left(\frac{x+y+z-xy/1-xy}{1-xy-xz-yz/1-xy}\right)$$
$$= tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$$

$tan^{-1}x + tan^{-1}y + tan^{-1}z = \pi \Longrightarrow tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right] = \pi$

$\Longrightarrow \frac{x+y+z-xyz}{1-xy-yz-zx} = tan\pi = 0$

$\Longrightarrow x + y + z - xyz = 0$

7.*Prove that* $tan^{-1}x + tan^{-1}\frac{2x}{1-x^2} = tan^{-1}\frac{3x-x^3}{1-3x^2}, |x| < \frac{1}{\sqrt{3}}$.

$tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)$

$tan^{-1}x + tan^{-1}\frac{2x}{1-x^2} = tan^{-1}\left(\frac{x+\frac{2x}{1-x^2}}{1-x\left(\frac{2x}{1-x^2}\right)}\right)$

$= tan^{-1}\left(\frac{x-x^3+2x/1-x^2}{1-x^2-2x^2/1-x^2}\right)$

$= tan^{-1}\frac{3x-x^3}{1-3x^2}$

8.Simplify: $tan^{-1}\frac{x}{y} - tan^{-1}\frac{x-y}{x+y}$.

$tan^{-1}x - tan^{-1}y = tan^{-1}\left(\frac{x-y}{1+xy}\right)$

$tan^{-1}\frac{x}{y} - tan^{-1}\frac{x-y}{x+y} = tan^{-1}\left(\frac{\frac{x}{y}-\frac{x-y}{x+y}}{1+\left(\frac{x}{y}\right)\left(\frac{x-y}{x+y}\right)}\right)$

$= tan^{-1}\left(\frac{x(x+y)-y(x-y)/y(x+y)}{y(x+y)+x(x-y)/y(x+y)}\right)$

$= tan^{-1}\left(\frac{x^2+xy-xy+y^2}{xy+y^2+x^2-xy}\right)$

$= tan^{-1}\left(\frac{x^2+y^2}{x^2+y^2}\right)$

$= tan^{-1}(1)$

$= tan^{-1}\left(tan\frac{\pi}{4}\right)$

$= \frac{\pi}{4}$

(OR)

: $tan^{-1}\frac{x}{y} - tan^{-1}\frac{x-y}{x+y} = tan^{-1}\frac{x}{y} - tan^{-1}\frac{1-y/x}{1+y/x}$

$= tan^{-1}\frac{x}{y} - \left[tan^{-1}(1) - tan^{-1}\left(y/x\right)\right]$

$= tan^{-1}\frac{x}{y} - \left[\frac{\pi}{4} - cot^{-1}\left(x/y\right)\right]$

$= tan^{-1}\frac{x}{y} + cot^{-1}\left(x/y\right) - \frac{\pi}{4}\ \left(\because tan^{-1}x + cot^{-1}x = \frac{\pi}{2}\right)$

$= \frac{\pi}{2} - \frac{\pi}{4}$

$= \frac{\pi}{4}$

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9.Solve $(i) sin^{-1}\frac{5}{x} + sin^{-1}\frac{12}{x} = \frac{\pi}{2}$

$(ii) 2tan^{-1}x = cos^{-1}\frac{1-a^2}{1+a^2} - cos^{-1}\frac{1-b^2}{1+b^2}, a > 0, b > 0$

$(iii) 2tan^{-1}(\cos x) = tan^{-1}(2cosecx)$ $(iv) cot^{-1}x - cot^{-1}(x+2) = \frac{\pi}{12}, x > 0$

$(i) sin^{-1}\frac{5}{x} + sin^{-1}\frac{12}{x} = \frac{\pi}{2}$

$\Longrightarrow sin^{-1}\frac{5}{x} = \frac{\pi}{2} - sin^{-1}\frac{12}{x}$

$\Longrightarrow sin^{-1}\frac{5}{x} = cos^{-1}\frac{12}{x}$ $\left(\because sin^{-1}x + cos^{-1}x = \frac{\pi}{2}\right)$

$\Longrightarrow sin^{-1}\frac{5}{x} = sin^{-1}\sqrt{1-\left(\frac{12}{x}\right)^2}$ $\left(\because cos^{-1}x = sin^{-1}\sqrt{1-x^2}\right)$

$\Longrightarrow \frac{5}{x} = \sqrt{1-\frac{144}{x^2}}$

$\Longrightarrow \frac{5}{x} = \sqrt{\frac{x^2-144}{x^2}}$

$\Longrightarrow \frac{5}{x} = \frac{\sqrt{x^2-144}}{x}$

$\Longrightarrow 5 = \sqrt{x^2-144}$

Squaring on both sides,

$\Longrightarrow 25 = x^2 - 144$

$\Longrightarrow x^2 = 169$

$\Longrightarrow x = 13$

(ii) *Let* $a = \tan\alpha, b = \tan\beta \Longrightarrow \alpha = tan^{-1}a, \beta = tan^{-1}b$

$2tan^{-1}x = cos^{-1}\frac{1-a^2}{1+a^2} - cos^{-1}\frac{1-b^2}{1+b^2}$

$$= cos^{-1}\frac{1-tan^2\alpha}{1+tan^2\alpha} - cos^{-1}\frac{1-tan^2\beta}{1+tan^2\beta}$$
$$= cos^{-1}(cos2\alpha) - cos^{-1}(cos2\beta)$$
$$2tan^{-1}x = 2\alpha - 2\beta$$
$$\Rightarrow tan^{-1}x = \alpha - \beta$$
$$\Rightarrow tan^{-1}x = tan^{-1}a - tan^{-1}b$$
$$\Rightarrow tan^{-1}x = tan^{-1}\frac{a-b}{1+ab}$$
$$\Rightarrow x = \frac{a-b}{1+ab}$$
$$(iii)2tan^{-1}(\cos x) = tan^{-1}(2cosecx)$$
$$\Rightarrow tan^{-1}\left(\frac{2cosx}{1-cos^2x}\right) = tan^{-1}(2cosecx) \qquad \left(\because 2tan^{-1}x = tan^{-1}\left(\frac{2x}{1-x^2}\right)\right)$$
$$\Rightarrow \frac{2cosx}{1-cos^2x} = 2cosecx$$
$$\Rightarrow \frac{2cosx}{1-cos^2x} = \frac{2}{sinx}$$
$$\Rightarrow \frac{\cos x}{sin^2x} = \frac{1}{sinx}$$
$$\Rightarrow \frac{\sin x}{\cos x} = 1$$
$$\Rightarrow \tan x = 1$$
$$\Rightarrow x = n\pi + \frac{\pi}{4}, n \in z$$
$$(iv)cot^{-1}x - cot^{-1}(x+2) = \frac{\pi}{12}$$
$$\Rightarrow tan^{-1}\left(\frac{1}{x}\right) - tan^{-1}\left(\frac{1}{x+2}\right) = \frac{\pi}{12}$$
$$\Rightarrow tan^{-1}\left(\frac{\frac{1}{x}-\frac{1}{x+2}}{1+\left(\frac{1}{x}\right)\left(\frac{1}{x+2}\right)}\right) = \frac{\pi}{12}$$
$$\Rightarrow tan^{-1}\left(\frac{x+2-x}{x^2+2x+1}\right) = \frac{\pi}{12}$$
$$\Rightarrow \frac{2}{(x+1)^2} = \tan\frac{\pi}{12}$$
$$\Rightarrow \frac{2}{(x+1)^2} = tan15 \rightarrow (1)$$
$$tan15 = \tan(45-30) = \frac{\tan 45 - \tan 30}{1+\tan 45 \tan 30}$$
$$= \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$
$$= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$
$$= \frac{3-2\sqrt{3}+1}{3-1}$$
$$= \frac{4-2\sqrt{3}}{2}$$
$$= 2-\sqrt{3}$$
$$(1) \Rightarrow \frac{2}{(x+1)^2} = 2-\sqrt{3} \Rightarrow (x+1)^2 = \frac{2}{2-\sqrt{3}}$$
$$\Rightarrow (x+1)^2 = \frac{2}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}$$
$$\Rightarrow (x+1)^2 = \frac{4+2\sqrt{3}}{4-3}$$
$$\Rightarrow (x+1)^2 = 4+2\sqrt{3}$$
$$\Rightarrow (x+1)^2 = 3+2\sqrt{3}+1$$
$$\Rightarrow (x+1)^2 = \sqrt{3}^2 + 2\sqrt{3} + 1^2$$
$$\Rightarrow (x+1)^2 = \left(\sqrt{3}+1\right)^2$$
$$\Rightarrow x+1 = \sqrt{3}+1$$
$$\Rightarrow x = \sqrt{3}$$

10.Find the number of solution of the equation $tan^{-1}(x-1) + tan^{-1}x + tan^{-1}(x+1) = tan^{-1}(3x)$

$$\Rightarrow tan^{-1}(x-1) + tan^{-1}(x+1) = tan^{-1}(3x) - tan^{-1}x$$
$$\Rightarrow tan^{-1}\left(\frac{x-1+x+1}{1-(x-1)(x+1)}\right) = tan^{-1}\left(\frac{3x-x}{1+(3x)(x)}\right)$$
$$\Rightarrow tan^{-1}\left(\frac{2x}{1-(x^2-1)}\right) = tan^{-1}\left(\frac{2x}{1+3x^2}\right)$$
$$\Rightarrow tan^{-1}\left(\frac{2x}{2-x^2}\right) = tan^{-1}\left(\frac{2x}{1+3x^2}\right)$$
$$\Rightarrow \frac{2x}{2-x^2} = \frac{2x}{1+3x^2}$$
$$\Rightarrow 2x(1+3x^2) = 2x(2-x^2)$$
$$\Rightarrow 2x + 6x^3 = 4x - 2x^3$$
$$\Rightarrow 8x^3 - 2x = 0$$
$$\Rightarrow 4x^3 - x = 0,$$ is a cubic equation and so it has three solutions.

EXAMPLE 4.16 *Prove that* $\frac{\pi}{2} \le sin^{-1}x + 2cos^{-1}x \le \frac{3\pi}{2}$.

$$sin^{-1}x + 2cos^{-1}x = sin^{-1}x + cos^{-1}x + cos^{-1}x = \frac{\pi}{2} + cos^{-1}x$$

$0 \le cos^{-1}x \le \pi \Rightarrow \frac{\pi}{2} + 0 \le \frac{\pi}{2} + cos^{-1}x \le \frac{\pi}{2} + \pi$

$\Rightarrow \frac{\pi}{2} \le sin^{-1}x + cos^{-1}x + cos^{-1}x \le \frac{3\pi}{2}$

$\Rightarrow \frac{\pi}{2} \le sin^{-1}x + 2cos^{-1}x \le \frac{3\pi}{2}$

EXAMPLE 4.17 Simplify: $(i) cos^{-1}\left(\cos\left(\frac{13\pi}{3}\right)\right)$ $ii) tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$ $(iii) sec^{-1}\left(\sec\left(\frac{5\pi}{3}\right)\right)$ $(iv) sin^{-1}(sin10)$

$(i) cos^{-1}\left(\cos\left(\frac{13\pi}{3}\right)\right) = cos^{-1}\left(\cos\left(4\pi + \frac{\pi}{3}\right)\right) = cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3} \in [0, \pi]$

$ii) tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right) = tan^{-1}\left(\tan\left(\pi - \frac{\pi}{4}\right)\right) = tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right)$

$= -\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$(iii) sec^{-1}\left(\sec\left(\frac{5\pi}{3}\right)\right) = sec^{-1}\left(\sec\left(2\pi - \frac{\pi}{3}\right)\right) = sec^{-1}\left(\sec\left(\frac{\pi}{3}\right)\right)$

$= \frac{\pi}{3} \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$

$(iv) sin^{-1}(sin10) = sin^{-1}\left[sin\left(3\pi + (10 - 3\pi)\right)\right]$

$= sin^{-1}\left[-sin\left((10 - 3\pi)\right)\right]$

$= -sin^{-1}\left[sin\left((10 - 3\pi)\right)\right]$

$= -(10 - 3\pi)$

$= 3\pi - 10 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

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EXAMPLE 4.18 Find the value of

$(i)\ \sin\left[\frac{\pi}{3} - sin^{-1}\left(-\frac{1}{2}\right)\right]$ $(ii)\ \cos\left[\frac{1}{2}cos^{-1}\left(\frac{1}{8}\right)\right]$

$(iii)\ \tan\left[\frac{1}{2}sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right]$

$(i)\ \sin\left[\frac{\pi}{3} - sin^{-1}\left(-\frac{1}{2}\right)\right] = \sin\left[\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right] = \sin\left[\frac{\pi}{3} + \frac{\pi}{6}\right] = \sin\left[\frac{\pi}{2}\right] = 1$

$(ii)\ \cos\left[\frac{1}{2}cos^{-1}\left(\frac{1}{8}\right)\right]$

$cos^{-1}\left(\frac{1}{8}\right) = \theta \rightarrow (1)$

$\Rightarrow \cos\theta = \frac{1}{8}$

$\Rightarrow 2cos^2\left(\frac{\theta}{2}\right) - 1 = \frac{1}{8}$

$\Rightarrow 2cos^2\left(\frac{\theta}{2}\right) = \frac{9}{8}$

$\Rightarrow cos^2\left(\frac{\theta}{2}\right) = \frac{9}{16}$

$\Rightarrow \cos\left(\frac{\theta}{2}\right) = \frac{3}{4}$

$\Rightarrow \cos\left[\frac{1}{2}cos^{-1}\left(\frac{1}{8}\right)\right] = \frac{3}{4}$ $(From\ (1))$

$(iii)\ \tan\left[\frac{1}{2}sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right]$

Let $a = \tan\theta$.

$\tan\left[\frac{1}{2}sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right]$

$= \tan\left[\frac{1}{2}sin^{-1}\left(\frac{2\tan\theta}{1+tan^2\theta}\right) + \frac{1}{2}cos^{-1}\left(\frac{1-tan^2\theta}{1+tan^2\theta}\right)\right]$

$= \tan\left[\frac{1}{2}sin^{-1}(\sin 2\theta) + \frac{1}{2}cos^{-1}(\cos 2\theta)\right]$ $\left(\because \frac{2\tan\theta}{1+tan^2\theta} = \sin 2\theta\ ; \frac{1-tan^2\theta}{1+tan^2\theta} = \cos 2\theta\right)$

$= \tan\left[\frac{1}{2}(2\theta) + \frac{1}{2}(2\theta)\right]$

$= \tan[\theta + \theta]$

$= \tan[2\theta]$

$= \frac{2\tan\theta}{1-tan^2\theta}$ $\left(\because \tan[2A] = \frac{2\tan A}{1-tan^2 A}\right)$

$= \frac{2a}{1-a^2}$

EXAMPLE 4.19 Prove that $\tan(sin^{-1}x) = \frac{x}{\sqrt{1-x^2}}$ for $|x| < 1$.

$sin^{-1}x = \theta \Rightarrow x = \sin\theta, -1 \le x \le 1.$

$\tan(sin^{-1}x) = \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sin\theta}{\sqrt{1-sin^2\theta}} = \frac{x}{\sqrt{1-x^2}}, |x| < 1$

EXAMPLE 4.20 Evaluate $\sin\left[sin^{-1}\left(\frac{3}{5}\right) + sec^{-1}\left(\frac{5}{4}\right)\right]$

$sec^{-1}\left(\frac{5}{4}\right) = \theta \Rightarrow \sec\theta = \frac{5}{4} \Rightarrow \cos\theta = \frac{4}{5}$

$\Rightarrow \sin\theta = \sqrt{1 - cos^2\theta} = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25}} = \frac{3}{5} \Rightarrow \theta = sin^{-1}\left(\frac{3}{5}\right)$

$\therefore sec^{-1}\left(\frac{5}{4}\right) = sin^{-1}\left(\frac{3}{5}\right)$

$\sin\left[sin^{-1}\left(\frac{3}{5}\right) + sec^{-1}\left(\frac{5}{4}\right)\right]$

$= \sin\left[sin^{-1}\left(\frac{3}{5}\right) + sin^{-1}\left(\frac{3}{5}\right)\right]$

$= \sin\left[2sin^{-1}\left(\frac{3}{5}\right)\right]$

$= \sin\left[sin^{-1}\left(2 \times \frac{3}{5}\sqrt{1-\left(\frac{3}{5}\right)^2}\right)\right]$ $\left(\because If\ |x| \le \frac{1}{\sqrt{2}},\ sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2sin^{-1}x\right)$

$= \sin\left[sin^{-1}\left(\frac{6}{5}\sqrt{\frac{16}{25}}\right)\right]$

$= \sin\left[sin^{-1}\left(\frac{6}{5} \times \frac{4}{5}\right)\right]$

$= \sin\left[sin^{-1}\left(\frac{24}{25}\right)\right]$

$= \frac{24}{25} \in [-1,1]$

(OR)

$sec^{-1}\left(\frac{5}{4}\right) = \theta \to (1) \quad \Rightarrow \sec\theta = \frac{5}{4} \Rightarrow \cos\theta = \frac{4}{5}$

$\Rightarrow \sin\theta = \sqrt{1-cos^2\theta} = \sqrt{1-\left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25}} = \frac{3}{5} \Rightarrow \theta = sin^{-1}\left(\frac{3}{5}\right) \to (2)$

$\sin\left[sin^{-1}\left(\frac{3}{5}\right) + sec^{-1}\left(\frac{5}{4}\right)\right]$

$= \sin[\theta + \theta]$ $From\ (1),(2)$

$= \sin[2\theta]$

$= 2\sin\theta\cos\theta$

$= 2 \times \frac{3}{5} \times \frac{4}{5}$

$= \frac{24}{25} \in [-1,1]$

EXAMPLE 4.21 Prove that $(i) tan^{-1}\frac{1}{2} + tan^{-1}\frac{1}{3} = \frac{\pi}{4}$

$(ii) 2tan^{-1}\frac{1}{2} + tan^{-1}\frac{1}{7} = tan^{-1}\frac{31}{17}$

$(i) tan^{-1}\frac{1}{2} + tan^{-1}\frac{1}{3}$ $\left(\because tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1\right)$

$= tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}\right) = tan^{-1}\left(\frac{5/6}{5/6}\right) = tan^{-1}(1) = \frac{\pi}{4}$

$(ii) 2tan^{-1}\frac{1}{2} + tan^{-1}\frac{1}{7}$

$= tan^{-1}\left(\frac{2\left(\frac{1}{2}\right)}{1-\left(\frac{1}{2}\right)^2}\right) + tan^{-1}\frac{1}{7}$ $\left(\because 2tan^{-1}x = tan^{-1}\left(\frac{2x}{1-x^2}\right), |x| < 1\right)$

$= tan^{-1}\left(\frac{1}{3/4}\right) + tan^{-1}\frac{1}{7}$

$= tan^{-1}\frac{4}{3} + tan^{-1}\frac{1}{7}$

$= tan^{-1}\left(\frac{\frac{4}{3}+\frac{1}{7}}{1-\left(\frac{4}{3}\right)\left(\frac{1}{7}\right)}\right)$ $\left(\because tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1\right)$

$= tan^{-1}\left(\frac{31/21}{17/21}\right)$

$= tan^{-1}\frac{31}{17}$

EXAMPLE 4.22 $If\ cos^{-1}x + cos^{-1}y + cos^{-1}z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$.

$cos^{-1}x = \alpha \Rightarrow x = \cos\alpha \Rightarrow \sin\alpha = \sqrt{1-cos^2\alpha} = \sqrt{1-x^2}$

$cos^{-1}y = \beta \Rightarrow y = \cos\beta \Rightarrow \sin\beta = \sqrt{1-cos^2\beta} = \sqrt{1-y^2}$ $\Big\} \to (1)$

$cos^{-1}x + cos^{-1}y + cos^{-1}z = \pi \Rightarrow \alpha + \beta + cos^{-1}z = \pi$

$\Rightarrow \alpha + \beta = \pi - cos^{-1}z \to (2)$

$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$From\ (1),(2),$

$\Rightarrow \cos(\pi - cos^{-1}z) = xy - \sqrt{1-x^2}\sqrt{1-y^2}$

$\Rightarrow -cos(cos^{-1}z) = xy - \sqrt{1-x^2}\sqrt{1-y^2}$ $(\because \cos(\pi-\theta) = -\cos\theta)$

$\Rightarrow -z = xy - \sqrt{1-x^2}\sqrt{1-y^2}$

$\Rightarrow -xy - z = -\sqrt{1-x^2}\sqrt{1-y^2}$

Squaring on both sides,

$\Rightarrow (xy+z)^2 = (1-x^2)(1-y^2)$

$\Rightarrow x^2y^2 + z^2 + 2xyz = 1 - y^2 - x^2 + x^2y^2$

$\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1$

EXAMPLE 4.23 $If\ a_1, a_2, a_3, \ldots, a_n$ is an arithmetic progression with common difference d, prove that

$tan\left[tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + \cdots . + tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right)\right] = \frac{a_n - a_1}{1+a_1a_n}$

$a_1, a_2, a_3, \ldots, a_n$ is an A.P.

$tan^{-1}\left(\frac{d}{1+a_1a_2}\right) = tan^{-1}\left(\frac{a_2-a_1}{1+a_1a_2}\right) = tan^{-1}a_2 - tan^{-1}a_1$

$tan^{-1}\left(\frac{d}{1+a_2a_3}\right) = tan^{-1}\left(\frac{a_3-a_2}{1+a_2a_3}\right) = tan^{-1}a_3 - tan^{-1}a_2$

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$$tan^{-1}\left(\frac{d}{1+a_n a_{n-1}}\right) = tan^{-1}\left(\frac{a_n - a_{n-1}}{1+a_n a_{n-1}}\right) = tan^{-1}a_n - tan^{-1}a_{n-1}$$

$$tan\left[tan^{-1}\left(\frac{d}{1+a_1 a_2}\right) + tan^{-1}\left(\frac{d}{1+a_2 a_3}\right) + \cdots . + tan^{-1}\left(\frac{d}{1+a_n a_{n-1}}\right)\right]$$

$$= tan[tan^{-1}a_2 - tan^{-1}a_1 + tan^{-1}a_3 - tan^{-1}a_2 + \cdots . + tan^{-1}a_n - tan^{-1}a_{n-1}]$$

$$= tan[tan^{-1}a_n - tan^{-1}a_1]$$

$$= tan^{-1}\left(\frac{a_n - a_1}{1+a_n a_1}\right)$$

$$= \frac{a_n - a_1}{1+a_1 a_n}$$

EXAMPLE 4.24 Solve $tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}tan^{-1}x$, for $x > 0$.

$$tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}tan^{-1}x \Rightarrow tan^{-1}1 - tan^{-1}x = \frac{1}{2}tan^{-1}x$$

$$\Rightarrow tan^{-1}1 = \frac{1}{2}tan^{-1}x + tan^{-1}x$$

$$\Rightarrow \frac{\pi}{4} = \frac{3}{2}tan^{-1}x$$

$$\Rightarrow tan^{-1}x = \frac{\pi}{4} \times \frac{2}{3}$$

$$\Rightarrow tan^{-1}x = \frac{\pi}{6}$$

$$\Rightarrow x = \tan\frac{\pi}{6}$$

$$\Rightarrow x = \frac{1}{\sqrt{3}}$$

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EXAMPLE 4.25 Solve $sin^{-1}x > cos^{-1}x$.

$$sin^{-1}x > cos^{-1}x \Rightarrow sin^{-1}x + sin^{-1}x > sin^{-1}x + cos^{-1}x$$

$$\Rightarrow 2sin^{-1}x > \frac{\pi}{2}$$

$$\Rightarrow sin^{-1}x > \frac{\pi}{4}$$

$$\Rightarrow x > \sin\frac{\pi}{4}$$

$$\Rightarrow x > \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = (\frac{1}{\sqrt{2}}, 1]$$

EXAMPLE 4.26 *Show that* $\cot(sin^{-1}x) = \frac{\sqrt{1-x^2}}{x}, -1 \le x \le 1$ and $x \neq 0$

$$sin^{-1}x = \theta \Rightarrow \sin\theta = \frac{x}{1} \Rightarrow \cos\theta = \sqrt{1 - sin^2\theta} = \sqrt{1-x^2}$$

$$\cot(\theta) = \frac{\cos\theta}{\sin\theta} = \frac{\sqrt{1-x^2}}{x} \Rightarrow \cot(sin^{-1}x) = \frac{\sqrt{1-x^2}}{x}$$

EXAMPLE 4.27 *Solve* $tan^{-1}2x + tan^{-1}3x = \frac{\pi}{4}, if\ 6x^2 < 1.$

$$tan^{-1}2x + tan^{-1}3x = \frac{\pi}{4}$$

$$\Rightarrow tan^{-1}\left[\frac{2x+3x}{1-(2x)(3x)}\right] = \frac{\pi}{4} \qquad \left(\because tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)\right)$$

$$\Rightarrow tan^{-1}\left[\frac{5x}{1-6x^2}\right] = \frac{\pi}{4}$$

$$\Rightarrow \frac{5x}{1-6x^2} = \tan\frac{\pi}{4}$$

$$\Rightarrow \frac{5x}{1-6x^2} = 1$$

$$\Rightarrow 5x = 1 - 6x^2$$

$$\Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow (6x-1)(x+1) = 0 \Rightarrow x = \frac{1}{6}\ ;\ x = -1 \text{ (Not a soln.)}$$

EXAMPLE 4.28 Solve $tan^{-1}\left(\frac{x-1}{x-2}\right) + tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

$$tan^{-1}\left(\frac{x-1}{x-2}\right) + tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$$

$$\Rightarrow tan^{-1}\left[\frac{\frac{x-1}{x-2}+\frac{x+1}{x+2}}{1-\left(\frac{x-1}{x-2}\right)\left(\frac{x+1}{x+2}\right)}\right] = \frac{\pi}{4} \qquad \left(\because tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)\right)$$

$$\Rightarrow \left[\frac{(x-1)(x+2)+(x+1)(x-2)\,/\,(x-2)(x+2)}{(x-2)(x+2)-(x-1)(x+1)\,/\,(x-2)(x+2)}\right] = \tan\frac{\pi}{4}$$

$$= \left[\frac{x^2+x-2+x^2-x-2}{x^2-4-x^2+1}\right] = 1$$

$$\Rightarrow \frac{2x^2-4}{-3} = 1$$

$$\Rightarrow 2x^2 - 4 = -3$$

$$\Rightarrow 2x^2 = 1$$

$$\Rightarrow x^2 = \frac{1}{2}$$

$$\Rightarrow x = \pm\frac{1}{\sqrt{2}}$$

EXAMPLE 4.29 Solve $\cos\left(sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right) = \sin\left\{cot^{-1}\left(\frac{3}{4}\right)\right\}$

$sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \alpha \Rightarrow \sin\alpha = \frac{x}{\sqrt{1+x^2}}$

$\cos\alpha = \sqrt{1-sin^2\alpha} = \sqrt{1-\left(\frac{x}{\sqrt{1+x^2}}\right)^2} = \sqrt{1-\frac{x^2}{1+x^2}} = \sqrt{\frac{1}{1+x^2}} = \frac{1}{\sqrt{1+x^2}}$

$\alpha = cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

$sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \alpha \Rightarrow sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

$\cos\left(sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right) = \cos\left(cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right) = \frac{1}{\sqrt{1+x^2}} \Rightarrow (1)$

$cot^{-1}\left(\frac{3}{4}\right) = \beta \Rightarrow \cot\beta = \frac{3}{4}$

$opposite\ side = 4,\ \ adjacent\ side = 3$

$Hypotenuse = \sqrt{4^2+3^2} = \sqrt{25} = 5$

$\sin\beta = \frac{4}{5} \Rightarrow \beta = sin^{-1}\left(\frac{4}{5}\right)$

$cot^{-1}\left(\frac{3}{4}\right) = \beta \Rightarrow cot^{-1}\left(\frac{3}{4}\right) = sin^{-1}\left(\frac{4}{5}\right)$

$\sin\left\{cot^{-1}\left(\frac{3}{4}\right)\right\} = \sin\left\{sin^{-1}\left(\frac{4}{5}\right)\right\} = \frac{4}{5} \rightarrow (2)$

$From\ (1), (2),$

$\frac{1}{\sqrt{1+x^2}} = \frac{4}{5}$

$\Rightarrow \sqrt{1+x^2} = \frac{5}{4}$

Squaring on both sides,

$\Rightarrow 1+x^2 = \frac{25}{16}$

$\Rightarrow x^2 = \frac{25}{16} - 1$

$\Rightarrow x^2 = \frac{9}{16}$

$\Rightarrow x = \pm\frac{3}{4}$

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